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Studies in the Greek Thematic Aorist:
Root Vocalism and Stem Distribution Patterns

A dissertation submitted in partial satisfaction
of the requirements for the degree
Doctor of Philosophy in Indo-European Studies

by

Chengzhi Zhang

2025

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ABSTRACT OF THE DISSERTATION

Studies in the Greek Thematic Aorist:
Root Vocalism and Stem Distribution Patterns

by

Chengzhi Zhang
Doctor of Philosophy in Indo-European Studies
University of California, Los Angeles, 2025
Professor Brent H. Vine, Chair

This dissertation examines two interconnected problems in the historical grammar of the Greek thematic aorist, a verbal category characterized by the zero-graded root and thematic inflection that, despite having a small reconstructable footprint in Proto-Indo-European, achieved remarkable productivity in Greek, especially in the Homeric Epics. The first study addresses the anomalous *o*-vocalism found in a small group of thematic aorists (ἔθορον 'leapt', ἔμολον 'went', ἔπορον 'gave', ἔτορον 'pierced') traditionally derived from *h_3-final roots of the shape *$CeRh_3$-. Against the standard analysis proposing metathesis (*-*eRo*- > -*oRe*-), I demonstrate that such accounts lack independent phonological motivation and face substantial empirical difficulties. Through careful examination of alleged counterexamples, particularly Pindar's πεπαρεῖν and the problematic ἐάλων, the dissertation proposes that the *o*-vocalism represents the regular Greek reflex of *$C\underset{\circ}{R}h_3V$- sequences, conditioned by preceding labial consonants. This phonological development eliminates the need for *ad hoc* metatheses and potentially provides a unified account of the "*o*-grade" thematic aorist and certain root present formations with *o*-vocalism. The second study investigates the stem distribution pattern exemplified by ἔλιπον-λιμπάνω-λείπω, where zero-grade thematic aorists of *$CeRT$- roots coexist with the so-called double-nasal presents and full-grade thematic presents. A detailed philological examination of seven verbs in this group reveals that the thematic aorist occupies the central position, the nasal presents show

weak early attestation despite comparative support for inheritance, and the thematic presents, though dominant in Homer, lack clear Indo-European parallels. I propose that this coordination pattern arose through analogical extension from *$CeRh_{1/3}$- roots, where early thematization of both aorist and nasal present stems provided the model for morphological restructuring in *$CeRT$- roots. The double-nasal formation is construed as a relatively late innovation emerging after the establishment of the pattern thematic aorist + νω-present. By situating these investigations within broader patterns of stem distribution and paradigmatic relationships, this dissertation contributes to understanding the mechanisms underlying the spread and formal development of the thematic aorist in the pre-history of Greek.

The dissertation of Chengzhi Zhang is approved.

Stephanie W. Jamison

Alexander Nikolaev

Anthony D. Yates

Brent H. Vine, Committee Chair

University of California, Los Angeles

2025

To my parents

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ABBREVIATIONS

Linguistic terms and symbols

*X	X is a proposed reconstructed form
†X	X is unattested
> X	becomes X by regular sound change
=> X	becomes X by analogy, levelling, or other non-phonological changes
>> X	analogically produces X
X̥	syllabic X
X̯	consonantal X
∅̂	zero-grade
acc.	accusative
act.	active
adj.	adjective
aor.	aorist
athem.	athematic
C	any consonant
dat.	dative
fut.	future
gen.	genitive
H	any laryngeal (*h*₁, *h*₂, *h*₃)
Hi…Ei…	the quality of the vowel *Ei* depends on the color of the laryngeal *Hi*
I	any glide (*i, u*)
imper.	imperative
impf.	imperfect
ind.	indicative
inf.	infinitive
inj.	injunctive
instr.	instrumental
intr.	intransitive
L	any liquid (*l, r*)
loc.	locative
mid.	middle
N	any nasal (*m, n, ŋ*)
n.	neuter
nom.	nominative
opt.	optative
part.	participle
pass.	passive
perf.	perfect
pl.	plural
pluperf.	pluperfect
pres.	present
R	any resonant (*i̯*, *u̯*, *l, r, m*, *n*)
R(…)	root in … grade
sg.	singular
subj.	subjunctive
T	any stop
them.	thematic
tr.	transitive
V	any vowel
voc.	vocative

Languages and dialects

Aeol.	Aeolic
Arm.	Armenian
Arc.	Arcadian
Att.	Attic
Boet.	Boeotian
Cret.	Cretan
CLuw.	Cuneiform Luwian
Dor.	Doric
Gk.	Greek
Goth.	Gothic
Hitt.	Hittite
HLuw.	Hieroglyphic Luwian
Hom.	Homeric
IIr.	Indo-Iranian
Ion.	Ionic
Khot.	Khotanese
Lac.	Laconian
Lat.	Latin
Lith.	Lithuanian
Lyc.	Lycian
MHG	Middle High German
MoE	Modern English
MPers.	Middle Persian
Myc.	Mycenaean Greek
OCS	Old Church Slavonic
OE	Old English
OFris.	Old Frisian
OHG	Old High German
OIr.	Old Irish
ON	Old Norse
Osc.	Oscan
PA	Proto-Anatolian
Parth.	Parthian
PGk.	Proto-Greek
PGmc.	Proto-Germanic
PIr.	Proto-Iranian
PIt.	Proto-Italic
PCelt.	Proto-Celtic
Skt.	Sanskrit
Sogd.	Sogdian
TA	Tocharian A
TB	Tocharian B
Thess.	Thessalian
Umbr.	Umbrian
Ved.	Vedic Sanskrit
YAv.	Young Avestan

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2024	Cross-morphemic laryngeal coloring in PIE: a short survey. The 43rd East Coast Indo-European Conference, University of Georgia, Athens, GA.
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Chapter 1
Introduction

1.1 The Greek thematic aorist and current issues

Within the Greek verbal system, the thematic aorist is a curious construction posing many questions about the diachronic grammar of Greek and Indo-European that have since the late 19th century attracted continuous scholarly attention. The canonical structure of this category in Greek is characterized by the simple non-ablauting zero-grade root followed by the thematic vowel (*-é/ó-*)[1] and the use of the secondary endings, in the shape:

**(e)-R(Ø)-é/ó-m, -s, -t*, etc., e.g. ἔ-λιπ-ο-ν 'I left' (: λείπω) < **lik*^{w}-

Much less frequently one also finds formations with the root in the *e*-grade, e.g. ἔτεκον 'I begot' (: τίκτω), *o*-grade, e.g. ἔμολον 'I went' (: βλώκσω), or with reduplicated stems, e.g. κέκλετο 's/he urged' (: κέλομαι). The discussions surrounding the thematic aorist so far have seen a strong focus on the debate of its status in the Proto-Indo-European (PIE) verbal system, but a relatively thin coverage on the analysis of the historical materials. The last treatment of the Greek data is still the single chapter in Cardona 1960 (supplemented by the more recent but partial coverage from Harđarson 1993 and Willi 2018), which is limited in scope and in many aspects needs an update and re-evaluation.[2]

One of the central issues surrounding the thematic aorist that is still not fully understood

[1] The accent is recessive in the finite forms in Greek, but the original accent on the thematic vowel is preserved in, e.g., imper. mid. 2sg. -οῦ (< **-é-o*), the infinitive ending -εῖν (< **-é-en*)

[2] The *reduplicated* thematic aorist has received extensive coverage (see, among others, Bendahman 1993, Beckwith 1996, and recently Willi 2018: 58-117) and will not be addressed in any detail in the current study.

is the striking contrast between its prolific profile in Greek, especially in Homer, and its marginal status in PIE with a relative poverty of reflexes in most daughter languages. The modern *communis opinio* recognizes only two reconstructable word equations for this category, namely **u̯id-é/ó-* 'see, know' (Skt. *vida-*, Gk. (ϝ)ιδε/o-, Arm. *egit*, etc.) and **h₁ludh-é/ó-* 'climb' (Gk. ἐλυθε/o-, TA *läc*, TB *lac*, OIr. *luid*), with the PIE antiquity of the thematic aorist heavily debated. Its productivity, however, is profoundly observed in Greek, especially in the Homeric epics, where 127 verbs attest a thematic aorist formation.[3] Thus, a series of questions central to the understanding of the Greek materials arise, including:

(1) Through what mechanisms did the small number of early types instigate the spread of a productive category, in so far as the patterns of the stem distribution of these verbs in Greek and across Indo-European can inform us?[4]

(2) How do the Greek thematic aorists end up with the formal properties (including the observed root vocalism and stem shapes) that they possess?

This study therefore takes a modest but, I hope, useful step toward addressing these questions related to how the thematic aorist behaves in the early (and pre-) history of Greek by concentrating on two targeted investigations. The first examines the peculiar root vocalism

[3] Based on Risch 1974:237-45, cf. Sandell 2015:110 which counts 117. Traces of thematic aorist are already found in Mycenaean, e.g. *-wi-de* /-wide/ 's/he saw' (PY Eq 213.1), *ra-ke* /lak^{h}e/ 'received by lot' (PY Eb 159.B), and a handful of other less secure instances (cf. Bartoněk 2003:340, Bernabé and Luján 2020:193-4; on potential traces of φυγέ/ό- in Mycenaean onomastics cf. fn.207). The category continues to be active and moderately productive in the post-Homeric era, with some 40 verbs in common usage in Attic-Ionic (Gignac 1985:49), before being substituted by – and, to some extent, merging with – the sigmatic/alpha-thematic formation in Koiné.

[4] A simple answer of "thematization" as the Neogrammarians would put it is surely insufficient and no more than a descriptive term. The trend observed in late Greek in fact points to athematization/alpha-thematization, cf. fn. 3. The spread of the category in the prehistorical times thus stands in contrast with its fate in the historical period.

in the small but significant group of "*o*-grade" forms (e.g. ἔπορον 'gave', ἔμολον 'went', etc.) and seeks to shed new light on ongoing debates surrounding the thorny phonological issues these forms illustrate, particularly the reflex of **CR̥HV-* sequences. The second offers a detailed philological examination of the coordination pattern exemplified by the type ἔλιπον-λιμπάνω-λείπω and explores its implications for understanding both a type of mechanism underlying the spread of the thematic aorist in **CeRC-* roots and its interaction with the nasal and thematic root present. Together, these focused studies contribute toward a more grounded understanding of the thematic aorist's place in the Greek verbal system and help situate long-standing debates within a firmer analysis of the actual historical materials. In doing so, the dissertation aims not at a full reconstruction of the category, but at refining specific empirical points that are central to its diachronic interpretation.

1.2 Previous scholarship

Before turning to the analyses themselves, the next section reviews how earlier scholarship has approached the origins and historical status of the thematic aorist in Greek and Indo-European and sets out the main lines of inquiry inherited by the present study.

1.2.1 From Brugmann to Kuryłowicz: early attempts at theorizing the origins of the thematic aorist

The formal properties of the thematic aorist – largely based on the Greek and Sanskrit materials, where the formation is best represented – have been recognized since the early handbooks of Brugmann (1916: 113-6) and Meillet (1937: 202), both establishing two major thematic stem types for PIE, i.e. *R(é)-e/o-* and *R(Ø)-é/ó-*, the former mostly imperfective as seen in the simple thematic present of the type **b^{h}ér-e/o-* and the latter mostly perfective, i.e. the thematic aorist. Similarly, Hirt (1928:164, 240-2) speaks of a verbal formation in PIE he

terms as "Aorist-Präsenssystem" whereby either an aorist or a present can be built to a root depending on the position of the accent (i.e. radical vs. on the thematic vowel). These early approaches to explain the thematic aorist as a pristine PIE verbal category in an *a priori* system where aspects are determined by the shift of accents (and thereby change of ablaut) are no longer maintained, as accent shift does not in general predict or affect aspectual properties of verbal stems in PIE, cf. Cardona 1960:4-5.

The attempt to explain the thematic aorist as a secondary or innovated formation began as early as Thurneysen 1894 (and later again in 1936: 116 fn.1). In a paper dealing with a separate topic (on the formation of the type λιμπάνω, cf. below 3.9), Thurneysen stated, as an example of independent innovation, that "der sog. starke thematische Aorist mit schwacher Wurzelgestalt im Griechischen und Indischen nicht altererbt ist – ausser dem einen Verbum **évidom ávidam* ἔϝιδον." (1894: 84). He further raises the possibility that the thematic aorist could be created from athematic root aorists such as in the pair βαλεῖν : ἔβλητο 'throw' and from reduplicated aorists, e.g. τετάρπετο : ταρπώμεθα 'enjoy'. There is no doubt that Thurneysen's brief observations had a profound influence on the field, as the secondary nature of at least some of the thematic aorists started to be evaluated and different routes of thematization explored. Hirt in his handbook (*op. cit.* 241; citing an earlier form of the thought from Fick 1878: 179-80) hinted at the possibility that the thematic vowel could have been added to an erstwhile ablauting root formation. The idea is pursued further by Meillet in an extensive article (1931: 201) on the thematic formation in Indo-European where he argues that the opposition exhibited by Gk. λείπω : ἔλιπον is not of Indo-European antiquity, citing a lack of comparative support, but instead developed largely in daughter languages. Following this initial observation, Meillet and Vendryes (1948: 204-5) further elaborated on the process of the thematization by pinpointing two pivotal forms that allegedly triggered the change in Greek, namely the 3pl. in -ον and the participle in -ων, whence, e.g., an athematic ἔκταν-ον 'they killed' could be reanalyzed as ἔκτανο-ν, and part. κτανών could provide an analogy with φέρων : ἔφερε whereby ἔκτανε could have been deduced. While Meillet's

analysis offered the first tangible explanation for the genesis (or the spread) of the thematic aorist in Greek, an approach that is essentially followed later by Cardona, some key issues remain to be accounted for, not the least of which is the assumption of an athematic 3pl. ending *-*on(t)* in Greek, which, if assumed, seems to incur circularity, as the creation of -ον in Greek (or *-*ont* in PIE) as opposed to the athematic -εν (*-*ent*) is an unsolved matter on its own and probably intertwined with the problem of thematization after all. Among the early reference works, a similar suggestion on the relatively young age of the Greek thematic aorist is also made by Schwyzer (1939: 746), who implicitly considered all Greek instances to be either secondarily thematized or simply poetic creations.

Against these views, however, one still finds early advocates for the inherited status of the thematic aorist, including Chantraine (1961: 171), who firmly states that "les aoristes thématiques…constituent un vieux type indo-européen" and that the type is relatively well *preserved* in Greek. Venturing into deep reconstruction, Kuryłowicz (1956: 71-4) theorizes the creation of the thematic aorist under a larger framework of the origin of the PIE thematic vowel. He contends that a primordial thematic aorist formation of the shape **R(Ø)-é-m, -s, -t* (with constant *e*-grade of the thematic vowel instead of the ablauting *-é/ó-*) must precede the simple thematic present of the type **R(é)-e-* because the unaccented *e-* in the latter is phonologically unexpected (cf. already Hirt 1898). In other words, the thematic vowel was originally limited to the aorist formation. This idea was later modified in Kuryłowicz (1964: 116-8, 121-2) where he treats the thematic aorist as "discarded imperfects" of the so-called *tudáti* presents (i.e. thematic present formations with the root in the zero-grade and originally suffixal accent) after they in their primary use are either converted to simple thematic presents with full-grade of the root or recharacterized in other ways, and thus Greek represents the end stage of this change whereas Vedic represents an earlier one. Although the *tudáti* presents are likely connected with the origin of the thematic aorist based on their formal similarities, the scenario proposed by Kuryłowicz is not supported by the philological data, which, as rightly pointed out by Willi (2018:288-9) citing Arnold (1897: 317, 319), reveal that

diachronically the number of *tudáti* presents indeed increased in the different layers of the Vedic texts and thus they are unlikely to be a relic formation.[5]

1.2.2 Cardona 1960: the first comprehensive treatment

Many of these early attempts at theorizing the historical background of the thematic aorists in Greek and Sanskrit are marred by a lack of close and comprehensive examination of the data, which often results in either making intuitive judgements or system-building at the PIE level based on a limited number of cherry-picked forms. This situation is mended to some extent by Cardona's 1960 dissertation which, as the first of its kind, offers a thorough analysis of the thematic aorist formations across the Indo-European family. The main conclusion reached by Cardona is that only two thematic aorists can be considered of Indo-European origin, namely **u̯id-é/ó-* (Skt. *vida-*, Gk. (ϝ)ιδε/o-, Arm. *egit*, etc.) and **h₁ludh-é/ó-* (Gk. ἐλυθε/o-, TA. *läc*, TB. *lac*, OIr. *luid*), while the abundant stock of thematic aorists found in Vedic and Greek must and can be demonstrated to be secondary formations innovated independently, a theory that has become the *communis opinio* and been codified in many modern handbooks, cf. Fortson 2010: 102 and Beekes 2011: 263.[6] Methodologically speaking, Cardona employs essentially the same approach taken by Meillet and Vendryes (1948: 171, cf. above) and Whitney (1889: 306), namely to identify the ambiguous form(s) in the athematic paradigm through which a reanalysis as thematic formations can be plausibly envisaged. As far as the Greek materials are concerned, Cardona follows Meillet closely and proposed that the trigger point is the 3pl. of the athematic root aorist (1960: 68, 76). Realizing the difficulty in assuming a gratuitous PIE 3pl. athem. **-ont* (or an otherwise unexplained

[5] Malzahn's recent treatment (2016: 234) comes to a similar conclusion that "the number of present indicative forms from *tudáti*-presents proper must have been practically zero in Very Early Proto-Indo-Iranian" and that they came into being via a reinterpretation of the old aorist stems (cf. the "*tēzzi*-principle", as in Malzahn 2010: 267-8).

[6] Patently excluding the thematic aorist as an inherited formation are Meier-Brügger 2021 (178f.) and LIV^2 (20f.).

Greek. -ον) as Meillet did, Cardona submits that the origin of Gk. -ον is to be sought directly in an analogy with the present participle in -ων, as in φέφων : ἔφερον :: λιπών : X where X is resolved as ἔλιπον.[7] The mismatch of the accents in this formula notwithstanding, a couple of questions may be raised regarding the scenario. The first, minor consideration concerns the antiquity of the athematic participle in -ων, -οντος, or a PIE *-*ont*-. Although it has been widely suspected (Bammesberger 1981, Szemerényi 1990: 345-7, Peters 2004, Widmer 2004: 135-8, Meier-Brügger 2021: 191) that the *o*-grade -*nt*-suffix *-*ont*- is used to build participles to athematic verbal stems on the strength of Gk. (ἐ)όντ- 'being', Hitt. *ašānt*- 'id.' (see Melchert 1994: 146f.), Lat. *sōns* 'guilty', PGmc. *-*sanþ* (OE *sōþ* 'true'), and maybe the word for 'tooth' ὀδών (*vis-à-vis* the zero-grade -*n̥t*- in Myc. *o-da-twe-ta* /odatwenta/, cf. Ved. gen. sg. *dataḥ* 'tooth'), the proposed amphi-/holokinetic paradigm, as in, e.g. *u̯é́kont- vs. *uḱn̥t-́ 'willing' (Ved. *uśánt*-, Gk. ἑκών), is nowhere attested and it remains possible that *-*ont*-participles in athematic verbs are *einzelsprachlich* innovations replacing the old hysterokinetic *-*ent-/-n̥t*- in, e.g., Ved. nom.sg. *sán*, gen.sg. *satás* (Rix 1992: 234).[8] Depending on the relative chronology, the formal similarity between the participial *-*ont*- (-οντ-) and the thematic 3pl. *-*ont* (-ον[τ]) also allows for the possibility of analogy in the other direction. Or perhaps more crucially is the question whether such category contact is really feasible at all, namely an analogical influence exerted by the participle on the 3pl. finite form, the latter being a more paradigmatically salient point in the verbal inflexion. The chain of morphological changes proposed by Cardona (from non-finite forms to finite forms, from 3pl. to 3sg.) arguably goes against the generally accepted directionality of analogy, namely

[7] This idea is endorsed and further expanded by Bammesberger (1984: 14), who views *all* thematic root formations – presents and aorists – across Indo-European to be created via this process.

[8] δρακείς in Pindar could be a relic form of the old hysterokinetic type as in *dr̥ḱ-ént-/dr̥ḱ-n̥t-́, as first analyzed thus by Forssman 1964. However, this formation is generally regarded as an old innovation, its vowel pattern taken over from the athematic 3pl. -*ent(i)*, which displaced the older -οντ- ending still visible in δράκων 'snake'. See Peters 2004 and Barnes 2013; more recently van Beek 2021: 375–7. Cf. also Morpurgo Davies 1978 on the Thessalian ἐντ- forms. See below fn.191.

from the more frequent forms to the less frequent ones (Albright 2008: 145 with references, cf. Cowgill 1965: 158, Hock 1991: 210-37). Prehistorically at least, the two paradigms should have no contact point, as the *o*-grade *-*ont*- would only occur in the strong stem, e.g. *$\mu\acute{e}\acute{k}$*ont*-, whereas the athem. 3pl. *-*ent* would have a zero-grade root, e.g. *uḱ-ént*, so the starting point of Cardona's scenario – participial *-*ont*- >> athem. 3pl. *-*ont* >> them. 3pl. *-*o-n(t)* >> them. 3sg. *-*e-(t)* – would have to happen after the zero-grade root was generalized for *-nt*-participles in Greek (while leaving relics such as ἑκών). Beyond this mechanism, Cardona also identifies the problems surrounding the thematic aorist stems of **CeRH*- roots (e.g. ἔταμον 'cut', ἔβαλον 'threw', etc.) and the possibility that some of the Greek thematic aorists continue old imperfects, although the treatment of these areas is of a more piecemeal nature (necessarily so in some cases given a lack of attested materials) and no bigger picture or theoretical framework is established other than the thematization based on 3pl. *-ov* as mentioned before.

1.2.3 Harđarson 1993 and the focus on the **CeRH*- roots

Building on Cardona's foundation, Harđarson in his monograph on the Greek and Indo-Iranian root aorist (1993: 155-70) lays out an important theoretical framework on the thematization of the root aorist of **CeRH*- roots in Greek. As already noted by Cardona (1960: 77), roots of this shape, such as *$g^{w}elh_1$- 'throw', will yield a quasi-thematic vowel by regular sound change in certain slots of their athematic root aorist paradigms, e.g. 3sg. *$g^{w}elh_1$-*t* > *g^{w}*ele-t*. Hence, the whole problem revolving around the fate of the root formations of the shape **CeRH*- is potentially important to the question of (one of the) origins of the Greek thematic aorist and is at the center of Harđarson's discussion as well as Willi's (2018: 332-42) response to that. Harđarson's core idea is that the different attested aorist stems of **CeRH*- roots are in some way relatable to the three phonologically regular outcomes of the root in different phonological environments in the root aorist paradigm. A conspectus is

provided below:

A. *$CeRh_1$- ROOTS, e.g. *g^welh_1-

i. root aor. **βλη**-

< *$g^wlē$- <= 1pl., 2pl. *$h_1é$-$g^w\underset{\circ}{l}h_1$-*me(s), -te* (augmented)[9]

ii. zero-grade them. aor. **βαλε**-

< *g^wale- <= 3pl. *(*é*)-g^wal-*ont* <= 3pl. *(*é*)-$g^w\underset{\circ}{l}h_1$-*ent*

iii. full-grade them. aor. Arc. **ἔζελε**

< *g^wele- <= 3sg. *($h_1é$)-g^welh_1-*t* and 2pl. *$g^w\acute{\underset{\circ}{l}}h_1$-*te* (unaugmented)

B. *$CeRh_2$- ROOTS, e.g. *$telh_2$- 'endure', *$\acute{k}emh_2$- 'toil'

i. root aor. e.g. **τλᾱ**- (Ion. **τλη**-)

< *$tlā$- <= 1pl., 2pl.*$h_1é$-$t\underset{\circ}{l}h_2$-*me(s), -te* (augmented)

ii. zero-grade them. aor. **καμε**-

< *kame- <= 3pl. *(*é*)-*kam-ont* <= *($h_1é$)-$k\underset{\circ}{m}h_2$-*ent*

iii. full-grade *s*-aor. **τελάσσαι**

< *tela- <= 1sg, 2sg, 3sg. *(*é*)-$telh_2$-*m, -s, -t*

iv. zero-grade *s*-aor. **ἐτάλασσε**

< *tala- <= 1pl, 2pl. *$t\acute{\underset{\circ}{l}}h_2$-*me(s), -te*, 3pl. *($h_1é$)-$t\underset{\circ}{l}h_2$-*ent*

C. *$CeRh_3$- ROOTS, e.g. *$melh_3$- 'come',

i. root aor. **βλω**-

< *$mlō$ <= 1pl., 2pl.*$h_1é$-$m\underset{\circ}{l}h_3$-*me(s), -te* (augmented)

ii. "*o*-grade" them. aor. **μολε-**

< *melo- <= 2sg, 3sg. *(*é*)-$melh_3$-*s, -t*

< *molo- <= 1pl. *$m\acute{\underset{\circ}{l}}h_3$-*me(s)* (unaugmented)

Table 1.1 Aorists to *$CeRH$- roots per Harđarson 1993

This framework offers an attractive structural explanation for the sources of the various aorist stems of these roots and must be considered correct in its basic conception. The phonological development in A (i, ii), B (i, ii), and C (i), namely *$C\underset{\circ}{R}H_iC$- > *$CR\bar{E}_iC$- and *$C\underset{\circ}{R}HV$- > *$CaRV$-, is generally accepted nowadays as the regular outcome of theses sequences (Rix 1992: 72-5, van Beek 2021:11-13). The particular issue involving *$C\underset{\circ}{R}h_3V$- will be treated in

[9] Note that Harđarson's assumption of zero-grade in the 1pl., 2pl. of the root aorist is contested. The Indo-Iranian evidence mostly points to *e*-grade throughout the paradigm except 3pl., e.g. *ákarma, ákarta, ákran*, etc., cf. Jasanoff 2003: 82f., Malzahn 2004.

detail in Chapter 2. A (ii) and B (ii) essentially involve a Cardonian type of development starting in 3pl. with secondarily substituted *-*ont*. The first part of A (iii), i.e. thematization based on 3sg. *g^{w}élh$_1$-t* etc., has been recognized since Wackernagel (1916: 14), but it is now supplemented by Harđarson's own proposal of another source that feeds into the process, namely the *unaugmented* root aorist forms in the 1pl. and 2pl. with (secondarily) retracted accent, which also partially underlies B (iv) and C (ii). This process critically assumes the disyllabic reflexes *-*E$_i$RE$_i$*- of accented *$C\acute{R}_{\circ}H_iC$- sequences (as in θάνατος 'death', κάματος 'toil', see Rix 1992: 73). The validity of such an account is still being explored and the reader is directed to the helpful discussions in Vine (1998: 66-9), Probert (2006: 233-6), van Beek (2021: 12f.), and more recently Höfler (2016-7).[10] Suffice it to say that even if one accepts the disyllabic reflexes as regular in these environments, they likely did not play a big role as the forms involved (unaugmented 1pl. and 2pl.) occupy relatively weaker slots in the paradigm. Overall, Harđarson's framework provides valuable insights into the potential routes of thematization of the *$CeRH$- roots. A few points still need to be investigated and accounted for. Firstly, the explanation of the "*o*-grade" forms heavily relies on the hypothesized *phonological* metathesis *-eRo- > -oRe-*. In Chapter 2 (esp. 2.2-3), the validity of this metathesis is evaluated (and refuted) and an alternative explanation based on the regular outcome of *$C\acute{R}_{\circ}h_3V$- sequences is proposed. Second, the spontaneous replacement of the original athematic 3pl. *-*ent* with *-*ont* is taken for granted by Harđarson, but the 3pl. of *$CeRh_3$- roots (i.e. *$C\acute{R}_{\circ}h_3$-*ent* > *$C\acute{R}_{\circ}h_3$-*ont*) must have provided crucial support. The significance of *$C\acute{R}_{\circ}h_3$-*ont* forms is discussed in 3.9 and the conditions for the coloring effect of laryngeals across morpheme boundaries are evaluated in 2.4.2.

[10] On the PIE "injunctive", see, e.g., Hoffmann 1967, Kiparsky 1968: 34-43, 2005, more recently Kloekhorst 2017.

1.2.4 Willi 2018, Watkins 1969, and the PIE status of the thematic aorist (Jasanoff 2017, Rau 2024)

The system of Harđarson is largely rejected by Willi (2018: 332-42), whose objection hinges on the chronology. Willi contends that the fact that there are many more cases of sigmatised aorist stems (of *e*- or *o*-grade) with roots in *-h_2- (e.g. τελάσσαι) and *-h_3- (e.g. ἐστόρεσε 'spread', cf. 2.7.3) than with roots in *-h_1- suggests that the thematization of *h_1-final roots (of the type A [ii] and A [iii] above) must have happened before the sigmatization (and hence bled it) and also before the alleged metathesis *-eRo-* to *-oRe-* (otherwise the *h_3-roots would look so similar to *h_1-roots as to have undergone thematization early on as well). But this would, according to Willi, create chronological issues because as long as the analogy based on 3pl. (type A [ii]) is available, the odd-looking forms (cf. type B [iii]), e.g. **e-tela-t*, should have been thematized after the model in A (ii) and therefore, should have left nothing to be sigmatized later (i.e. forms like τελάσσαι should not exist), unless one assumes that the 3pl.-based thematization happened after sigmatization, but this would create further issues. In 3.3.4 (fn. 241), this objection of Willi's will be assessed in more detail. As an alternative, Willi (2018: 342-7) opts for a scenario where the bulk of the thematic aorists in Greek (presumably not just of the **CeRH*- type) is to be projected back to PIE as a synchronic competitor of root aorists (296) and not created through analogy as envisaged by Harđarson and Cardona. Without embarking on a full assessment of Willi's reconstruction of the PIE verbal system and the position of the thematic aorist therein,[11] one could argue that the virtual absence of simple thematic formations with alternating *-*e/o*- in Anatolian (Lehrman 1985, Yoshida 2022: 122; *contra* Willi 2018: 189) at least suggests Proto-Nuclear-Indo-

[11] The thematic aorist is, according to Willi (2018: 345-51), an offshoot of the reduplicated aorist (in his system the earliest perfective formation in PIE) built to *h_1-initial roots, whereby the reduplicant *h_1e- was generalized and became an all-purpose reduplicator. For more comprehensive evaluations of Willi 2018, cf. Kazansky 2019, Kölligan 2019, van Beek 2019, Pinault 2021.

European (the ancestor of the IE family excluding Anatolian) as a *terminus post quem* for the creation of the thematic aorist. From the view of economy, it seems untenable to assume a significant number of inherited thematic aorists with so few equations attested in the daughter languages.

An important aspect of the problem that Willi points out but does not pursue further is the functional difference between the thematic aorists and the root aorists – whether there is any, and if so, whether any traces of it can still be captured in the attested materials. In fact, the semantic/functional observation that many Greek active thematic aorists are coordinated with medio-passive presents, e.g. δέρκομαι vs. ἔδρακον 'see', ἀπο-πέρδομαι vs. ἀπ-έπαρδον 'break wind', and τρέφομαι vs. ἔτραφον 'be nurtured', played a central role in Watkins's (1969: 100f.) influential theory that connects the thematic inflection with the PIE perfect and mediopassive endings.[12] More specifically, Watkins submits that Ved. *ávidat* and Gk. ἔ(ϝ)ιδε continues a pre-PIE 3sg. middle form **u̯id-é* (primary **u̯id-éi*, cf. Ved. *vidé*) which is later recharacterized as **u̯idé-t* with a "leeres *t*" borrowed from the athematic inflection. This would then serve as a pivotal form for the entire thematic paradigm to be created.[13] Following essentially the same path, Jasanoff (2017) later revised the starting point of the change from a pre-PIE middle **u̯id-é* to an "**h₂e*-conjugation" active **u̯id-é.* In a recent contribution focusing on the investigation of the functional properties of the thematic aorist, Rau (2024) observes, for Greek and also OCS and Indo-Iranian, that the thematic aorist has a "propensity to be intransitive – even in the active" and that such an intransitive thematic aorist often stands in contrast with a transitive *s*-aorist or reduplicated aorist, e.g. ἔτραπον

[12] Similarly, Villanueva Svensson (2011, esp. 40-3) observes a systematic link between the thematic aorist and anticausative-inchoative verbs, which again points to a middle origin, with comparative support from pairs such as OCS *legъ* 'lay down' : ἔλεκτο, OCS *sědъ* 'sat' : Ved. *ásādi*, etc.

[13] Cf. Peters 2004: 268 fn.10; Watkins's wider theory on the nominal origin of the thematic conjugation in PIE is endorsed with restrictions by Mottausch (2003), Willi (2018: 186-8 with references), Villanueva Svensson (2021). See Yoshida (2022: 120-2) for a recent synopsis of Watkins's theory.

(intr. in κ 469) : ἔτρεψα ‘turn’, ἤριπον (intr.) ‘fall’ : ἤρειψα ‘throw’, etc. Such observations are no doubt crucial to the determination of the original function – if a distinct one can be identified – of this category in P(N)IE, and a separate, full-scale study on the topic is needed.

1.3 Methodology and organization of the dissertation

The main methodology adopted throughout the project is traditional comparative analysis, supplemented with internal Greek morphological reconstruction, and critical re-examination of earlier scholarly theories.

In Chapter 2, I examine the “*o*-grade” thematic aorists and reassess the traditional explanations for their unexpected vocalism. I first review the prevailing accounts: the hypothesis of an *-eRo-* > *-oRe-* metathesis (2.2) and the proposal that these aorists were remodeled by analogy with *CRā* : *CaR* patterns from **h₂*-roots (2.3). Given their shortcomings, I then propose that the *o*-vocalism directly reflects the regular Greek outcome of **CR̥h₃V-* sequences (2.4-5). I further re-examine items such as πεπαρεῖν and ἐάλων and show that they do not contradict the regularity of the *CoR-* outcome. The chapter closes with additional discussions of the “*o*-grade” *s*-aorists such as ἐλόεσα, ἐστόρεσα, etc. (2.7).

In Chapter 3, I turn to the coordination patterns involving the zero-grade thematic aorist (e.g. ἔλιπον), the double-nasal present (e.g. λιμπάνω), and the full-grade thematic present (e.g. λείπω). The first part of the chapter (3.2) is a detailed philological and etymological examination of each verb in this group. The second part (3.3) then develops a coherent diachronic account of how this stem-distribution pattern arose, using the findings from the philological section as evidence.

While the two chapters occasionally intersect and certain conclusions from Chapter 2 are cited in Chapter 3, each chapter is designed as an independent, self-contained study. Readers may therefore approach them separately, and each can be read on its own without requiring prior familiarity with the other.

Chapter 2
The "*o*-grade" thematic aorists and *$C\underset{\circ}{R}h_3V$ sequences

2.1 Preliminaries

The Greek thematic aorist typically uses the zero-grade root (e.g. ἔλιπον 'left' < **é-lik^w-o-m*), with occasional *e*-grade formations (e.g. ἔτεκον 'begot' < **é-teḱ-o-m*). There are, however, four thematic aorists found from Homer onward that exhibit an entirely unexpected *o*-vocalism on the surface, namely ἔθορον 'leap, spring', ἔμολον 'go, come', ἔπορον 'give, offer', and ἔτορον 'pierce'.[14] A synopsis of their stem distribution is provided below:

- **ἔθορον** (Hom.+) 'leapt', the simple verb is only poetic, but with ἀνα- and ὑπερ- in Hdt., Xen., etc.:
  - pres. θρῴσκω (Hom.+), θόρνυμαι (Hdt. 3.109, Nic.), Hsch. θ 110 θάρνυται· μετεωρίζεται 'lifted up', thematic stem in θ 109 θαρνύει· ὀχεύει 'mounted'
  - aor. ἔθρωξα (Opp. *Hal.* 2.293)
  - perf. τεθορυῖα (Antim. fr. 144 Matthews)
  - fut. (ὑπερ-)θοροῦμαι (Θ 179, A. *Supp*. 873)

- **ἔμολον** (Hom.+) 'went', mostly poetic, once 3pl. subj. μόλωσιν Xen. *Anab.*7.34:
  - pres. βλώσκω (Hom.+; in Homer only compounded with κατα-, προ-, ἐκ-, etc.)
  - aor. Hsch. ε 89 ἔβλω· ἐφάνη 'appeared', ᾤχετο 'gone', ἔστη 'stood'
  - perf. μέμβλωκα (Hom.+), Hsch. β 424 βέβλωκεν· ἠρεμεῖ 'is at rest', φύεται 'grows'
  - fut. μολοῦμαι ([A.] *PV* 689, S. *OC* 1742, A.R. *Ar.* 1.1174)

- **ἔπορον** (Hom.+) 'gave':
  - aor. Hsch. π 1495 πεπορεῖν· δοῦναι 'give' (cf. below 2.4.1 on Pi. πεπαρεῖν)
  - perf. πέπρωμαι (Hom.+), only impersonally as "it is fated…"
  - fut. πόρσω (*EM.* 683.54)

[14] Not included here are ὠλόμην 'ruin' and ὦρετο 'rise', whose *o*-vocalism is caused by an initial *h₃* in the expected zero-grade stems (i.e. **h₃l̥h₁-e/o-*, **h₃r̥-e/o-*), cf. LIV^2 *s.v.* **h₃elh₁-* and **h₃er-*. The *hapax* γόον 'lament' (Z 500) is obscure and may be an imperfect, see recently Nussbaum 2025.

part. πρωτός (Hdn. Gr. 1.215 Lentz)

- **ἔτορον** 'pierced' (Λ 26)[15], mostly poetic, frequently with ἀντι-:
  - pres. (ἀντι-)τοροῦντα (*h. Mer*. 4.283)
  - aor. (ἀντ-)ετόρησεν (Ε 337, Κ 267), τετορήσας (*h. Mer.* 4.119), Hsch. τ 606 τέτορεν· ἔτρωσεν 'damaged'
  - fut. (ἀντι-)τορήσων (*h. Mer.* 4.178), τετορήσω (Ar. *Pax* 381)
  - part. τορητός (Lyc. 456)

Relevant to the following discussions are two descriptively sigmatic "*o*-grade" aorists ἐκόρεσα 'satisfy, satiate' and ἐστόρεσα 'spread'. Their attestations are summarized here, but see 2.7 for detailed analysis:

- **ἐκόρεσα** (Hom.+) 'satisfied', with ὑπερ- (Thgn. 1158):
  - pres. κορέω (Nic.), κορέσκω (Nic.), κορέννυμι, κορίσκομαι in late grammatical and medical works.
  - aor. ἐκορέσθην (Hom.+), on κορήσατε 'sweep out' in υ 149, cf. fn.136.
  - perf. κεκόρημαι (Hom.+)
  - fut. κορέω (Hom.+), κορέσω (Hdt. 2x)
  - part. ἀκόρητος (Hom.+, poetic)

- **ἐστόρεσα** (Hom.+) 'spread':
  - pres. στόρνυμι (Hom.+, *hapax* in Homer with κατα- ρ 32), late στορέννυμι, στρώννυμι
  - aor. ἔστρωσα (A.+),
  - perf. ἔστρωμαι (Hom.+, *hapax* in Homer K 155), ἐστόροται in late grammatical work, late ἔστρωκα
  - fut. (κατα-)στορῶ (Ar. *Eq.* 481), στρώσω (E.+)
  - part. στρωτός (Hes.+)

These forms have been standardly reconstructed as belonging to **h₃*-final roots of the shape **CeRh₃*-, on the basis of the *CRō*- stems (absent in ἐκόρεσα), which reflect the regular outcome of the **CR̥h₃*- sequences before consonants in Greek, e.g. θρῴσκω < **dʰr̥h₃-ské/ó*-, βλώσκω < **ml̥h₃-ské/ó*-, πέπρωμαι < **pé-pr̥h₃-mai* (cf. LIV² *s.vv.*). According to the current *communis opinio* on the reflex of **CR̥H*- sequences before vowels (Peters 1980a: 30 fn.19,

[15] More below (2.4.3.2) on connection with τίτρωσκω 'wound', etc. Cf. Peters 1980b: 349f., Harðarson 1993: 219-22.

Rix 1992: 74, van Beek 2021: 11f.), the thematic aorist forms of the canonical zero-grade type as such are expected to appear as *CaR-e/o-* (< **C̥R̥h₃-é/ó-*; originally thematized from 3pl. **C̥R̥h₃-ónt*, see 1.2.3, 3.3.3.1), cf. ἔβαλον 'threw' < **gʷl̥h₁-é/ó-*, ἔκαμον 'toiled' < **ḱm̥h₂-é/ó-*. The *o*-vocalism thus requires an explanation.

2.2 Previous solution 1: **-eRo- > -oRe-*

The now-standard theory for these aberrant *o*-grade formations, going back to Ruipérez 1950 and adopted in many modern treatments (cf. Cardona 1960: 78-9, Harðarson 1993: 169f.), attributes the phenomenon to a metathesis *-eRo- > -oRe-*.

2.2.1 Metathesis originating from the future

The original proposal by Ruipérez posits that the process was first effected in the future, where, e.g., *θερό-(σ)ομαι would have – supposedly at a stage when οω-inflection was still not fully established in Greek – metathesized to θορέ-(σ)ομαι (Θ 179), whence the *o*-vocalism would spread to the aorist and produce ἔθορον. Several key objections may be made that render this scenario highly implausible. First, the morphological motivation for this metathesis, namely the desire to remodel the future stem ending in όω, is insufficiently founded. Although rare in Greek and clearly an innovation, the Linear B materials (unbeknownst to Ruipérez at the time of his writing) now indicate that denominatives of the -οω type are already established at Mycenaean times, attested in forms such as *e-re-u-te-ro-se* (PY Na 395.A, etc.) = aor. ἐλευθέρωσε 'freed' : ἐλευθερόω (cf. Chantraine 1961: 242) and *qe-qi-no-me-na* (PY Ta 713.1) = perf. part. *βεβινώμενα/δεδινώμενα 'provided with *qi-no-*' : *βινόω/δινόω (see DMic *s.v.*).[16] Though admittedly small in number in Homer, stem

[16] On (non-)connection with Hom. δινωτός 'turned, rounded', see García-Ramón 1999, Nikolaev 2025: 116 fn.61. The Linear B materials also conform to the tendency observed in Homer that οω-stems are predominantly attested outside of the present, cf. Schwyzer 1939:

formations in -όω have been attested since the earliest materials, and if forms like *θερό-(σ)ομαι ever existed, the need to eliminate them from a purely paradigmatic point of view should have been minimal. An original -όω future, in fact, has been recognized in Cret. OMOTAI 'swears' (< **omo-etai* from the root **h₃emh₃-*,[17] see Galavotti 1977: 130-5, Forssman 2002) and Hom. ὀμοῦμαι (4x; never †ὀμεῦμαι or †ὀμέομαι) also points to *ὀμό-ομαι with the -όω future intact (Cowgill 1965: 158).[18] Even if any remodeling did happen, as must be the case for many -έω futures to disyllabic roots in -α (< **h₂*) such as θανοῦμαι 'die' and καμοῦμαι 'toil' as well as Hom. 3sg. ὀμεῖται (reflecting *ὀμέ-εται), the observed pattern is best explained as a simple replacement of the stem final vowel, probably in analogy with liquid stems such as βαλέω.[19] On an empirical level, Ruipérez's proposal also faces the problem that the patterns of the attestation of future stems in *CoRé-*, crucial to his analyses, do not support the idea that they are the starting point of the *o*-vocalism of these paradigms – only θοροῦμαι (*hapax*) and κορέω (3x in a formula) are found in Homer, whereas the *o*-grade aorist forms are all Homeric and have a much stronger footprint (with the exception of ἔτορον, more below 2.4.3.2). Moreover, even allowing for a scenario where future stems *θερο-, *μελο-, *περο-, etc. existed at one point and metathesized into *θορε-, *μολε-, *πορε-, etc., the creation of a new aorist stem – in particular with respect to its root vocalism – under the analogical influence of the future is decisively rare in Greek, if existent at all (Cowgill 1965: 157f). To provide a true parallel to this type of "future-to-aorist" analogy

728, Tucker 1990: 274-328, Sihler 1995: 522.

[17] I follow the reconstruction in EDG (*s.v.* ὄμνυμι), but LIV^2 (*s.v.*) sets up a **h₂emh₃-* , in which case *ἀμοσ- becomes ὀμοσ- via vowel assimilation (cf. fn. 20). Nussbaum (2002: 190 fn.38, fn.39) correctly points out that *a-o* > *o-o* is unusual in Greek and instead posits a root **h₁emh₃-* via an assimilation of *eNo-* > *oNo-* in comparison with the word for 'name', as in Lac. Ἐνυμα- > ὄνομα/ὄνυμα elsewhere.

[18] A detailed argument for the historic primacy of 1sg. ὀμοῦμαι *vis-à-vis* 3sg. ὀμεῖται can be found in Nussbaum (2002: 187-93).

[19] Granted, forms such as θανοῦμαι are *compatible* with the hypothesis that *θενά-(σ)ομαι metathesized into θανέ-(σ)ομαι, but one then has to assume **-eRa-* > *-aRe-*, which is untenable for various reasons.

envisaged by Ruipérez, one looks at cases where an *s*-future shows the expected *e*-grade of the inherited desiderative stem, whereas at least one of the corresponding aorist stems could have shown an unexpected *e*-grade that cannot be explained on the basis of the present. Such conditions exist in a set of roots with nasal present and thematic aorist, such as χανδάνω : ἔχαδον : χείσομαι 'hold' and τυγχάνω : ἔτυχον 'meet, chance upon' : τεύξομαι make ready', and in πάσχω : ἔπαθον : πείσομαι 'suffer' – in all cases no innovation has been made on the aorist stem based on the future (e.g. aor. †ἔτευχον). Thus, an analogy of, e.g., ἔθαρον/ἔθερον >> ἔθορον based on θορέ-(σ)ομαι, must be regarded as highly unlikely. The not infrequent matching of the root vocalisms between the aorist and the future stems in Greek (see examples given in Ruipérez 1950: 399), in particular of the disyllabic roots, as seen in, e.g., ἔκαμον : καμοῦμαι 'toil', ἔβαλον : βαλῶ 'throw', but also elsewhere, cf. δόξω : δοκέω 'seem', must predominantly be the result of the future analogically created after the aorist (and/or the present), as is generally accepted (Rix 1992: 223), and not the opposite.

What follows from our brief examination of Ruipérez's proposal is that a scenario where ἔθορον and the like first acquired their *o*-vocalism from the future faces many difficulties.[20]

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[20] Ruipérez's suggestion (1950: 399) that the rise of the *o*-vocalism in στορῶ and θοροῦμαι could have been facilitated by the present στόρνυμι and θόρνυμαι is no doubt plausible, but in this case the idea of an actual "metathesis" seems irrelevant when the change is no more than an analogy. This still leaves open the question where the *o*-vocalism in στόρνυμι etc. comes from, for which Ruipérez's claim (1949) that -ορ- reflects IE *$*\bar{r}$* is hardly convincing. Meißner's attempt (1998: 41f.) to explain the *o*-vocalism in θόρνυμαι, στόρνυμι, etc. by positing a sound change *-eR(R)u-* > *-oR(R)u-* in Greek is also unattractive because of the numerous counterexamples, cf. Vine 2005: 27f. Rather a streamlined account for the *o*-timbred present θόρνυμαι is that it is built under the influence of the aorist θορε- (van Beek 2021: 414), similar to what must be the case in πέρνημι : ἐπέρασα 'sell'. στόρνυμι will be discussed below at 2.7.3. Opposed to this group are the sources of the *o*-vocalism in ὄρνυμι 'rise' and ὄλλυμι 'ruin', which reflect root initial *$*h_3$* as in *$*h_3er$*- and *$*h_3elh_1$*- (cf. fn.14). This could be the case for ὄμνυμι as well (as is the view in EDG *s.v.*), but LIV^2 reconstructs *$*h_2emh_3$*- citing a connection with Lat. *amāre* 'love' etc., in which case ὄμνυμι may be analogical after ὄρνυμι etc. ὀμόργνυμι (Hom.+) 'wipe' poses a somewhat difficult case. Operating with *$*h_3merǵ$*- (Forssman 1995: 252), one may explain the *o*-vocalism as a sort of vowel assimilation from *ὀμάργνυμι or of Aeolic origins. Many handbooks (DELG, GEW *s.v.*, LIV^2 *s.v.* *$*h_2merǵ$*-) connect ὀμόργνυμι with ἀμέργω 'pluck', both going back to

Available here is a more economical explanation, namely that the future of the shape *CoRe-* is the relatively younger formation, created under the converging influence of (1) the prevalent descriptively *o*-grade aorist, which is presumably older based on the distribution of the forms in the epics and occupies a more central position in the paradigm (as is the tendency in Greek in general, cf. Garrett 2008: 138), and (2) the spread of -έω future to disyllabic roots in -α and -o (and elsewhere also to *aniṭ* roots ending in resonant). Therefore, instead of the development envisaged by Ruipérez:

fut. *θερό-(σ)ομαι => fut. θορέ-(σ)ομαι >> aor. ἔθορον (replacing *ἔθαρον)
fut. *στερό-(σ)ω => fut. στορέ-(σ)ω >> aor. ἐστόρεσα (replacing *ἐστέροσα)

the process that gave rise to the *CoRe-* futures should be construed more directly as:

aor. ἔθορον >> fut. θορ-
fut. θανέομαι, κερέω, ὀλέομαι, etc. >> -εω future
} => θορέομαι (replacing *θερόομαι[?])

2.2.2 Metathesis originating from the root aorist

Despite the shortcomings of using the future as the starting point, Ruipérez's conception of an actual metathesis has become the working hypothesis for many treatments of this topic that followed. As represented in Peters (1980a: 30 fn.19) and later Harðarson (1993: 169f.) and – with reservation – Willi (2018: 332-4),[21] a modified version of this solution supposes the metathesis took place in the original root aorist paradigm, e.g.:

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*$h_2merǵ$-, in which case the *o*-vocalism is even harder to account for. Peters's (1980a: 24fn18) 'zweifache Vokalharmonie' seems entirely *ad hoc*. The vocalism of the aorist ὤμορξα is probably analogical to the present.

[21] Cf. Sihler 1995: 107 fn.2 for suggestion of an even broader metathesis that affects any -*eCo*- sequences.

aor. 2sg. *ἔ-θερο-ς, 3sg. ἔ-θερο-τ, etc.	>	aor. 2sg. ἔ-θορε-ς, 3sg. ἔ-θορε-τ etc.
aor. 2sg. *ἔ-περο-ς, 3sg. ἔ-περο-τ, etc.	>	aor. 2sg. ἔ-πορε-ς, 3sg. ἔ-πορε-τ, etc.

This process is generally thought to be phonological in nature, see Peters (*ibid.*) who speaks of "einer rein lautlichen, nicht morphologisch bedingten griech. Metathese von *-eLo-* zu *-oLe-*". In fact, if one considers ἐστόρεσα 'spread' and ἐκόρεσα 'satiated' to have undergone the same development, the metathesis *has to* be phonologically/phonetically motivated, because the pre-metathesized forms *ἐστέροσα and *ἐκέροσα would have been under no morphological pressure to change, cf. aorist forms ending in -οσα such as ὤμοσα 'swore' and ἤροσα 'ploughed'. Several factors, however, speak against the validity of such a phonetic change.

The date of the supposed *-eRo-* > *-oRe-* metathesis needs to be post-Mycenaean, given that the only corroborating evidence, namely Myc. *re-wo-te-re-jo* 'for bathing' (PY Tn 996) vs. Hom. λοετρόν and Myc. *re-wo-to-ro-ko-wo* 'bath attendants' (PY Aa 783, etc.) vs. Hom. λοετροχόος, suggests that the process – if real – had not yet taken place in Mycenaean. Before turning to chronology, a couple of points need to be made about the Mycenaean evidence in terms of its probative value on this matter. First, the vocalism in Hom. λοετρόν etc., though formally admissible as the outcome of a metathesis of **leu̯o-* reflected in Myc. *re-wo-*, may simply reflect a productive derivation based on the present λοέω 'to bathe'.[22] It is known that -τρον derivatives are predominantly deverbative (cf. Schwyzer 1939: 532), usually built directly to the present stem less any thematic vowel, e.g. λέκτρον 'couch' << (*)λέγω 'lay', φέρτρον 'bier' << φέρω 'carry', etc, whereby the creation of λοετρόν from λοέω would hardly be a novelty. One may expect such a derivative from λοέω to appear as †λοητρον, as is standard for most contract verbs such as θέᾱτρον 'theatre' << θεάομαι 'see' or ἴᾱτρα 'doctor's fee' << ἰάομαι 'heal'. λοετρόν could be understood as following the standard structure seen in many deradical nouns of disyllabic roots such as ἄροτρον 'plough' << ἀροω

[22] See already briefly in Peters 1980a: 30 fn.19.

'id.' (vs. †ἄρωτρον), κρεμάθρα 'means of hanging' (Ar.) << κρεμάω (Pi.+) 'hang', ἔμετος 'vomiting' (Hdt.) << ἐμέω 'vomit' (vs. †ἔμητος), etc., where the short presuffixal vowel is preserved in derivations that usually call for a base in -η/-ᾱ/-ω.[23] The present λοέω itself (impf. λόεον, δ 252), although not abundantly attested in Homer, need not be a backformation from ἐλόεσα (so EDG *s.v.*, cf. LIV[2] **leu̯h₃-* fn.1). Its archaic credentials as an iterative-causative formation going back to **lou̯h₃-éi̯e/o-* are supported by Lat. *lavāre* (Rix 1999: 519)[24] and the inherited radical *o*-grade is also paralleled in Arm. *loganam* 'bathe'.[25] Therefore, instead of invoking a metathesis, the Homeric forms λοετρόν and λοετροχόος can and probably should be regarded as an analogical innovation under the influence of the

[23] The sporadic occurrence of -ε/α/ο- bases in contract verbs (outside the present where only they are synchronically appropriate) against the productive pattern ultimately reflects the two competing derivational models, one provided by inherited disyllabic roots as in ἀρόω : ἤροσα 'plough', καλέω : ἐκάλεσα 'call', γελάω : ἐγέλασα 'laugh', etc., and the other by the numerous *-i̯e/o-* denominatives such as τιμάω : ἐτίμᾱσα 'honour', οἰκέω : ᾤκησα 'inhabit', etc. This distribution is disrupted (largely at the cost of the inherited pattern) when Greek speakers are no longer able to distinguish between the two types, whence aberrant forms such as Hom. ἐκότεσα 'grudge, hate' from the denominative verb κοτέω (cf. κότος m. 'grudge') and Hom. καλήτωρ 'crier' from disyllabic καλέω were created, the latter being the more prevalent type of generalization of the two. Similar considerations may account for the irregular base in -ε- in certain -μων stems, e.g. ἡγεμών 'leader' : ἡγέομαι, and ἀγρεμών 'hunter' : ἀγρέω, although the details require further investigation, cf. Yates 2022.

[24] Cf. Meiser 2002: 85; on the vocalism, see Vine 2006: 238f.

[25] The precise morphological and phonological backgrounds of Arm. *loganam* (and its possible connection with *lowanam* 'wash') are debated, see Martirosyan 2010: 310f. with references, and more recently Kocharov 2019: 138f, Kim 2021: 376 fn.21. The *o*-grade in *loganam* (a secondary nasal stem of sorts) is disputed by Winter (1965: 108) and Klingenschmitt (1982: 116f., cf. 277), both setting up **leu̯a-* as the pre-form of *loga-*. But there is little evidence for the assumed rounding of **-eu̯-* to **-ou̯-* preceding a vowel (unlike the well-established rounding of **-eu̯-* to *-oy-* in preconsonantal positions, e.g., *loys* 'light' < **leu̯k-os*), and as Cowgill (1965:159) points out, *areg-akn* 'sun' (*areg* from gen. sg. **h₂reu̯i-os*, cf. Schmitt 2007, Kim 2021: 375 fn.18) constitutes a counterexample. For the one plausible case Klingenschmitt adduced, namely *č^{c}ogan* 'they went' < **ki̯eu̯-n̥to*, Olsen (1999: 32 fn.58) suggests the possibility of a dereduplicated perfect, but it is also possible to connect the *o*-grade with Greek causative σοέω 'set in motion' (mid. 'run, hasten') and Ved. *cyāváyati* 'set in motion', in which case *č^{c}og-* would have kept the *o*-grade from the inherited **k$^{(w)}$i̯ou̯-éi̯e/o-* and the middle meaning but was remade into a radical present.

present stem. Such an innovation is motivated by the unmatching vocalism between the nominal formation *lewotro-* (as it were in Myc. *re-wo-te-re-jo*[26]) from **leu̯h₃-tro-*, and the inherited iterative-causative present *loweye-* (in λοέω) from **lou̯h₃-eie/o-*. [27] This distribution is levelled by the time of Homer in favor of the vocalism in the present, replacing *lewotro-* with *lo(w)etro-* analogically after λοέω. The *s*-aorist ἐλόεσα could also be a youngish creation (i.e. post-Mycenaean) formed in the same way as λοετρόν after λοέω. Another possible scenario is that, given the pervasive *o*-vocalism in the attested forms of λοέω and its derivatives and of this root in other languages, λοέω was understood as a disyllabic root in its own right (of the καλέω : ἐκάλεσα type, cf. fn. 23 and 2.7.1) from the beginning, with the replacement of what would be the inherited *s*-aorist, i.e. **lēu̯h₃-s-* > PGk. **lēu̯os-* => λε(ϝ)οσ- , by λο(ϝ)εσ- taking place at an early stage in the prehistory of Greek.[28] Crucially, the changes discussed here are analogical and morphological in nature, not phonetically via λεϝοτρο- > λο(ϝ)ετρο- or (*)λεϝοσ- > λο(ϝ)εσ-. In fact, a cursory look at the Mycenaean materials reveals counterexamples to the supposed metathesis where the *-eRo-* sequence is kept intact, e.g. Myc. *re-wo-te-jo* 'with (heads) of lion' (PY Ta 722) next to Hom. λεόντειος, Myc. *i-je-ro-wo-ko* 'sacrificing priest' (PY Ep 613) next to Att. ἱερουργός, Myc. *qe-ro-me-no* 'being (rowers)' (PY Ad 697) next to πέλομενος.[29] One may point to the

[26] The antiquity of the Mycenaean forms is supported by the general preference for *e*-grade in **-tro-* formations, e.g. Ved. *kṣétram* 'land' <= *kṣay-* 'live' (**tḱei̯-*), Lat. *spectrum* 'image' <= *speciō* 'see' (**speḱ-*), cf. Leumann 1977: 312-4, Weiss 2020: 302f., although *o*-grade forms are not entirely absent, see Cowgill 1965: 159.

[27] The iterative nature of the actionality of 'wash' may find a parallel in Hitt. *ārri* 'washes' < **h₁orH-eie-*, cf. Oettinger 1979: 437f., LIV² *s.v.* **h₂erH-*. Cf. below fn.127.

[28] For possible connection of this root with Hitt. *lāḫ(ḫ)u-* 'pour (a liquid)' (cf. fn.68) and the limited retention of medial **h₃* in Anatolian, see Melchert 2011. Despite the fact that the analysis above hinges on λοέω, it does not seem feasible – given the Myc. *re-wo-* forms – to entirely get rid of the **h₃* and reconstruct **leu̯h₁-/lou̯h₁-* as Olsen (1999: 32, 788) stipulates, unless one assumes a derivation of *-tro-* formation based on a thematic nominal stem, i.e. **leu̯h₁o-tro-*, to which there appears to be no parallels in Greek.

[29] Thus, *contra* Peters' claim (1980a: 30 fn.19) that "Metathese von *-eLo-* zu *-oLe*…hat kein Gegenbeispiel von bewahrtem *-eLo-* wider sich". In principle, the metathesized forms †λοέντειος and †ιορουργός could exist, as they create no morphological or paradigmatic

common understanding that metatheses are by nature irregular and sporadic and thus cannot be considered exceptionless in the neogrammarian sense, as is the view of, e.g., Rix 1992: 17, where metathesis is listed under "nicht-lautgesetzlicher Lautwandel", and has been argued in theoretical linguistics, e.g., Spencer 1996: 68. However, aside from the fact that if taken as sporadic the *-eRo-* > *-oRe-* metathesis loses most of its explanatory power and becomes essentially an *ad hoc* solution tailored for the *o*-grade aorists, it has been shown by recent scholarship (Hume 2001, Canfield 2016, etc.) that metathesis in fact can be accounted for in a systematic and regular manner.[30] One way to avoid the problem of the inconsistent operation of the supposed metathesis is to tailor-fit the sound change so that it only affects *o* from vocalized *$*\underset{\circ}{h}_3$* (and not the apophonic *o*), i.e. *$*-eR\underset{\circ}{h}_3-$* > *$-\underset{\circ}{h}_3Re-/-oRe-$*, as all the cited forms are from roots with a final *$*h_3$* (or alleged to be so).[31] To me this is a very difficult

complications (unlike e.g. the suffix -τερος > †-τορες). Their absence could be explained away by the influence of the base forms, i.e. λέων 'lion' and ἱερός 'sacred'. But in a similar way the vocalism in Hom. λοετρόν etc., as shown above, is likely to have been affected by the base verb λοέω. The point here is that the Mycenaean materials are ambiguous at best and do not constitute independent evidence to support the change *-eRo* > *-oRe-*. On Myc. *qe-ro-me-no*, cf. fn. 105.

[30] For instance, metathesis has been shown to result from ambiguous perception of consonant coarticulation (Blevins and Garrett 2004), natural language constraints (Hume 2001), and partial morphemic reduplication (Harris and Halle 2005). That is not to say genuine sporadic metatheses do not exist (cf. fn.109), but in the specific case of the *o*-grade thematic aorist, the alleged targets of the metathesis are restricted to one specific functional category with clear inherited stem structure. The metathesis supposed also tampers with root vocalism, which is morphologically sensitive in Greek. A further typological point to consider here is that an actual change of *-eRo-* > *-oRe-* would involve *non-adjacent* segments. Although existent, this type of long-distance ordering reversal is known to be rare crosslinguistically, cf. the typological study of Hume (2004: 229) that concludes "metathesis overwhelmingly involves adjacent sounds", see also Buckley 2011 with more references. All these factors speak against a spontaneous metathetical process.

[31] Among the proponents of this theory, the idea that this development only affects *$*h_3$* is only notationally acknowledged by Cowgill (1965: 158). Incidentally, Höfler (2016-7: 180) suggests a suffix -νη- (*$*-neh_2-$*; rather than *$*-eneh_2-$*) in Gk. βελόνη 'needle', connecting it with *$*g^welH$* 'pierce' (cf. LIV^2 *s.v.*) as *$*g^wélh_3-neh_2$*, which would be a case showing the preservation of *-eRo-* even with the specification of *$*o < *\underset{\circ}{h}_3$*, although with this specific word the labial reflex of *$*g^w$* and the productivity of the -ονη suffix in tool-related vocabulary

case to argue, as it creates an almost impossible scenario where post-Mycenaean Greek still possessed a phonemic distinction between PIE/Pre-Greek *o and *h_3, when all the evidence available points to an early merger of the two across the board in Greek with no other cases where *h_3 and *o are treated differently.[32]

Despite Peters' insistence on its phonetic nature, the metathesis must be taken as at least partially morphologically conditioned if it were to be defended at all. Although not explicitly formulated as such by Peters, Harðarson et al., the change of *-eRo-* to *-oRe-* can be thought of as a remedial device that is motivated by the paradigmatic need to "fix" the odd-looking root aorists like *ἔθερο < *(e)-t^h*éro-t* < *(e)-t^h*érh₃-t*. But if we are dealing with a prehistoric root aorist paradigm, as is likely, where the singular forms would have looked like *ἔθερον, -ος, -ο, it is hard to see a purely morphological drive to level the paradigm in the direction of *ἔθορον, -ες, -ε (on the effect of the plurals, see below), instead of a simple step toward †ἔθερον, -ες, -ε or (perhaps less likely) a direct thematization of the 3sg. into †ἐθέροε (cf. 3sg. οὖτα 'wounded' : imper. οὔταε), unless there are *o*-graded forms already existing somewhere in its paradigm (3pl. *$CR̥h_3$-ent* > *CoR-ont*, see 2.4) or certain interparadigmatic influence in play that favors the *o*-vocalism (cf. 2.3). It is worth noting that Harðarson's suggestion (1993: 169f.) that the metathesis may have been facilitated by the *CoRo-C* outcome of (secondarily) accented *$CŔ̥h_3$-C* sequences in the 1pl. and 2pl., e.g. *tóro-me(s)*, *-te* < *$tŕ̥h_3$-me(s), -te* 'pierced', is technically plausible (*if* the disyllabic reflexes are regular in these

undermines this etymology.

[32] Distinction between apophonic *o's and laryngeal-induced *o's (< *h_3e, *eh_3, and in Greek *h_3) is rare elsewhere too, if any at all. Kortlandt (1980: 127f.) first made a case for different treatments of *h_1o- (> *Ø-* as in Arm. *orb* 'orphan') and *h_3e- (> *h-* as in Arm. *hot* 'scent') in Armenian. This is followed by Lubotsky (1990), who claims that Brugmann's Law fails to apply to *h_3e- in Ved. *ápas* 'work', *aratní* 'elbow', and in the weak cases of *gó* 'cow', thus yielding a different outcome from apophonic *o. One may also add the so-called "Saussure effect" in Greek which seems to only target the apophonic *o in ablauting roots. But overall, these phenomena are not well understood and the existence of *h_3 in the relevant forms often debated (e.g. the word for 'cow' is standardly reconstructed as *$g^wou̯$- without *h_3), and for our purpose, clear evidence suggesting a phonemic difference between *h_3 and *o is surely absent.

environment, cf. 1.3.1) but does not provide support to the assumption of the (phonological) metathesis that allegedly happened independently in the singular (e.g. 3sg. *(*e*)-*tero-t* > *(*e*)-*tore-t*). Such **CoRo-* forms perhaps contributed to the rise of the *o*-vocalism wherever it originated from, although the effect must be minimal considering their marginal status (i.e. 1pl. and 2pl.) in the verbal paradigms.[33] Furthermore, the metathesis theory presupposes the genesis of the type in the strong stem of the original root inflection, i.e. **CéRh₃-s, -t,* etc., which at least superficially operates against the general pattern that Greek standardly derives its thematic aorists from the zero-grade for **CeRH-* roots, e.g. ἔταμον, ἔβαλον, ἔκαμον, etc., as it does with roots of all other shapes.[34]

2.3 Previous solution 2: analogy after *CRā- : CaR-*

Thus, through the examination above, it is shown that a metathesis of *-eRo- > -oRe-* in Greek lacks justification on both phonetic and morphological grounds. We now turn to the other solution, which resorts to an interparadigmatic analogy with **CeRh₂-* roots. Originally proposed by Kuryłowicz (1956: 207f.),[35] this approach seeks to explain the *o*-vocalism in ἔπορον, ἔθορον etc. via an analogical process whereby *CRō-C-* forms (< **CR̥h₃-C-*) would have exerted their vocalism on what should be **CaR-V-* (**CR̥h₃-V-*), thus yielding *CoR-V-* forms, after the ablaut model set by roots that show the pattern *CRā-* : *CaR-*:

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[33] The existence of such **CoRo-* forms relies on the assumption that **CŔ̥h₃C* gives the disyllabic outcome *CoRoC* when accented (in parallel to the better supported reflex of **CŔ̥h₁C* > *CeReC* and **CŔ̥h₂C* > *CaRaC*, as in θάνατος, γένεσις, etc.), but this remains controversial and lacks independent evidence. The only cited example ἐστόροται (Ioann. Gr. 245; cf. fn.141) is a gloss from a late grammatical work and alleged to be an Aeolic form. See Höfler 2016-7 for an updated discussion on the treatment of **CŔ̥HC* sequences in Greek. On the uncertainty regarding the zero-grade of 1pl. and 2pl. in the PIE root aorist paradigm, cf. fn.9.

[34] Even given occasional full-grade forms such as ἔτεμε, Arc. ἔζελε, etc. (cf. 2.4, fn.225), one still wonders why there are no traces of any relic forms such as *ἔθαρε, *ἔπαρε, etc.

[35] Similar ideas are already suggested by Brugmann in the second edition of the *Grundriss* (1906: 478), cf. Ruiperez 1950: 401, Willi 2018: 342f.

(Dor.) **κμᾱ**τός	:	ἔ**καμ**ον	⟹	**βλώ**σκω	>>	ἔ**μολ**ον
(Dor.) **θνᾱ**τός	:	ἔ**θαν**ον		πέ**πρω**μαι	>>	ἔ**πορ**ον
				θρώσκω	>>	ἔ**θορ**ον

Table 2.1 The analogical explanation of the "*o*-grade" thematic aorists

It must be noted in favor of this hypothesis that such an analogy is in principle a reasonable development to resolve the otherwise somewhat uncommon stem alternation between *CRō* and **CaR*. As Cowgill puts it (1965: 148), this would be an example of "the impoverishment of subsidiary ablaut patterns", a phenomenon that is also seen in a series of long-vowel perfects such as Dor. ἕστᾱκα 'stand', λέλᾱθα 'escape notice' etc., where ω (< **oh₂*) is expected but analogically replaced with ᾱ to avoid the pattern *ă̄* : *ō* in the same paradigm. This type of secondary ablaut – discussed in some details in Kimball 1988 – where an inherited qualitative distinction is reworked into a mere contrast of vowel quantity is a known innovation in Greek. Aside from the perfect of roots with internal **h₂*, such a development also underlies the stem alternation in, for instance, the replacement of **-neu̯-* : **-nu-* with νῡ : νυ in *nu*-presents after the model of νᾱ : να. Hence, the possibility of a pure morphological change from inherited **CaR-e/o-* to *CoR-e/o-* cannot be ruled out and, as will be shown below, it probably played a contributing role to the rise of the type.

2.3.1 Remarks on Greek secondary ablaut

It is perhaps appropriate to start with a few general observations about this type of ablaut rearrangement. Firstly, in the case of ἕστᾱκα, which – together with βέβᾱκα – probably forms the earliest source of the *a*-vocalism in the perfect, the long ᾱ could have originated from the inherited 1sg. **se-stoh₂-(h₂) a*[36] or have been analogically borrowed from the accompanying

[36] Perceived as endingless and with the later attachment of -κ-, cf. Willi 2018: 216.

aorist ἔστᾱν or ἵστᾱμι.[37] In other words, the substitution of ω with ᾱ there has a genuine phonetic and/or morphological basis beyond a spontaneous apophonic levelling,[38] a condition that is not present in the case of the *o*-grade aorists.[39] Secondly, most of the innovative ablaut patterns in Greek appear to create artificial full-grades/strong stems while leaving the original zero-grades/weak stems intact (as the basis). Such is the case in the long-vowel perfects (**ō* > *ā*) and the *nu*-presents (**-neu̯-* > *-nū-*), as shown earlier. Further examples can be found in Att. δήξομαι (replacing *δένξομαι), δέδηγμαι (replacing *δέδογγμαι) to δάκνω 'bite' (**denḱ-*, cf. Ved. *dáṃśa-* 'bite'; on the present δάκνω 3.3.4), comp. μᾶλλον 'more' (replacing *μέλλον, cf. Lat. *melior*), Att. μέμηνα to μαίνομαι 'rage' (cf. μέμονα), Att. εἵληχα, ἐλήχθην, λῆξις etc. to λαγχάνω 'obtain by lot', and so on (cf. for more discussion of secondary ablaut in the perfect Hackstein 2002: 168-238). The only instances where a secondary ablaut grade with a short vowel are generated involve the sequences ῡ : υ and ῑ : ι, as in, for instance, ἔφῡν 'grow' : φυτόν 'plant', τῡ́φω 'smoke' : τυφῆναι, τρῑ́βω 'rub' : τριβῆναι, etc.[40] What to my knowledge Greek does not have a good parallel of is a secondary contrast of ω : ο (with ο being the innovated part), as would be the case here.[41] Lastly, although *o*-vocalism from **-oh₂-* (that is, perfects of roots with internal **-eh₂-*) has no doubt been mostly levelled out under the influence of ἕστᾱκα etc., in Homer one still finds examples

[37] See Kimball 1988: 248.

[38] No doubt the already existing alternation of *ā* : *a* in the present (ἵστᾱμι : ἵσταμεν) facilitated the process.

[39] The only places where *o* appears naturally in these verbs according to the current theories are 1pl., 2pl. in the root aorist with accent retraction and without the augment (**CŔ̥h₃-me(s)* > **CóRo-me(s)*), but these forms are marginal in the paradigm and their existence is questionable, see above, 1.2.3.

[40] At least two of the three examples provided (ἔφῡν and τῡ́φω) are from roots that are often considered "non-apophonic" even in PIE (Jasanoff 1997), with zero-grade persisting also in nominal formations such as the *s*-stem, see Vine 2022. In this case, the creation of an artificial short υ seems to be the only viable option to make apophonic contrast in these verbs.

[41] Schwyzer (1939: 363) suggests a secondary ο in ποτός, πέποται (χ 56, cf. LIV^2 *s.v.* **peh₃(i)-*), etc. next to the ω in Aeol. Dor. πώνω 'drink', but this could reflect the inherited alternation **peh₃*(*-C*) : **ph₃*(*-C*), cf. Ved. *pītá-* 'drunk'.

like λαγχάνω : λέλογχα, which indicates that *a/o* ablaut is still permitted to a limited extent even within the same verbal paradigm. Further, if the supposed apophonic analogy (which eliminates *all* **CaR*- variants for the **h_3*-final verbs in question) is a real historical process, it should apply to **CeRh$_1$*- roots as well, since they would otherwise exhibit a similarly "irregular" ablaut pattern of *CRēC*- : *CaRV*-. But *CaR*- aorists are standard for these roots even in Doric,[42] e.g. παρέβαλε (IG IV² 1.121.15, etc.). For **temh$_1$*- 'cut', Dor. τάμνω *vis-à-vis* Att. τέμνω indeed is the opposite of what is predicted, with *a*-vocalism preserved in Doric next to τμη- forms such as τέτμηνται (Pi. *I.* 6.22, cf. Forssman 1966: 158-60.) without being levelled to τεμ-.[43] Making further comparisons with verbs that show *ē/ō* alternation such as ῥήγνυμι : ἔρρωγα 'break', the conclusion we draw is that given the great variety of verbal ablaut alternations Greek still allows for in the historical period (including *a* : *o*, *a* : *ē*, and *ē* : *ō*), in theory there should be no synchronic constraints that prohibit pairs like †ἔθαρον : θρώσκω or †ἔπαρον : πέπρωται from being preserved as such. Of course, there is no doubt that some of the ablaut sequences involving *a* were no longer *productive* (in an etymologically *correct* sense) at an early stage given the synchronic ambiguity of its full-grade (*ā* [*-*eh$_2$*-], *en*, *em*), as will be discussed in more detail in 3.3.2.1.

2.3.2 Other empirical problems

Moreover, a spontaneous analogy from **CaR*- to *CoR*- as Kuryłowicz suggested is also challenged by the fact that in early epics θορ-, μολ-, and πορ- allomorphs all vastly outnumber those of θρω- (pres. θρῴσκω), βλω- (pres. βλώσκω, perf. μέμβλωκε), and πρω-

[42] In Att.-Ion., one assumes such a pattern is acceptable, given that dialectal **-ā-* > *-ē-* disrupts the original ablaut even for **CeRh$_2$*- roots, e.g. κέκμηκα : ἔκαμον.

[43] For more on **temh$_1$*-, see Forssman 1967, Beekes 1969: 221-4, esp. 221, Harðarson 1993: 157-66, Willi 2018: 340-2; τόμοντες in Alcaeus fr. 129.15 is likely a hyper-Aeolic form (and not the regular outcome of **tm̥h$_1$-e/o*- as per Strunk 1970: 117-9), cf. Beekes 1969: 224, Peters 1980a: 27 fn.19, van Beek 2021: 12 with references.

(perf. πέπρωται) forms, on the order of 36:12, 12:5, and 71:7.[44] If one follows the conventional wisdom that analogy – considered typologically – tends to be exerted from more frequent forms to less frequent ones,[45] such a distribution raises the question whether there would have been enough analogical pressure on the thematic aorists (allegedly in the form of **CaR*-), which are more abundantly attested and in general a more central tense-stem in the Greek verbal system (cf. 2.2.1, Chantraine 1961: 203). Similarly unattractive is the attempt to derive the *o*-vocalism in ἐστόρεσα and ἐκόρεσα (both well attested in early epics) from στρω- and *κρω- forms: the former occurs only 2x (pluperf. ἔστρωτο K 155 and verbal adj. στρωτοῖς Hes. *Th.* 799) and the latter is not attested. Semantically, the connection between the models (κμᾱτός : ἔκαμον 'toil' and θνᾱτός : ἔθανον 'die') and their analogical targets (ἔθορον 'leapt', ἔμολον 'went', ἔπορον 'gave', and ἔτορον 'pierced') also seems tenuous. Taking a closer look at the individual items, one identifies other paradigmatic and phonological obstacles as well. For ἔθορον, the existence of Hsch. θάρνυται and θάρνυσθαι, which must be old (cf. 2.4.3.1) and once part of the *averbo*, in theory would have provided considerable support for the preservation of an earlier *ἔθαρον. For ἔμολον besides βλώσκω/μέμβλωκα, the phonetic differences resulting from the treatment of the -μλ- cluster mar an exact formal equivalence with the models above. In the case of ἔπορον, the perfect πέπρωται (cf. 2.4.1) is not only much rarer but also somewhat semantically and functionally distant, used only in the impersonal sense of 'it is fated, it is destined…'.[46] Its ability to analogically alter *ἔπαρον is highly doubtful. ἔτορε poses a special case in its semantic (and possibly etymological) association with τετραίνω 'pierce, perforate' (**terh₁*-) etc. and will be

[44] Especially in the case of ἔπορον, where the rare perfect πέπρωται is supposed to be the analogical pivot, cf. Strunk 1970: 111f. The analogical explanation may work well only for ἔτορε (a *hapax* artificial form in Λ 236), assuming a confusion with τίτρωσκω, ἔτρωσα, τέτρωμαι (Pi.), etc.; see more discussion of this form in 2.4.3.2.

[45] See Albright 2008: 145 with references (although Albright himself is critical towards overgeneralizing this tendency), cf. also Cowgill 1965: 158, Hock 1991: 210-37.

[46] Admittedly, the general semantic connection between 'give, grant' and divine providence is of course *diachronically* justifiable, cf. Ved. *pūrdhí* 'grant!' (often used in prayers to gods), Nikolaev p.c.

discussed later in 2.4.3.2.

2.4 **CoRV* as the regular outcome of **CR̥h₃V*

In light of the various difficulties faced by the two current theories on the *o*-grade thematic aorists, one relying on a metathesis that starts from the *e*-grade stem **CeR-e/o-* and the other an analogical change from the zero-grade stem **CaR-e/o-*, I believe we will find a better solution in a third option, namely that the *o*-vocalism in **CoR-e/o-* continues a *phonologically regular* outcome of the sequence **CR̥h₃V-*, originating in the 3pl. **CR̥h₃-ont*. The idea that the epenthetic vowel before **R̥h₃* in prevocalic position may reflect the color of the laryngeal is by no means a new one: often dubbed "laryngeal umlaut", the broader phenomenon of **CR̥HᵢV* becoming *CEᵢRV* has been proposed by Beekes (1969: 216f., 221-7, 231-4), Monteil (1978: 147-53), Ruijgh (1997: 270-4), among others, and its mechanism can be demonstrated via the examples below:

(1) **CR̥h₁-V* > *CeR-V*: **gʷl̥**h₁**-e/o-* > **gʷₑlh₁-e/o-* > Arc. ἔζ**ε**λε
(2) **CR̥h₂-V* > *CaR-V*: **km̥**h₂**-e/o-* > **kₐmh₂-e/o-* > ἔκ**α**μον
(3) **CR̥h₃-V* > *CoR-V*: **dʰr̥**h₃**-e/o-* > **dʰₒrh₃-e/o-* > ἔθ**ο**ρον

The theory has not gained wide acceptance, with the current agreement on the behavior of these sequences in Greek being that they uniformly became *CaRV-* regardless of the color of the laryngeal.[47] The effective criticism, I think, has mostly focused on (1), namely that there is no way to exclude the possibility that ἔζ**ε**λε, ἔτεμε, etc. are simply morphologically *e*-grade forms (so Peters 1980a: 27 fn.19, cf. also van Beek 2021: 11 fn.33), wherever the *e*-grade may have originally come from. The process involving **R̥h₃* is often rejected as part of the "whole package" under the assumption that **h₁* and **h₃* should behave in the same way. However, with our critical analysis of the two existing theories that do without any inherited reflex of **CoR-*, I submit that, all evidence considered, operating with a regular **CR̥h₃V-* >

[47] See Rix 1992: 74, Harðarson 1993: 159f, van Beek 2021: 11f.

CoRV- still offers the most economical and coherent account for the materials at hand.[48] To preface the discussion below, we are reminded of the fact that the *o*-grade aorists are the only known sources where the sequence *$C\underset{\circ}{R}h_3V$ can be relatively securely established. In other words, ἔθορον, ἔμολον, ἔπορον, and ἔτορον are precisely the testimony of this cluster in Greek. It seems to me, therefore, unnatural to operate under an *a priori* premise that the original forms should be *ἔθαρον, *ἔμαλον, *ἔπαρον, and *ἔταρον, and then to seek explanation for their conversion into the attested forms, instead of accepting these as phonologically regular and testing this assumption against the broader data. To my knowledge there are only two forms that have been alleged to display an *a*-vocalism outcome of *$C\underset{\circ}{R}h_3V$, namely ἐάλων 'was caught' and the Pindaric *hapax* πεπαρεῖν (*P.* 2.57). We will now examine their reliability as truly diagnostic forms to this claim.

2.4.1 Alleged counterexample 1: πεπαρεῖν

For the isolated Pindaric πεπαρεῖν[49], the alignment with ἔπορον and Hsch. πεπορεῖν· δοῦναι remains problematic for a number of reasons. The passage in question is *P.* 2.56-8:[50]

> ...τὸ πλουτεῖν δὲ σὺν τύχᾳ
> πότμου σοφίας ἄριστον.
> τὺ δὲ σάφα νιν ἔχεις ἐλευθέραι φρενὶ **πεπαρεῖν**,
> πρύτανι κύριε πολλᾶν μὲν εὐστεφάνων ἀγυι
> ᾶν καὶ στρατοῦ...[51]

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[48] A less believable system is proposed by Strunk (1970), who suggests even *$C\underset{\circ}{R}h_1V$ becomes *CoRV* on the strength of πολύς 'many', πόλις 'city', δολιχός 'long', Aeol. τόμοντες, etc., cf. Vine 1993: 55.

[49] With *varia lectio* πεπορεῖν (of inferior authority), see Turyn's *app. crit.*; not mentioned in ed. Snell-Maehler. πεπαρεῖν is also the reading of the scholia vetera, see *P*. 2.105 Drachmann.

[50] This interpretation of πεπαρεῖν, regarded as a relic by-form of ἔπορον from a historical point of view, is favored among linguists (see, e.g., Peters 1980a: 30 fn.19, Harðarson 1993: 219, Willi 2018: 73, 343), but generally objected to commentators, cf. Gentili et al. 1995: 388, Most 1985: 92 fn.87, to be held against Kirkwood 1982: 153.

[51] References to Pindar follows the text of Snell-Maehler 1987.

In a 1971 treatment of this word, Floyd rejected the conventional interpretation and made an argument that the word πεπαρεῖν should be understood as 'give' or 'offer', and he translates the passage as:

> To be wealthy, with the fortune
> of fate, is best of wisdom.
> You clearly have this to **offer** with liberal spirit,
> Prince, lord over many well-crowned streets
> and people….

This reading, however, does not strike me as cogent as (or offering any advantage over) the standard translation of the word as 'show' or 'display',[52] which follows the meaning given in the Hesychian gloss π 1436 πεπαρεῖν· ἐνδεῖξαι, σημῆναι 'show, point out', π 1435 πεπαρεύσιμον· εὔφραστον, σαφές 'easy to understand, clear'. According to Floyd's analysis, v.57 would be a commendation of Hieron's accomplishments and his generosity by stating that he "provide[s] material for the poet's praise", picking up on σοφίας ἄριστον in the preceding line which he takes to mean the "best topic" for poetry. The latter point is certainly a valid interpretation, as σοφία in Pindar frequently refers to poetic crafts and wisdom or the product thereof (e.g. a poem) by way of the usual Greek metonymy.[53] But from the perspective of encomiastic strategy, it does not seem appropriate here to think that Pindar commences the "crescendo" of this ode, namely the main accolades of Hieron's wealth and his many political achievements (as immediately emphasized by the elaborate vocative in the following line), by anchoring the laudation to his patronage to the poet himself. Rather than

[52] See, e.g., Lattimore (1947: 50) "You, in freedom of your heart, can make this plain", Lloyd-Jones (1973: 122) "You can in truth point to this with your generous mind", Race (1997a: 237) "You clearly have it to display with a liberal spirit", Most (1985: 92) "You are capable of manifesting this certainly with your free soul", Verity (2007: 48) "You clearly possess this, and can display it with a liberal spirit".

[53] So Thummer (1972: 297), "Das Beste der Dichtung, d.h. der beste Gegenstand der Dichtung, ist Reichtum und Erfolg", cf. Miller 1981: 142, Hubbard 1986: 58, also Slater 1969 *s.v.* σοφία.

'bestowing' or 'offering' either τὸ πλουτεῖν (riches)[54] or σοφίας ἄριστον (best [topic] of [poetic] wisdom), Hieron, who many times in Pindar is lauded as being rich with blessing (e.g. *O*. 1.10-11, *P*. 1.48-50), is the *manifestation* of the gnomic statement in v.56 or specifically the subject of τὸ πλουτεῖν δὲ σὺν τύχαι πότμου.[55] The rhetorical purpose of v.57 is to acclaim Hieron as an exemplar of someone who has achieved wealth with "the fortune of the fate"[56] – in other words, the focus must be on the properties that Hieron possesses and the model that he sets up,[57] much more so than on his capacity in either granting riches or offering subject matters to poets. The conventional understanding of πεπαρεῖν as 'show' or

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[54] Such is the view of, e.g., Bowra (1937: 23), who suggests that νιν refers to τὸ πλουτεῖν in its literal sense and that v.57 is about "the wealth which Hieron bestows on Pindar" and how the poet "values this payment for his work". This in my opinion creates a rather unbefitting encomium in this context that limits Pindar's ostensibly more profound praise of Hieron to mere poetic patronage. The supposed contrast between Pindar's well-earned wealth and Archilochus's poverty (v.54 τὰ πόλλ' ἐν ἀμαχανίαι) is nebulous, cf. Gerber 1960: 101, Gantz 1978: 24, Most 1985: 90. There is certainly a tension between the self-referent agenda in v.52-56, where Pindar uses Archilochus as the negative foil to admonish against slanderous blames, and the victor-praise agenda starting from v.57, but the emphatic τὺ δὲ shows a strong urge to swiftly switch the poetic focus from Pindar himself to the *laudandus* Hieron, cf. Miller 1981: 141.

[55] The particular analysis of this verse is much debated, see, for instance, Gerber 1960 for the five different ways of parsing the syntax. With Gerber, I think the sentence should be understood as τὸ πλουτεῖν δὲ | σὺν τύχαι πότμου | σοφίας ἄριστον, translated roughly as "to be prosperous with the fortune of fate is the best (kind) of wisdom". Here, the σοφία may be of the poetic nature as many commentators opine (cf. fn.53), but equally possibly it is wisdom in the general sense that applies to Hieron as well (cf. Kirkwood 1982: 153). Pindar's deliberate ambiguity marks a natural transition from the subjective reflection (v.52 ἐμέ δὲ...) back to the ode's addressee Hieron (v.57 τὺ δὲ...).

[56] The interpretation of σὺν τύχαι πότμου beyond its literal sense is a point of debate. Kirkwood (1982: 152f.) suggests "by the dispensation of fate"; similarly, Miller (1981: 142, cf. fn.30 there) "with the dispensation of destiny". Lloyd-Jones (1973: 122) paraphrases the expression as "granted by divine will and not gained unjustly", cf. also Henry 2000: 295 fn.4. The general sense of being in accordance with fate (and thus with honor and excellence) and not through *hubris* is clear.

[57] Against the negative paradigms set by Ixion the ingrate (v.25-41), who defiles Zeus's initial benevolence, and Archilochus (v.54-6), who falls in distress (ἐν ἀμαχανίαι) for his supposedly slanderous poetry.

'display' is thus perfectly adequate in that regard and I see no compelling reason to translate the word as 'give' or 'offer'.[58]

Could it be that πεπαρεῖν indeed means 'show' or 'display' but this meaning is covered by the semantics of ἔπορον, as Floyd seems to suggest towards the end of his paper? This looks unlikely as a quick examination of the use of ἔπορον in Pindar – a commonplace lexical item there – shows that it has a fairly well-defined meaning of to 'give' or 'offer', either taking a direct object (e.g. χρυσόν 'gold' *O*. 13.78, κίθαριν 'cithara' *P*. 5.65, στέφανον 'crown' *I*. 7.49, ὕμνον 'hymn' *I*. 4.21, etc.) or with an infinitive (in the sense of 'assign [a task]' or 'grant [the ability to…]') as in *P*. 3.45, *P.* 4.259, and so on. The sense of 'show' or 'display' is difficult to reconcile with these general denotations of ἔπορον.[59]

Returning from this brief literary digression, we now take a holistic view of the stem distribution of ἔπορον should πεπαρεῖν be accepted as an archaic reduplicated aorist. The current theory posits, as discussed earlier, that the thematic aorist was originally *ἔπαρον, which is now to be joined by πεπαρεῖν. In addition to these, on the basis of Ved. *pṛṇā́ti* 'gives' and OIr. *ern(a)id* 'id.'[60] and the fact that Greek otherwise lacks a present stem to this verb, it is reasonable to assume the existence of an inherited nasal present **pr̥-n(e)-h₃-* that may

[58] Even if one opts to understand the context of v.56-57 in the sense of poetic topics or materials for praise, the opening of Isthmian 4 still offers a parallel where the verb φαίνω 'show, make known' is used in such contexts: ἔστι μοι θεῶν ἕκατι μυρία παντᾷ κέλευθος, / ὦ Μέλισσ᾽, εὐμαχανίαν γὰρ **ἔφανας** Ἰσθμίοις, / ὑμετέρας ἀρετὰς ὕμνῳ διώκειν "I have, through the gods' favor, countless roads in every direction, O Melissus – since at the Isthmian games you **revealed** an abundant resource – to pursue in song your family's achievements" (translation from Race 1997b: 165). If εὐμαχανίαν 'capability, resource' here can be understood as "suitable subject-matter" as Miller (1981: 140) rightly suggests, the passage then further proves that in Pindar's diction the patron does not so much "grant" or "offer" the poetic materials as he "reveals" or "displays" them.

[59] It is also worth noting that ἔπορον in Pindar is very strongly bivalent in accordance with the meaning 'give', as in all but one of its 13 occurrences (excluding the separate idiom πέπρωται 'it has been destined…') the verb takes up an explicit recipient in the dative case.

[60] See LIV^2 *s.v.* **perh₃-*. On the *e*-grade in *ern(a)id*, see Watkins 1958, Joseph 1982: 47, cf. Strunk 1967: 53f., McCone 1991: 27-37, Gordon 2012: 193f., 196.

have been continued in Greek as an unattested *πάρνῡμι.[61] Thus, in this scenario, the starting point for ἔπορον in pre-Greek would be as follows:

Nasal Pres.:	*παρνῡμι	<	*$p_{\text{ḷ}}$*r-n(é)-h₃-*
Them. Aor.:	*ἔπαρον	<	*$p_{\text{ḷ}}$*rh₃-é/ó-*
Redupl. Aor.:	(*)πεπαρεῖν	<	**pe-p_{ḷ}rh₃-é/ó-*
Perf.:	πέπρωται	<=	**pe-p_{ḷ}rh₃-toi*

Table 2.2 The stem distribution of πεπαρεῖν if from the root **perh₃-*

It should be apparent that πεπαρεῖν only exacerbates the problem of a lack of analogical drive to create *CoR*- stems out of a paradigm that has overwhelming *a*-vocalism.[62] Adding to the difficulties is the very isolated status of πέπρωται. As mentioned earlier (cf. 2.3.2), it is only used impersonally in the 3sg. as a fossilized idiom "it has been destined/fated…" and for all practical purposes amounts to being a separate lexeme.

Therefore, πεπαρεῖν cannot be associated with ἔπορον with any certainty and incorporating it makes the analogical change that is supposed to produce the attested stem distribution even less likely. The actual etymology of πεπαρεῖν remains obscure. An old proposal since Döderlein (1849: 53), followed by Curtius (1880: 34) and Saussure (1879: 60) and still mentioned in GEW (*s.v.*) and LEW (*s.v. pāreō*), assumes a connection with Lat. *pāreō* (equally obscure; see EDLIL *s.v.*), but DELG (*s.v.*) raises the problem of the long *ā*, although this could be reconciled if a putative root **peh₂r*- is reconstructed for both πεπαρεῖν (**pe-p_{ḷ}rh₂-e/o*-) and *pāreō* (**peh₂r*-). Dietrich (1975) offers a novel interpretation that relates πεπαρεῖν to πέπανος (see LSJ *s.v.*), a rare collateral form of πέπων 'ripe, mature', on the basis of the heteroclitic behaviour of this word (cf. feminine πέπειρα [S., Ar.]). To be said in favor of this idea, Dietrich's supposition of an adjective *πέπαρος (which he deems to be the basis

[61] A remodeling from *πάρνωμι (< **p_{ḷ}r-ne-h₃*-) may be assumed, as is usual for nasal stems to **h₃*-final roots, cf. Hsch. θάρνυται (<= **d^h_{ḷ}r-ne-h₃*), ὄμνυμι (<= **h₃m̥-ne-h₃*-), etc.

[62] It does not help if one follows Harðarson's framework and starts from 1sg. *ἔπερ-ον, 2sg. -ος, 3sg. -ο(τ), in which case the short *o*-vocalism is still all but elusive.

for a denominative πεπαρεῖν ‘become ripe; be prosperous’) and/or a (neuter) noun *πέπαρ ‘ripeness, maturity’(?) is not unfathomable, if πέπων (: πέπειρα) follows the pattern of πίων (: πίειρα) ‘fat’ besides the neuter noun πῖαρ (cf. Clayton 2023: 59, 73) and the adjective πιαρός. However, the semantics is not convincing, especially considering the word πέπων twice in the Iliad (B 235, N 120) is used in the pejorative sense of ‘soft’ or ‘weak’, making it a risky choice of word in this context.[63]

The conclusion we arrive at from the assessment above is that the evidence for the interpretation of Pi. πεπαρεῖν as going back to **pe-pr̥h₃-é/ó-* (or being a secondary form based on a putative †ἔπαρον) is questionable and the form does not provide a reliable instance of the *CaRV-* reflex of PIE **CR̥h₃V-*.

2.4.2 Alleged counterexample 2: ἐάλων

We now proceed to ἐάλων ‘be taken’, which presents a stronger case for a genuine *CaR-* outcome of **CR̥h₃V-* but in my opinion still falls short of being convincing.[64] The crucial point of debate here is the morphological makeup of ἐάλων, namely whether it is to be understood as an active root aorist or an aorist passive/**eh₁-*stative. The idea that ἐάλων may continue a stative **u̯l̥h₃-éh₁-* is first elaborated in detail in Francis’s 1970 dissertation (65-76, esp.71),[65] which is followed by Harðarson (1993: 208), Hackstein (1995: 302), LIV^2 (*s.v.*

[63] To offer another possibility, though perhaps still not compelling, one may attempt to trace πεπαρεῖν to πείρω ‘pierce, pervade’, περάω ‘go beyond, pass through’, and ultimately to **per-* ‘across’, if (1) formally πεπαρ- (substituting the expected *πεπρεῖν < **pe-pr-é/ó-*) can be accepted as influenced by aor. ἐπάρην and/or perf. πέπαρμαι and (2) a causative semantics (befitting the reduplicated aorist) that follows the path of ‘make across, cause to reach’ => ‘send, project’ => ‘display, signal, show’ (cf. MoE *get* (*one’s point*) *across*) is deemed within the limits of reasonable conjecture.

[64] By many authorities on this matter including Rix (1992: 74), Beekes (1988: 74f.), and van Beek (2021: 11 fn.3), ἐάλων (or ἁλίσκομαι) is listed as the only testimony of the *aR* reflex of prevocalic **R̥h₃*.

[65] Earlier in Chantraine (1961: 165f.) ἐάλων is already compared to other -η/-θη aorists though not in laryngealistic terms. For a summary of earlier views that ω/η may be apophonic,

u̯elh₃-), among others; this is to be held against the reconstruction of an active root aorist **u̯̥leh₃-* (with the Lindeman-variant of the initial cluster) supported by Hettrich (1973), Peters (1980a: 31 fn.19), and more recently Balles (2007: 19 fn.16). The reasons for positing a stative stem **u̯̥lh₃-éh₁-*, which must be regarded as highly archaic and irregular with the suffix allegedly colored by the laryngeal and not restored (see below 2.4.2.1), are twofold, as neatly summarized by Harðarson (*ibid.*), namely that (1) it explains the passive/intransitive semantics and (2) a state II full-grade is not attested for this root elsewhere. However, recent progress in our understanding of this root, especially in light of the Anatolian evidence, in my view significantly undermines these two premises.

Semantically, the argument for **u̯̥lh₃-éh₁* stems from the assumption that the root **u̯elh₃-* originally means 'strike, hit' (cf. LIV^2 *s.v.*), which shifts into the opposite diathesis in Greek by aid of the stative suffix. However, the difficulty to reconcile, on the one hand, Lat. *vellō* 'pluck, tear' (cf. *vulnus* 'wound'), Hitt. *walḫ-* 'strike', and perhaps Gk. οὐλή 'scar, wound' with, on the other hand, CLuw. *walant(i)-/ulant(i)-* 'dead', HLuw. *wa/i-la-* 'die', Lyc. *la-* 'die', TA *wläs* 'died',[66] ON *valr* 'the slain', and Gk. ἐάλων 'be taken, killed' under a unified root **u̯elh₃-* has been increasingly recognized.[67] This, compounded with phonological problems including the assumed retention of a medial **h₃* in Hitt. *walḫ-^{zi}*,[68] has led to a

see Francis 1970: 66-8.

[66] On the formation of the Tocharian forms, see Hackstein 1995: 302, Malzahn 2010: 893.

[67] See, e.g., Weiss 2001: 359 fn.47, Lamberterie and Perpillou 2004, Balles 2007: 19, Oreshko 2023: 242 fn.14, cf. also Vine 1999: 563 on the nasal suffix and the Saussure's effect in οὐλή (*contra* GEW *s.v.*). The identification of the underlying root (or roots) of the forms in the first group (ones that show the active meaning 'tear' or 'strike') is extremely controverted: see, in particular, Oettinger 1979: 261-5 on connection between Hitt. *walḫ-* and *hulle-* 'fight' under **h₂u̯elh₂-*, followed by Melchert (1984a: 115 fn.72) but subsequently rejected (1994: 169), cf. Hackstein 1995: 302 for a similar connection but under a root **h₂u̯elh₁-* and Puhvel 1991: 367f. under **h₁welh₂-*. Alternatively, Schrijver (1991: 180f.) and Weiss (2001: 359 fn.47) opt to set up an *aniṭ* **u̯el-* 'pluck' for Lat. *vellō*. The idea of an *aniṭ* root **u̯el-* 'die' has also been entertained by Melchert (1994: 254) for CLuw. *walant-/ulant-* and TA. *wäl-*, and more recently by Oreshko (2023: 242) with additional materials in Germanic and Balto-Slavic.

[68] Note, however, that Melchert 2011 argues for a limited preservation of medial **h₃* in

separation of these forms in the latest addenda to LIV[2] into two different roots *$\underset{\wedge}{\mu}elh_2$- 'rupfen, reißen' and *$\underset{\wedge}{\mu}elh_3$- 'sterben', which is no doubt fully justified. If aligned with Gk. ἐάλων, as it seems probable, the Anatolian and Tocharian materials should make it abundantly clear that the intransitive semantics is intrinsic to the root and need not be furnished by the stative suffix *-eh_1-.[69]

On the formal side, the sum of the Anatolian evidence suggests that the root does have state II full-grade (or Schwebeablaut of some sort). In particular, Lycian *lati* 'dies' – attested twice in grave inscriptions at TL 88.2, 112.2, and possibly at TL 75.4 – should go back to PA *$\underset{\wedge}{\mu}laH$- (so Melchert 1994: 290, 316, cf. 2004 *s.v. la-*), with a stem shape comparable to other *seṭ* roots such as *adi* 'does, makes' < *$(H)\underset{\wedge}{\imath}eh_1$-, *sttati* 'stands' < *$steh_2$-, *tadi* 'puts' < *$d^heh_1$-, etc.[70] Naturally, *lati* cannot continue directly *$\underset{\wedge}{\mu}lóh_3$- (< *$\underset{\wedge}{\mu}léh_3$-) which in Lycian should

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Anatolian (cf. fn.28), citing Hitt. *lāḫ(ḫ)u-* 'pour (a liquid)' from *leh_3u- and CLuw. *tatarḫ-* 'break' from *$terh_3$-, but admits the precise conditions are still unclear, cf. Kloekhorst 2018, Kümmel 2022. For *lāḫ(ḫ)u-*, see various proposals in favor of a root *leh_2- by García Ramón (2000: 66 fn.9), Katz (2004: 205–6), Nikoloudis (2006: 231), cf. Zhang (2025: 37). Also to be noted is the possibility raised by Kloekhorst that the alleged preservation of *h_3 in *lāḫ(ḫ)u-* is due to the sequence *$h_3\underset{\wedge}{\mu}$ becoming a unitary "labiolaryngeal" in Proto-Anatolian.

[69] In Greek, the suppletive function of ἁλίσκομαι to αἱρέω without connotation of dying or destruction is demonstrably a secondary development, appearing only in the post-Homeric era, e.g. Xen. *Hel.* 1.1.23 γράμματα ἑάλωσαν εἰς Ἀθήνας "the letters fell into the hands of the Athenians". In the epics, ἐάλων is almost exclusively associated with death and/or other fatal predicaments and the connection is often explicitly expressed via such formulae as ἁλοῦσά τε περθομένη τε ("fallen and laid waste" B 374, Δ 291, N 816), λευγαλέῳ θανάτῳ εἵμαρτο ἁλῶναι ("fated to be caught by a miserable death" X 281, ε 312, with οἰκτίστῳ in ω 34), and in the antithesis of ἠὲ κατακτάμεν ἠὲ ἁλῶναι ("to kill or be killed", M 172). The alignment of ἁλίσκομαι with αἱρέω (and the acquisition of a meaning 'be taken' more generally) may have started in cases like ἕλοιμί κεν ἤ κεν ἁλοίην ("whether I slay or be slain", X 253 at the onset of the duel between Hector and Achilles) where ἁλοίην must be regarded as the opposite of ἕλοιμί in its sense of "kill" or "slay", but such collocations may have prompted a matching in semantics of ἁλίσκομαι with other, less specific usages of αἱρέω. Thus, the pristine Homeric use of ἐάλων is in line with its cognates elsewhere, especially in Germanic, where ON *valr* etc. also carries a strong martial and violent tone. The root *$\underset{\wedge}{\mu}elh_3$- may be more properly glossed as 'die in battle, fall to the enemies', which in Anatolian and Tocharian has undergone semantic bleaching and denotes death in general.

[70] The simplification of the initial cluster *$\underset{\wedge}{\mu}l$- to *l*- is likely regular, see Melchert 1994: 316,

appear as *(*w*)*le*- (on *$\check{o}$ > Lyc. $\check{e}$, see Hajnal 1995: 91-6, 98, 116fn126), and thus the *a*-vocalism must have been borrowed from the weak stem *$\underset{\breve{}}{u}\underset{\breve{}}{l}h_3$- of the original athematic root present. An earlier alternation between *le- : la- (< *$\underset{\breve{}}{u}$léh$_3$- : *$\underset{\breve{}}{u}\underset{\breve{}}{l}h_3$-$\acute{}$), i.e. with a state II full-grade, offers a more compelling starting point than *wel- : la- (< *$\underset{\breve{}}{u}$élh$_3$- : * $\underset{\breve{}}{u}\underset{\breve{}}{l}h_3$-$\acute{}$) to arrive at the generalized stem *la*-.[71] The Lycian form nicely matches CLuw. *ulant(i)*- 'dead' (KUB 35.45 ii 20, 35.48 ii 26) which properly reflects an *-nt*- participle *$\underset{\breve{}}{u}\underset{\breve{}}{l}h_3$-é/ónt- and is formally equivalent to Lyc. *lãta*- 'dead' (N 306.4, 309d.13, cf. Neumann 2007: 183). Because the weak stem is at home in *-nt*- participles, one should then not regard the (rare) variant spelling of *ulant(i)*- as *walant(i)*-[72] as reflecting a full-grade *$\underset{\breve{}}{u}$elh$_3$-é/ónt- with subsequent syncope (*pace* Melchert 1994: 276).[73] Instead, the initial sign probably spells a silent vowel and points to an underlying cluster /*wl*-/ (*contra* van Windekens 1961), which presumably is applicable to HLuw. *wala*- 'die'- as well, assuming a stem correspondence with Lyc. *la*-.[74] The spelling of an initial cluster /*wl*/ as *wal*- finds support in, e.g., HLuw. *walilid*- 'land,

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cf. Schürr 2016: 163.

[71] The association of Lyc. *lati* with Ved. *láyate* 'hides (oneself)' under LIV^2 1.*leiH*- 'sich anschmiegen' as an iterative formation *loi̯H-éie*- seems far-fetched semantically and has been rightly removed in LIV$^{2\ Add.}$. In light of the Tocharian and Greek evidence, it also seems less preferrable to follow the suggestion by Hawkins (1980) and (with reservation) Melchert (1994: 254, 259) that the Luwic forms may reflect *g^wel*- (cf. OE *cwelan* 'die'), see also LIV^2 1.*g^welH*-.

[72] [*wa*]-*a-la-an-ti-in-z*[*i*] KUB 35.13 (see Melchert ftcm: 318), cf. *u-wa-la-an-ta-<al>-li-ya-an* KUB 32.81 'of a mortal', *wa-la-an-t*[*i-ya-ti*] KBo 29.21.4 'of the dead'.

[73] Also worth noting is that a hypothetical *$\underset{\breve{}}{u}$el*- should be susceptible to the change described by Melchert (1994: 262, cf. 144) whereby PA *e would undergo raising (> *i*) between a labial glide and a dental consonant, albeit perhaps sporadically. That Luwian *l* is most likely a dental sound is indicated by the regular affrication of *-sa* to *-za* (/*tsa*/) in *l*-stem neuters such as HLuw. MALUS-*za* (SULTANHAN §34, etc.), phonetically from *adduwal-sa* (: CLuw. *a-ad-du-u-wa-la*), cf. Melchert 2003: 183, Yakubovich 2014a: 285f.

[74] One potential objection here concerns the rhotacized stem ("MORI")*wa/i+ra/i-ha-a* in KULULU 2, A4B §3, which may suggest a pronunciation /*wara*/. But it is unclear whether rhotacism in Hieroglyphic Luwian is strictly confined to intervocalic position. As Morpurgo Davies (1982: 248) admits, at least one secure example, namely ARHA *ha-ta+ra/i* in KARABURUN 1, which syntactically must be a 3pl. in *-nti*, shows *-t*- to *-r*- in a sequence *R_V*.

territory', if its connections with Hitt. *ulili-* 'green (land), field' (see Tischler 2010: 38f.) and further with Palaic heteroclite **ulatar* 'thicket' (attested in dat./loc. *ulanna*, cf. Melchert 1984b: 38f.), both of which seem semantically and formally appealing, are to be maintained.[75] Alternatively, it remains a possibility to me that the initial */wl/* may be realized with anaptyxis as */wəl/* (> */wal/*). In other words, the alternation between *ul-* and *wal-* could reflect a genuine pair of Lindeman variants, i.e. **u̯l-é/ónt-* and **u̯l̥-é/ónt-*, in which case the stem *walant-* would be an exact formal equivalence of Gk. ἀλοντ- (cf. ἀλόντε Ε 487). The effect of Lindeman's (and Sievers's) Law is surely observed in Anatolian, although the precise conditions of its operation are still undetermined.[76] Thus, in my opinion, a PA **u̯lo(h₃)-/u̯l(h₃)-* provides the best overall account for the avatars of this root in Anatolian.

Through the discussion above we have demonstrated that the two motivations for positing a stative stem ἐάλων, namely to justify its intransitive semantics and the state II full-grade, are no longer well-grounded.

2.4.2.1 Excursus: cross-morphemic laryngeal coloring in PIE

Another problem with the reconstruction **u̯l̥h₃-eh₁-* is the need to assume laryngeal coloring across morpheme boundary, which has been argued for (e.g., Peters 1980a: 192 fn.149, Beekes 1996: 6 fn.9, EDHIL 260, *passim* Francis 1970, etc.) and against (e.g., Weiss 2009: 56 fn.12, Byrd 2015: 12 fn.15) parenthetically in various places, although a comprehensive treatment of the phenomenon in Greek (or in Indo-European) is to my knowledge still wanting. Sparing a full study, we now provide a short excursus on this issue, focusing mainly

[75] Sasseville (eDiana *s.v.* Palaic *ulātar/n-*) reconstructs **ul-éh₂-* as the basis for the Palaic word and **ul-i-* for Hittite *ulili* and for Luwian *walilid-* where it is remade into a dental stem.
[76] Specifically with liquids, Kimball (1986: 99) argues convincingly that Hitt. *parā* (*pa-ra-a*) 'forth', in light of Lyc. *pere* 'id.', shows there is an anaptyctic vowel before the resonant *r* and thus points to a Lindeman variant, cf. Gk. πρό etc. < **pr-*. For cases involving *w* including the Hittite first plural ending *-umeni* from a Lindeman variant of the old dual *-weni* (and subsequent dissimilation of *-uw-* > *-um-*), see Melchert 1984a: 24f.

on the Greek materials. First, as a general observation, certain types of cross-morphemic coloring appear to be regularly blocked or restored at a very early stage in PIE including:

1. most inflectional endings with an initial **e* such as perf.3sg. **e*,[77] athem.3pl. *-ent*,[78] gen.sg. **-es*, dat.sg. **-ei̯*, nom.pl. **-es*, etc.; an exception here is with **-eh₂-* stems, if their gen./dat./nom.pl. are to be reconstructed as effectively **-ah₂as*, **-ah₂ai̯*, **-ah₂as*.
2. the thematic vowel *e* in virtually all environments, e.g. ἔθανε 'die' (< **h₁e-dʰn̥h₂-e-t*; not †ἔθανα), πορεῖν (< **porh₃-é-s-en*; not †ποροῦν), Lat. *bibit* 'drink' (< **pí-ph₃-e-ti*; not †*bibut*[?]), OIr. *melid* 'grinds' < **melh₂-e/o-*, etc.; there are also no discernible allomorphs for voc.sg. **-e* (as **a* [< **-h₂-e*] or **o* < [**-h₃-e*]) or thematic n.pl. **-e-h₂* (as **ō* < **-h₃-e-h₂*, if **h₃* wins in such environments, cf. fn.122).
3. primary *s*-stem suffix **-es-*, with no traces of allomorphic **-os-* or **-as-* in, e.g., gen.sg. πλάτους 'breadth' (Hdt.+) <= **πλέτεος < **pleth₂-es-os* (not †πλατως[79]), n.sg.adj. ὑγιές 'healthy' < **h₁su-/h₁iu-gʷih₃-es* (not †ὑγιος).
4. primary *i*-stem suffix **-ei̯-* and *u*-stem suffix **-eu̯-*, with no traces of allomorphic **-ai̯/oi̯-* or **-au̯/ou̯-* in, e.g., βαρεῖς 'heavy' < **gʷr̥h₂-éu̯-es* (not †βαράες > †βαρᾶς), πλατεῖα 'wide' < **pl̥th₂-éu̯-ih₂* (not †πλαταῖα[80]).[81]

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[77] Other than the lack of **a* and **o* variants in the perfects of *seṭ* roots, compare also Hitt. 3sg.pres. *malli* 'grinds' which continues **mólh₂-e(i)* without coloring (on the *molō*-type present, cf. Jasanoff 2003: 71 and *passim*); the "thematic" *mallai* is better regarded as secondary (see Puhvel 2004 *s.v.*, cf. Sasseville 2021: 390 fn.82, 394) despite the apparent early attestation (cf. Melchert 1984a: 16-8), *contra* EDHIL140.

[78] Not strictly probative, in my opinion, are ἔβαν, ἔδιδον, and so forth as examples of colored *-ent* (Weiss 2020: 410): ἔβαν could be from **gʷeh₂-nt-* with generalized *e*-grade (cf. ἔβημεν) and shortening of the Ostoff's type, if not analogical after the participle βαντ-. Whatever the original vocalism in ἔδιδον was, the short *o* is readily explicable through 1pl. and 2pl.

[79] Technically, the weak stem of κρέας 'meat' (gen.sg. κρέως, etc.) could be taken as **kreuh₂-es-* if one assumes an ablauting **-s-/-es-*, although this is unparalleled in Greek and the comparison with Ved. *kravís-* (gen.sg. *kravíṣas*, etc.) could suggest a constant suffix here, cf. Stüber 2002: 20f. See Litscher (2007) for the analysis that considers both Gk. -ας and IIr. *-iṣ* to be secondarily derived.

[80] See Neri 2017a: 110 on its possible preservation in the toponym Πλαταῖα; There might be several cases of **h₂*-colored **-eu̯-* in secondary derivation according to Zair (2012: 205, 210, 212), e.g., OIr. *Letha* < **pl̥th₂-eu̯-i̯eh₂-*, Breton *divalav* 'ugly, odious' < **ml̥h₂-eu̯-i̯eh₂-*, less plausibly OIr. *tanae* < **t(e)nh₂-eu̯(-)o-* (cf. ταναός 'long').

[81] Good examples of *i*-stem based on *seṭ* roots are difficult to locate, cf. Weiss 1998: 46f on ἔρις 'strife' from **h₁erh₂-i-*, but this particular word is remade into a *d*-stem from the earliest attestations. The etymology and analysis for βοῦς 'cow' and ναῦς 'ship' do not guarantee -

5. suffix *-*eh₂*-, with no traces of allomorphic *-*oh₂*- (< *-*h₃-eh₂*-, if **h₃* wins in such environments, cf. fn.122) in, e.g., βορά 'food' (Pi.+) < **g*^{w}*orh₃-éh₂* (not †βορώ).

Not all colored variants postulated above are categorically unacceptable from a synchronic perspective, nor do they necessarily preclude the possibility of surviving as relic forms; thus, that they do not occur suggests they are either blocked in PIE or undergo an early restoration. In contrast, some instances of cross-morphemic coloring appear to have been preserved fairly well into individual branches. The following are among the more convincing examples to me:

6. nasal infix *-*ne-h₂*- > *-*nā*- as Gk. -νᾱ-/-νη- in, e.g., δάμνᾱμι 'subdue' <= **dm̥-né-h₂-mi*.
7. iterative-causative suffix *-*h₂-éi̯e/o*- > *-*ái̯e/o*- as in Lat. *domā-(re)* 'tame' < **domh₂-éi̯e/o*-, OLat. *votā-(re)* 'forbid' < **u̯oth₂-éi̯e/o*-;[82] this may also underlie Gk. ποτάομαι 'fly' < **poth₂-éi̯e/o*-, (rare) πτο(ι)άω 'frighten' (next to the common πτοέω) < **pioh₂-éi̯e/o*- (Hackstein 1992: 151-3.);[83] curiously no **h₃*-colored variant, namely *-*h₃-éi̯e/o*- > *-*oi̯e/o*-, e.g., λοέω <= **lou̯h₃-éi̯e/o*- (not †λοόω), Slov. *molíti* 'extends' <= **molh₃-éi̯e/o*- (LIV^{2} *s.v.* **melh₃*-),[84] Hitt. *aniya*- 'work' *if* from **h₁onh₃-éi̯e/o*- (not †*anaya*-; cf. fn. 85), more on this below.
8. factitive suffix *-*h₂*- coloring preceding *-*e*- in the "*newaḫḫi*-type" forms in, e.g., the eponymous Hitt. *newaḫḫ*- 'renew', Gk. νεάω 'plough', Lat. *novāre* 'make new' < **neu̯e-h₂-(i̯e/o-)*, etc.
9. *-*h₂-etó*- > *-*atό*- in θάνατος 'death' < **d*^{h}*n̥h₂-etó*-, κάματος 'toil' < **ḱm̥h₂-etó*-; possibly *-*h₃-etó*- in βίοτος 'life' < **g*^{w}*ih₃-etó*-, see Waanders 1974, Vine 1998: 12, 66-70.
10. *-*h₂-es*- > *-*as*- in some secondary derivations, e.g.: παλαστή 'palm' (Alc.+) < **pl̥h₂-es-t-éh₂*, cf. Hitt. *palḫasti*- 'breadth' and OCS *-ostъ*, see Höfler 2018: 125; γαληνός 'calm' (Ibic.+) < **gl̥h₂-es-nó*-, cf. γαλήνη/Dor. γαλᾱνᾱ 'stillness', see EDG *s.v.*,

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h₃/h₂-eu- sequences, cf. fn19.

[82] Vine p.c; for more examples of this type in Latin, see Schrijver 1991: 111.

[83] Willi (2018: 273) mentions the possibility of deriving σχάω 'slit open' and ἀρόω 'plough' respectively from **sḱh₂-éi̯e/o*- and **h̥₂rh₃-éi̯e/o*-. But zero-grade *-*éi̯e/o*- formations are in general untypical. Rather, the traditional reconstruction as **sḱh̥₂-i̯é/ó*- and **h̥₂rh̥₃-i̯é/ó*- is preferrable. On possible effects of 'Pinault's Law' in these environments, see Willi 2018: 275 fn.168. Further, γοάω should probably be analysed as **gou̯h₂-éi̯e/o*- (cf. Yamazaki 2009), but a denominative from **gou̯h₂-e-i̯e/o*- is equally plausible.

[84] On the phonology of Slavic *i*-presents, see Andersen 2014: 60-9.

Höfler 2017: 75.[85]

11. *-h_2-ero- > *-aro- in βριαρός ‘strong’ (Hom.) < *$g^{w}rih_2$-ero- (so EDG *s.v.*), cf. βρῑ́θω ‘be full of’ (Hom.+); less securely *-h_2-eno- > *-ano- in βάλανος ‘acorn’ < *$g^{w}l̥h_2$-eno- (Hom.+; see EDG *s.v.*), cf. Arm. *kalin* ‘id.’ with uncolored *-eno- (?)
12. some archaic lexical items such as *san- in Lat. *sanguis* ‘blood’ reflecting *h_1sh_2-en- (if not *h_1sh_2-n-?), Hitt. *išḫan-* (obl.) ‘id.’ < *h_1sh_2-(e)n-, etc., cf. EDHIL *s.v. ēšḫar/išḫan-*, Yates 2024; the word for ‘salt’ (Lat. *sāl*, Gk. ἅλς, TA *sāle*, etc.) could contain an ablauting *l*-stem *seh_2-l-/sh_2-el- if one follows the Leiden reconstruction.[86]

Being cautious not to overinterpret, the materials above suggest a couple of tendencies. Superficially, cross-morphemic coloring appears more likely to be maintained when it occurs *away from* the right edge of the word, perhaps when the affected *e is at least one syllable apart from the inflectional ending, as in *-*éi̯e/o-* (7), *-*etó-* (9), *-*es-teh₂-* (10), and *-*ero-/-eno-* (11). This may explain why the thematic vowel (2), basic *i*- and *u*-stems *-*ei̯-*/-*eu̯-* (4), *-*eh₂-* stems (5), and *s*-stems *-*es-* (3) do not have laryngeal-conditioned allomorphs.[87] On a structural level, this tendency may be understood in terms of faithfulness constraints, namely that the preservation of a colored variant depends on whether the colored ending and/or suffix can still be interpreted or mapped onto some part of the morphology in the grammar.[88] Thus, the reason why *éi̯e/o-, *-etó, and *-ero-/-eno- could survive as *-ái̯e/o-,

[85] Lat. *onustus* ‘loaded’ may belong here (as *h_1enh_3-es-to-) if one adopts a root *h_1enh_3-, so Stüber 2002: 88, cf. EDLIL *s.v. onus*. A base with *-os-* is also possible, so Weiss 2020: 256. κάρηνα (n.pl.) ‘heads’ (Hom.+) / Dor. κάρᾱνος ‘chief’ is another possible case (if < *$k̑r̥h_2$-es-no-), but given the accent this may reflect the *palma*-rule outcome.

[86] For an updated discussion on the PIE word for ‘salt’, see de Vaan 2022.

[87] This may also help account for the often-assumed coloring of the 3pl. *-*ento* reflected indirectly in (Hom.) πρίατο ‘buy’ (cf. Myc. *qi-ri-ja-to* KN Ai 1037+; if not < *$k^{w}ri(i̯)h_2$-to*) <= πρίαντο < *$k^{w}rih_2$-énto* if one considers the alteration of *-*ento* to *-anto* is farther from the right edge than, e.g., the perfect 3sg. **e*. Cf. Francis 1970: 87-92 on the idea that *$CRih_2C$- regularly becomes Gk. *CR*ια*C*-.

[88] This is already hinted by Peters (1980a: 192 fn.149), “wenn sie dadurch gestützt wurden, daß in den betreffenden Formkategorien bereits andere Formen mit gleichem oder vergleichbarem Ausgang existierten”. I differ from Peters crucially in that a significant part of the restoration in my view must have taken place in PIE, cf. fn.92,

*-*ató*-, and *-*aro*-/-*ano*- (and theoretically as those with *o*-vocalisms as well) perhaps lies in the fact that there exist the more basic *-*i̯e/o*, *-*tó*, and *-*ro*-/-*no*-, whereby the change of the vocalism from *e* to *a* is simply understood as part of the presuffixal stem. Similarly, the coloring effect of the factitive *-*h*$_2$- (8) on the preceding *e[89] resulting in a quasi-suffix *-*e*-*h*$_2$- > *-*ah*$_2$- finds support in its similarity with the denominative formation based on *-*eh*$_2$-stems (of the τιμάω type).[90] The preservation of the nasal infix *-*ne*-*h*$_2$- > *-*nah*$_2$- but the loss of *-*ne*-*h*$_3$- > *-*noh*$_3$- (e.g. ὄμνυμι) may also be explained if one considers *-*eh*$_2$- a more recognizable predesinential element whereas *-*oh*$_3$- is not.[91]

With these observations, we now assess the case at hand, namely the coloring in the supposed *$u̯_{\text{ŭ}}l̥h_3$-*eh*$_1$- > *$u̯_{\text{ŭ}}l̥h_3$-*oh*$_1$-. In my opinion, it is unlikely that the colored variant in this formation would have been kept intact, because *-*oh*$_1$- (followed by athematic endings -*m*, -*s*, -*t*, etc.) would not be a recognizable suffix in the synchronic grammar assuming the form had to continue being analyzable as a stative formation by the speakers of PIE. Unlike *-*nah*$_2$-, *-*á̯i̯e/o*-, *-*a*-*h*$_2$-, etc., an *-*oh*$_1$- is paradigmatically unsupported and completely isolated. The sequence *-*h*$_{2/3}$-*eh*$_1$- should behave in the same way as the other monosyllabic suffixes in post-laryngeal environment (*-*h*$_{2/3}$-*es*-, *-*h*$_{2/3}$-*ei̯*-, *-*h*$_{2/3}$-*eh*$_2$-, etc.) and would have restored (or generalized) the *e*-vocalism at an early stage. A comparison of ἐάλων with synchronic forms such as ἔγνων, ἔβρων, etc. as a proof of the acceptability of *-*ō*-(*m, -s, -t*, etc.) is anachronistic, as the type of restoration discussed here must be thought of as a process in PIE,[92] at which time *-*oh*$_1$- and *-*oh*$_3$- remain distinct. Empirically, there are simply no

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[89] On the original association of the *h_2-factitive formation with thematic adjectives, see Jasanoff 2003: 139f.

[90] Probably originally athematic, cf. the athematic conjugation of contracted verbs in Aeolic. More discussions of the thematization of originally athematic conjugation of Greek contracted verbs can be found in Rau 2009b.

[91] This certainly does not contradict the view that the merger of *-*nō/no*- with *-*neu/nu*- may be triggered by the operation of Cowgill's Law in *-*no*- > *-*nu*-, cf. Sihler 1995: 526f, (somewhat differently) Milizia 2004; a more recent discussion can be found in Sturm 2021: 90-2.

[92] Compare the examples in (1)-(5). That certain uncolored allomorphs of *-*h*$_2$-*e*- and *-*h*$_3$-

instances across Indo-European of *-*oh₁*- or *-*ah₁*- as an allomorph of *-*eh₁*-, as demonstrated by Gk. ἐδάμην 'conquer' <= **demh₂-eh₁*-,[93] OCS *zъrěti* 'ripen' < **ǵ̑rh₂-eh₁*-, Lith. *kedéti* 'burn' < **(s)kedh₂-eh₁*- (LIV[2] *s.vv.*), all with restored *-*eh₁*-.[94] Hence, if **ů̯lh₃-eh₁*- ever existed, one would reasonably expect †ἐάλην. That is not to say that the reconstruction of the form as a stative is impossible, but my general suggestion is that the assumption of cross-morphemic coloring in **ů̯lh₃-eh₁*- and its subsequent resistance to analogical levelling from (pre-)PIE up to historical Greek as the sole relic of this allomorph is not a trivial one.[95] Combining with the comparative evidence discussed above, the form rather finds a better explanation as a root aorist **ů̯leh₃*-.[96]

2.4.3 Alleged *a*-vocalism in the "*o*-grade" thematic aorists

If the outline of the analyses above should be accepted, πεπαρεῖν and ἑάλων can no longer

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e- are shared across all branches suggests coloring (of an apophonic *e*, cf. fn.119) has ceased to be a synchronic rule in the reconstructable stage of PIE (or perhaps "late" PIE), in much the same way that rhotacism is not synchronically operative in Classical Latin or the intervocalic loss of *s* in Greek. In other words, laryngeal coloring must be construed differently from synchronic phonological rules in PIE such as the double-dental rules, Stang's and Szemerenyi's Laws, etc., to which there are no shared exceptions in the daughter languages that can be projected back to PIE.

[93] See Francis 1970: 74f. on a putative δαμᾶναι preserved in the name Ἀδαμαντ-, but cf. Chantraine 1933: 269, Lubotsky: 1998 for alternative explanations. To the homonymous **demh₂*- 'build' may belong Hitt. *dam(m)eta* 'property' according to Watkins (1979); cf. Nikolaev 2010 for a critical assessment of the reconstruction of the root-final **h₂*. See Jasanoff 2022b for reconstructing **damh₂*- 'tame' with *a*-vocalism.

[94] If *-*á̯i̯e/o*- (next to the loss of *-*o̯i̯e/o*-) is informative, *a*-colored variants appear to be more resilient than *o*-colored ones, possibly due to the fact that *seṭ* roots with **h₂* are more abundant in number than those with **h₃*. This makes the survival of *-*oh₁*- in the absence of *-*ah₁*- even more suspicious.

[95] For the same reason, I fail to see the appeal in analyzing ἐβίων as a stative **g*^{w}*ih₃-eh₁*- (so Francis 1970: 83-5, Klein 1988: 270-2, Harðarson 1993: 208-10) rather than a plain root aorist **g*^{w}*i(i̯)eh₃*- following Lindeman 1985.

[96] Alternatively, -αλ- (instead of the expected *-υλ- from **e-ů̯leh₃*-) could have been imported from the present, see Rix 1992: 74.

stand as reliable testament to the **CaR-* outcome of inherited **CR̥h₃V* sequences in Greek. However, before one could conclude that **CR̥h₃V* regularly becomes Greek *CoR-*, as is the proposed development here, the same scrutiny must also be applied to the *o*-grade thematic aorists to identify and account for any traces of *a*-vocalism there. In that regard, two relevant points should be made.

2.4.3.1 θάρνυται

For the present of ἔθορον, next to θόρνυμαι one finds Hsch. θάρνυται· μετεωρίζεται and θάρνυσθαι· ὀχεύειν, which given the predominant *o*-vocalism of this verb must reflect an archaism in one way or another. The *a*-vocalism is most naturally explained as the expected outcome of an inherited nasal stem **dʰr̥-ne-h₃-* (remade into a νυ-present of the same type as ὄμνυμι 'swear' [<= **h₃m-neh₃-*]) and thus does not directly pertain to the issue here. Less plausibly, θάρνυται etc. could have been modeled on an earlier *ἔθαρον, if the nasal present itself is to be deemed a late innovation of this verb (with θρώσκω as the inherited present) and θάρνυται was created using the stem of an unattested aorist *θαρ- in the same way as θόρνυμαι that of θορ- (cf. fn.20). This conforms to the earlier attestation of θρώσκω (Hom.+) in comparison to θόρνυμαι (*hapax* Hdt. 3.109, Democr. A 153.10 Diels-Kranz, otherwise rare and poetic), but is complicated by the fact that the nasal presents have the specialized meaning of 'mate, impregnate' *vis-à-vis* θρώσκω with the more general meaning of 'jump (on), leap (upon), spring'.[97] Without reliable comparative evidence,[98] the relative antiquity

[97] In the main, inherited pairs of nasal presents and **-ske/o-* presents are rare in Greek. A relevant formal and semantic comparandum here is perhaps (συμ)μίγνυμι in its sense of 'have intercourse with', whose coordination with (συμ)μίσγω and semantic closeness may have inspired the creation of θαρνυ- and θορνυ-. *Contra* Sturm 2021: 95, semantic narrowing itself is not indicative of either archaism or innovation as the development is possible in either direction.

[98] Outside of Greek, OIr. *no-dair* 'mounts' may point to a **-i̯e/o-* present (< **dʰr̥h₃-i̯e/o-*), so McCone 1991: 30, cf. LIV² *s.v.* **dʰerh₃-*.

of the two present formations cannot be determined with certainty and a phonologically regular *$d^{h}r̥$-*ne-h₃*- > θαρνυ- still offers the most straightforward explanation.

2.4.3.2 *ἔταρον

Speculatively, *a*-vocalism has also been postulated for the prehistory of the form ἔτορε, albeit with much uncertainty. The thematic aorist itself is a *hapax* in Λ 236 (οὐδ᾽ ἔτορε ζωστῆρα παναίολον… "he did not pierce the sparkling girdle…") and ostensibly related to (or serves as the basis for) (ἀντ-)ετόρησε, τετορήσας, (ἀντι-)τορήσων (all Homeric), τετορήσω (Ar.), etc. Whether a poetic nonce word or not, ἔτορε as such is unlikely to directly continue *$tr̥h_3$-*e/o*-, given that its meaning 'pierce, transfix' aligns semantically better with *$terh_1$- (τέρετρον 'borer', τετραίνω 'pierce, perforate' [A.+], OCS *тьрjǫ* 'rub', cf. LIV^2 *s.v.*) as opposed to *$terh_3$- (τιτρώσκω, τρώω, etc. 'wound, harm'). The two roots show signs of contamination but are still kept apart etymologically and synchronically in Greek. Peters (1980b: 349 fn.53) followed by Harðarson (1993: 219f.) opines that the whole set of τορ- stems originates from a denominative τορέω (surviving in the *hapax* ἀντιτοροῦντα at *h. Mer.* 4.283) based on nominal τορ- formations of *$terh_1$- including τόρος 'borer', τορός 'shrill', διάτορος 'piercing', τόρμος 'socket, peg', τόρνος 'carving knife', etc. In Peters' view, ἔτορε could originally be the thematic aorist of *$terh_3$- 'wound' (presumably for him after *ταρ- => τορ-) and is later associated with τορέω modeling after pairs such as στυγέω : ἔστυγον 'hate', κτυπέω : ἔκτυπον 'crack, rumble'.[99] Formally, this provides a plausible explanation for the attested stem distribution (in particular the τορη- forms), although a noticeable drawback is that all such nominal forms are post-Homeric.[100] Alternatively, according to Willi (2018: 343 fn.79), the contact point between *$terh_1$- and *$terh_3$- is rather a prehistoric thematic aorist

[99] Similar thoughts already in Schwyzer 1939: 747 fn.1, cf. Ruipérez 1950: 405.

[100] That being said, one may reasonably surmise the existence of a simplex τορός 'piercing' *vel sim.* on the strength of ῥινοτόρος 'shield-piercing' (*hapax* Φ 392). A derivation based directly on a compounded -τορος is not impossible, cf. Tucker 1990: 78 and *passim*.

*ἔταρον, which, following the standard theory, would be the outcome of both *$t_{\mathring{r}}h_1$-*e/o*- and *$t_{\mathring{r}}h_3$-*e/o*-. The form retains the semantics of *$terh_1$- but is formally exclusively associated with *$terh_3$- (τιτρώσκω etc.) and later remade into ἔτορε as part of the analogical process discussed above (see 2.3). While both scenarios are possible, the zig-zag that ἔτορε must have travelled under these assumptions to acquire its inconsistent semantic and formal identities seems dubious to me.[101] While that aor./fut. τορησ- and τετορησ- are built upon a denominative τορέω should be basically correct, the attempt to join an alleged *ταρ (=> τορ-) here is an unnecessary complication. As essentially a *hapax legomenon* across the entire Greek corpus outside of lexicographical works,[102] the most parsimonious solution in my

[101] *Pace* Willi (*ibid.*), the formal association of the supposed *ἔταρον with τετραίνω appears no more or less recognizable than that with τιτρώσκω. That *ἔταρον would fall with the latter (but somehow still be semantically aligned with *$terh_1$-) is arbitrary. A similar objection applies to Peters' account: if τορ- analogically replaced *ταρ-, as it were, the affiliation of *ἔταρον to the τρω- forms under a word meaning 'wound' must have been strongly felt (and further strengthened after the analogy), in which case its later coalescence with τορέω is curious.

[102] The only other possible instance of the word is in A.R 2.113: the received text is ὑπὸ ζώνην θόρε χαλκός, but Platt (1920: 75) suggests a reading of τόρε. This emendation is followed in ed. Fränkel and by Vian in the Budé edition (1974: 181). Since there is a thematic connection (both passages involving descriptions of a non-lethal strike upon the belt/girdle) and phrasal similarity (κατὰ ζώνην, ὑπὸ ζώνην) between the Homeric and the Apollonian passages, it is possible that Apollonius' word choice might be influenced by the model in the Iliad:

Λ 234-6 Ἰφιδάμας δὲ κατὰ ζώνην θώρηκος ἔνερθε
νύξ', ἐπὶ δ' αὐτὸς ἔρεισε βαρείῃ χειρὶ πιθήσας
οὐδ' ἔτορε ζωστῆρα παναίολον, ἀλλὰ πολὺ πρὶν
But Iphidamus stabbed the belt under the belt
and himself leaned in with confidence of his strong hand
but could not pierce the bright warrior belt, before…

A.R. 2.112-3 ἀλλά μιν οὐ κατέπεφνεν, ὅσον δ᾽ ἐπὶ δέρματι μοῦνον
νηδυίων ἄψαυστος ὑπὸ ζώνην τόρε χαλκός.
But he did not strike him down; the bronze only cut on the skin
beneath the belt not touching the flesh of the belly.

opinion is to treat ἔτορε as an artificial creation, either in the sense of an *Augenblicksbildung* (possibly a *metri causa* creation replacing ἐτόρησε, see below) or somewhat productively from ἔστυγον besides στυγέω, ἔκτυπον besides κτυπέω, and ἔχραισμον besides χραισμέω 'help' (all Hom.+ and mostly poetic). This type of coordination (them. aor. + έω-present + *o*-stem noun)[103] requires a separate examination, but suffice it now to say that (1) for ἔστυγον and ἔκτυπον at least, metrical accommodation is readily a motivation to replace †στυγησ- and †κτυπησ- if their respective -έω denominatives are to be considered primary and (2) in the limited prose attestation of these verbs the thematic aorists do not occur. Thus, a poetic creation of ἔτορε from τορέω is plausible.[104]

2.4.4 βόλομαι

The examination above shows there is no positive evidence in the *o*-grade thematic aorists that ἔθορον, ἔμολον, ἔπορον, and ἔτορον represent the expected forms *ἔθαρον, *ἔμαλον, *ἔπαρον, and *ἔταρον. Even if one cautiously removes ἔθορον and ἔτορε as probative examples on the basis of the very tenuous suggestion of *a*-vocalism, ἔμολον and ἔπορον still offer two strong exemplars of inherited *CoR*- stems from *$C\underset{\circ}{R}h_3$-*e/o*-:

*$m\underset{\circ}{l}h_3$-*é/ó*-	>	**mol-é/ó*-
*$p\underset{\circ}{r}h_3$-*é/ó*-	>	**por-é/ó*-

To these one may adduce two more tentative examples outside of the realm of the thematic aorist that may find new interpretations with the proposed phonological development. One concerns the Homeric root thematic present βόλομαι 'wish', besides Att. βούλομαι, Dor.

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[103] Cf. Chantraine 1948: 347, Tucker 1990: 89, 189 fn.247, 247. As a parallel to a rare thematic aorist besides a more common denominative in -*i̯e/o*- and a thematic noun, one may draw the case of γόος, γοάω, γόον, where the thematic aorist (but cf. Nussbaum 2025 on the interpretation as imperfect) is also a Homeric *hapax*, cf. Chantraine 1948: 392.

[104] Under this account, the thematic aorist of τιτρώσκω (whether *ἔταρον or (*)ἔτορον) as such is not attested, cf. the generalization of *CRō*- in βιβρώσκω, ἔβρων, etc. 'devour'.

βώλομαι, δήλομαι, etc. The *o*-grade is difficult to account for and has been attributed to the influence of an assumed perfect *βέβολα (cf. προβέβουλα 'prefer' *hapax* A 113) and/or the nominal βουλή 'will' (cf. GEW and EDG both *s.v.* βούλομαι). A radical explanation of the *o*-vocalism is offered in LIV^2 which, following Peters (1986: 310f.), reconstructs *$g^{w}elh_3$- and invokes the *-eRo-* to *-oRe-* metathesis in a supposed "Narten-present", i.e. *$g^{w}el̥h_3$*-to(i)* > *$g^{w}ole$*-to(i)*.[105] The overall problem with the metathesis has been discussed earlier (see above 2.2).[106] As a way of offering another possibility, the thematicity of this verb in its most archaic form βόλομαι might point to an inherited thematic stem, of which one finds two radical formations in PIE, namely *R(é)-e/o-* and *R(z)-é/ó-*. The *o*-vocalism and the subsequent spread of a quasi-root *$g^{w}ol$- could have started from the latter, namely a zero-grade thematic present of the "*tudáti*-type", i.e. *$g^{w}l̥h_3$*-é/ó-*, which becomes *$g^{w}ol$*-é/ó-* by the regular development proposed here. Alternatively, at a perhaps much greater time depth the 3sg. middle of an athematic paradigm *$g^{w}l̥h_3$*-or* (> *$g^{w}ol$*-or* => *$g^{w}ol$*-to(i)*) could also be a possible pivotal form, but the relative chronology might be problematic depending on one's view on when the substitution of *-*or* by *-*to(i)* took place in pre-historic Greek.[107]

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[105] The reconstruction of *$g^{w}elh_3$- and the subsequent metathesis is followed by Minamimoto (2012). He further draws Myc. *qe-ro-me-no* /g^{w}elomenoi/ (PY Ad 697.a) as a possible pre-metathesized form of *$g^{w}elh_3$-, but as he admits this is speculative and an equally possible reading /k^{w}elomeno-/ : πελόμενος is preferred in DMic *s.v.*

[106] Admittedly, the Pamphylian form βολεμενυς 'willing' (3.13 Brixhe, Sillyon, 4th cent. BC) would be difficult to explain without invoking the metathesis (< *βελομενυς, as did Peters 1980b: 349 fn.52; 1986: 310). An alternative interpretation, as suggested by Brixhe (1976: 118f.), is to take βολεμενυς as phonetically βολε͂μενυς, an athematic denominative based on βολᾱ́ (Att. βουλή), i.e. *βολάομαι (with *a* > *e* before *o* per Brixhe 1976: 13 and athematization) => *βολε͂μαι, with the base verb still observed in the aorist ἐβολασετυ.

[107] It must be noted that Jasanoff (2003: 75) views βόλομαι as one of the few inherited "*molō*-type" presents that preserve the original acrostatic *o/e* ablaut, which may be a more elegant explanation for the Greek materials if one operates under the broader framework of a PIE "*h_2e-conjugation". Whatever the solution to the root formation is, the problem involving the extended stems *$bōl$- (Att.-Ion. βουλ-, Cret. βωλ-) and *$dēl$- (Thess. βελλ-, Boet. βειλ-, Heracl. δηλ-, Delph. δειλ-) continues to be a topic of controversy, specifically with regard to the debate over its interpretation as an inherited nasal formation vs. an *s*-aorist subjunctive, see Slings 1975, more recently Minamimoto 2012, Savić 2018: 23-6, Willi 2018: 446 fn.63

2.4.5 ὄρομαι

Similar lines of thought may apply to Hom. ὄρονται (ξ 104), ὄροντο (γ 471) 'keep watch', Myc. *o-ro-me-no* 'guarding' (PY Ae 134.a), usually connected with ὁράω 'see', Aeol. ὄρημμι (Sapph. fr. 31.11), etc. Because the attested forms show both signs of a digamma (cf. impf. ἑώρων) and an initial sibilant (cf. the aspiration, plausible connection with Lat. *servus*, see EDLIL *s.v.*), the root has been variously reconstructed as **ser-* (LIV^2 *s.v.* 1.**ser-*, DMic. *s.v.*, Rix 1994: 77f.), **u̯er-* (GEW and EDG *s.v.* ὁράω), or a mixture of the two **su̯er-* (DELG *s.v.* ὁράω, Bader 1978 *apud* Rix 1994: 78 fn.87). The prevailing *o*-vocalism is again problematic. While a denominal origin may lie behind ὁράω, the radical ὄρονται in light of the Mycenaean evidence is clearly old, nor does a substitution of the stem **ἐρ-* by ὀρ- under the influence of supposed **sor-ó-*, **sor-ā́-* (so LIV^2) seem convincing. Addressing these issues, a more compelling interpretation is advanced by Hackstein (2002: 123-31), who incorporates ἔρῡμαι 'ward, protect' (cf. them. ἐρύομαι, without vocalic anlaut ῥῦ́ατο, ῥύομαι [all Hom.+]) into this family of words and reconstructs a *seṭ* root **su̯erh₃-* on the basis of the long quantity in ἐρῡ- and, in his view, the disyllabic reflex *-oro-* in ὄρονται.[108] The critical development from **su̯erh₃-* to the doublet **su̯r̥h₃-* (ὀρο-) and **sruh₃-* (ῥῡ) in Hackstein's theory hinges on a PIE metathesis **u̯L̥* > **Lu* before consonant or at word-end. This is a long recognized phenomenon since Wackernagel (1896: 206f.) and Brugmann (1897: 260f.) supported with clear examples such as **u̯ĺ̥kʷos* 'wolf' > **lúkʷos*, e.g. Gk. λύκος, **kʷ(e)tu̯r̥* 'four' > **kʷ(e)tru*, e.g. Lat. *quadru-*, etc.[109] A minor additional specification required by Hackstein, to my

with literature.

[108] A key obstacle in this proposal (and a central point of debate in the discussion of this root otherwise) is the absence of the *u̯*-glide in the Mycenaean form, but Hackstein (2002: 129f. with lit.) makes a strong case for an earlier drop of *u̯* before *-o-*, with evidence from both dialectal and cross-linguistic comparisons, cf. Solmsen 1901: 140-2, Bernabé and Luján 2020: 204.

[109] Cf. Mayrhofer 1986: 161f., Lubotsky 1994: 98-100., Meier-Brügger 2021: 109, recently Clayton 2023: 9f., 22-4 with more literature. As a contrast to the *-eRo-* to *-oRe-* metathesis discussed earlier, **u̯r̥* > **ru* is both typologically more plausible and supported by

knowledge not in the standard formulation of this sound change elsewhere, is the possible addition of a laryngeal, i.e. **u̯L̥(H)*. This to me looks plausible. For a positive example of this expanded formula, one finds Ved. *śvaśrū́-* 'mother-in-law' < **su̯akrúH-* < **su̯ekú̯r̥-h₂-* (cf. unmetathesized *śváśura-* 'father-in-law'),[110] which probably is generalized from the nom.sg. **su̯ékú̯r̥-h₂(-s)* and acc.sg. **su̯ékú̯r̥-h₂-m*. The non-application of the metathesis when **u̯L̥H* is prevocalic is reflected in Ved. *urú-* 'wide' < **(h₁)u̯r̥H-ú-* (also *úras-* 'breast' <= **(h₁)u̯r̥H-és-*), CLuw. *ulant(i)-* 'dead' < **u̯l̥h₃-e/ont-* (cf. above 2.4.2), Lat. *īn-solēscō* 'become overbearing' and Hitt. *šulle-* < **su̯l̥H-eh₁-* (cf. Melchert 2005, LIV² *s.v.* **su̯elH-*), and so forth,[111] and in the general absence of metathesized variants in such environments.[112] Under this assumption, the stem ῥῡ in ῥῦατο etc. continues the metathesized root **sruh₃-(h₂e, -to*, etc.*)*, whereas ὀρο- in ὄρονται etc. is built from, according to Hackstein (2002: 127, 129), the restored unmetathesized root **su̯r̥h₃-* with retracted accent and the disyllabic reflex of the **CŔ̥h₃C* sequence, which is later thematized via the same mechanism described by Harðarson (1993: 169-70, cf. fn.19) for the *o*-grade thematic aorists. In my opinion, rather than an

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unambiguous evidence that cannot be otherwise explained.

[110] Cf. Peters 1980a: 184 fn.138, Mayrhofer 1986: 162, Meier-Brügger *ibid*. and 101, NIL *s.v.* *su̯ekú̯r̥/u̯er-*, recently Olsen 2025: 316-20. The proof of **u̯L̥H* > **LuH* is in a way unnecessary, since in the original formulation of the rule the sequence **u̯L̥H* automatically constitutes **u̯L̥* in a preconsonantal position.

[111] According to Peters (1980a: 23-7), aor. (ἐπ-)αυρεῖν 'share, enjoy' (Hom.+) is to be connected with εὗρον (following DELG *s.v.* ἐπαυρίσκομαι; on εὗρον, cf. fn. 236) and goes back to a zero-grade **u̯ar-e/o-* < **u̯r̥h₁-e/o-,* in which case the form shows no metathesis as expected (cf. **u̯r̥h₁-e/o-* > †*ruh₁-e/o-* > †(ἔ)ρυε *vel sim.*). But Peters' account requires further remodeling involving yet another type of metathesis, i.e. **u̯aC-* > *au̯C-* and *u̯eC-* > *eu̯C-*, the details of which must be left aside for now (see Balles 2007).

[112] The precise nature of the conditioning of this metathesis is curious. The underapplication of **u̯L̥H* > **LuH/_V* goes against the prediction of the standard version of this rule (i.e. **u̯L̥* > **Lu* before consonant). This may be due to the undesirable resulting sequence **LuH-V-* > **LuV-* with vowel hiatus. Clayton (2023: 9) suggests syllable boundary as a conditioning factor (i.e. **u̯L̥]σ* > **ru*), but this is also contradicted by a lack of **u̯L̥H* > **LuH/_V* and by *su̯ekú̯r̥-h₂-(s)*. The best way to formulate the metathesis here is precisely as Hackstein does, namely to treat **u̯L̥(H)* as one unit, with the same conditioning factors, i.e. before a consonant or at absolute auslaut.

unmotivated restoration, a better way to account for the distribution of the metathesized *$sruh_3$- vs. the unmetathesized *$s\underset{\circ}{u}rh_3$- is to take ὀρο- as a zero-grade thematic formation to begin with, in which case the stem *$s\underset{\circ}{u}rh_3$-*e/o*-, based on the discussion above, would (1) not undergo *$\underset{\circ}{u}LH$ > *LuH because it is prevocalic and (2) regularly become *$suor$-*e/o* under the proposed reflex of *$C\underset{\circ}{R}h_3V$. The existence of a zero-grade thematic present in this verb is, in fact, directly observable in ῥύομαι (Hom.+) < *$sruh_3$-*e/o*-, built from the metathesized root with thematization modeled after *$s\underset{\circ}{u}rh_3$-*e/o*-.

The structural advantage of viewing βόλομαι and ὄρομαι as *tudáti*-presents is that it removes from the very small set of *o*-grade thematic presents (*R(o)-e/o*-) two important instances which cannot be explained either through an initial *h_3 or as denominal formations.[113] Thus, βόλομαι and ὄρομαι are to be classified instead together with λύω, γράφω, and so forth as a present of the *R(Ø)-é/ó*- type, exhibiting superficial *o*-vocalism only by laryngeal-conditioned sound changes, a situation precisely paralleled by the *o*-grade aorists. In this way, by supposing a regular outcome of *$C\underset{\circ}{R}h_3V$ sequences as *CoR-, one potentially finds a unifying account for both the aberrant *o*-grade forms in the thematic aorist and the (simple) thematic present, without the need to invoke large-scale analogical levelling (in the case of the thematic aorists) or denominative explanations.

2.5 Labial conditioning of *$C\underset{\circ}{R}h_3V$ > *$CoRV$

Taking a conspectus of the evidence so far (with the two undetermined cases listed in brackets, cf. 2.4.3; n.b. the proposed *$C\underset{\circ}{R}h_3V$ > *$CoRV$ is considered a phonological process):

[113] Of the other four *R(o)-e/o*- presents in Homer (based on the list in Risch 1974: 262), ὄπλεσθαι ‘prepare’ is readily denominative from ὅπλον ‘tool, instrument’ (on the type, see Schwyzer 1939: 722f.); οἴχομαι ‘depart’ and (ἀν-)οίγω ‘open’ points to *h_3eig^h- (LIV^2 *s.v.*, on the potentially corrupted Hsch. ε 1211 εἴχεται· οἴχεται, see Latte and Cunningham 2020 *s.v.*) and *$h_3\underset{\circ}{u}ei\underset{\circ}{g}$- (esp. in view of Aeol. inf. ὀείγην, cf. Forssman 2005, EDG *s.v.* οἴγνυμι, differently Woodhouse 2015b: 281); ὄθομαι is obscure, cf. below 2.7.1.

**ml̥h₃-é/ó-*	>	**mol-é/ó-*
**pr̥h₃-é/ó-*	>	**por-é/ó-*
**gʷl̥h₃-é/ó-*	>	**gʷol-é/ó-*
**su̯r̥h₃-é/ó-*	>	**suor-é/ó-*
[**dʰr̥h₃-é/ó-*	>	**dʰor-é/ó-/dʰar-é/ó-*?]
[**tr̥h₃-é/ó-*	>	**tor-é/ó-*?]

Table 2.3 Conspectus of examples showing **CR̥h₃V > *CoRV*

it is observed that the sequence **R̥h₃* is preceded by a labial consonant (*m-*, *p-*, *gʷ-*, and *u̯-*) in the four more reliable examples. One suspects that the change **CR̥h₃V > *CoRV* is indeed conditioned by the presence of a preceding labial segment. Such a conditioned development is natural from a phonetic point of view. An intermediate stage may be construed as **CəRh₃V* with an anaptyctic vowel that ultimately merges with α in most cases (see van Beek 2021: 11), but under the condition discussed here the combined effect of the labial gesture of the preceding consonant and the anticipatory coarticulation with the following **h₃* could plausibly influence the acoustics of /əR/ enough that the sequence is perceived as *oR*. The rule may be formulated in this way:

PIE **R̥* > Gk. *oR* / *[+labial, -syllabic] _*h₃V*

Within Greek, the labial conditioning of the *o*-timbred reflexes of the syllabic liquids (in non-laryngeal contexts) has been recognized in certain dialects since Morpurgo Davies (1968) and has recently been discussed extensively by van Beek (2021 *passim*)[114] and Scarborough (2023: 92-107). Although details are still debated, it is generally agreed that there is a strong tendency in Arcadian, Cypriot, and potentially in Mycenaean[115] and Cretan for *oR/Ro* reflexes to be accompanied by a labial environment, besides the regular, unconditioned *a-*

[114] See, for instance, 21-38 for general remarks, 63-66 on Mycenaean, 107-10 on Cretan, 111-3 on Laconian, 120-9 on Aeolic, 352-4 on epic.

[115] For a concise summary on the discussions surrounding the Mycenaean materials, see Bernabé and Luján 2020: 193-7, cf. also Hajnal and Risch 2006: 206-9, Thompson 2010: 191f., Sabattini 2021a.

timbred outcomes. Comparatively, the effect of the labiality of the surrounding consonants on the reflexes of the syllabic liquids is also observed in Balto-Slavic (Vaillant 1950: 172, followed by Kortlandt 2009: 39-41) where *$\underset{\circ}{R}$ becomes *uR when following a labiovelar stop but *iR elsewhere. Offering a more direct parallel, in Indic the distribution of $\breve{u}r$ (as opposed to $\breve{i}r$) < *$\underset{\circ}{R}H$- is also, among other factors, largely dependent on the presence of a neighboring labial element (see recently Clayton 2022). Thus, the type of conditioned reflex of *$C\underset{\circ}{R}h_3V$ > *$CoRV$ is both familiar to Greek and supported with comparative evidence.[116]

By extension, it is hypothesized that when a labial environment is absent (as in *$d^h\underset{\circ}{r}h_3$- and *$t\underset{\circ}{r}h_3$-), the regular outcome should be *$CaRV$-, i.e. the same as the standard theory predicts for *$C\underset{\circ}{R}h_1V$- and *$C\underset{\circ}{R}h_2V$-. Within the group of the *o*-grade thematic aorist, ἔμολον and ἔπορον would thus form the historical core of this type. Since ἔτορε is no longer to be aligned with τιτρώσκω etymologically (see 2.4.3.2), only ἔθορον (if indeed from *ἔθαρον) requires an analogical solution, probably via βλώσκω : ἔμολον :: θρώσκω : X where X is resolved as ἔθορον. Such an analogy is much easier to maintain than one after the model of $CR\bar{a}$- : CaR- (ἔκαμον : κμᾰ́τος etc.) as Kuryłowicz et al. proposed (cf. above 2.3), for (1) it would be modeling after forms of the same vowel series and (2) βλώσκω ‘go, come’ and θρώσκω ‘leap, spring, rush’ both denote some kind of physical motion and thus are semantically relatable.

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[116] It may be said that the *o*-reflex of *$\underset{\circ}{R}$ in *$C\underset{\circ}{R}HV$ is doubly conditioned by a preceding labial and a following *h_3 (cf. below on chronology). The effect of *h_3 in inducing the *o*-vocalism here (either through an “umlauting” effect on the anaptyctic *schwa* or by labializing the original syllabic liquid) is no doubt related to its commonly assumed rounding character, cf. Beekes 1995: 148 ([ʕʷ]), Rasmussen 1999: 77 ([ɣʷ]), Kloekhorst 2018 ([qʷ]), Weiss 2020: 55 ([ʕʷ]), differently Kümmel 2022, who assumes only backing but no rounding. It is, therefore, technically possible that the rule may be formed in a more generalized way as *$\underset{\circ}{R}$ > oR / [+labial, -syllabic] _ [+labial, -syllabic] (i.e. when the syllabic liquid is flanked by labial [and/or rounding] features on both sides, not limited to *h_3), as in, e.g., *to-pe-za* ‘table’, cf. τράπεζα, < *$(k^we)t\underset{\circ}{u}\underset{\circ}{r}$-*ped*- (but see Sabattini 2021b on the alternative “three-footed” etymology, i.e. *$t\underset{\circ}{r}$-*ped*-). But since the vocalization in *$C\underset{\circ}{R}C$ postdates that of *$C\underset{\circ}{R}HV$ and the dialectal reflex of *$C\underset{\circ}{R}C$ > *CoR does not require such strict conditioning, the vocalization of *$\underset{\circ}{R}$ in *$C\underset{\circ}{R}C$ and *$C\underset{\circ}{R}HV$ may follow different rules.

With respect to the chronology, the phonologically regular outcome *CoR*- must have appeared by common Greek, since the *o*-vocalism in the forms discussed so far are (1) pan-Hellenic where dialectal materials are available and (2) characterized by a $*h_3$ after $*\underset{\circ}{R}$. Because an intermediate stage $*C_{ə}\underset{\circ}{R}h_3V$ is postulated, the *terminus post quem* for the development proposed is the realization of $\underset{\circ}{R}$ with a prop vowel *before* the liquid (in the specific context of $*C\underset{\circ}{R}HV$). There are good reasons to believe this happened at a relatively early stage. The uniformity itself of the *CaR*- (or *CoR*-) reflex of $*C\underset{\circ}{R}HV$ across all dialects suggests that the vocalization of the syllabic liquids in this particular environment is probably a pre-Greek process and certainly precedes their vocalization in other phonetic contexts, which differs dialectally both in the position of the prop vowel (*_R/R_)* and in its timbre (*a/o*) (cf. García-Ramón 1985: 206-8, van Beek 2021: 11). More tentatively, the anaptyxis in $*C_{ə}\underset{\circ}{R}h_3V$ may have already begun (perhaps sporadically) in late PIE, considering the fact that in all branches where the outcome of $*C\underset{\circ}{R}HV$ can be established the reflex is consistently $*_{ə}R$, i.e. with a short prop vowel (in most cases $*a$) preceding the liquid.[117] This pattern stands in contrast to the treatment of unaccented $*C\underset{\circ}{R}HC$ which shows a considerable variety in the daughter languages, cf. Lat. *Rā*, Ved. *īR/ūR*, OIr. *Rā*, Germ. *uR*, Gk. *Rē/ā/ō*, Arm. *aR/aRa*(?)[118], etc. That only $*h_3$ (together with the preceding labial) induces the *o*-timbred reflex stipulates the loss of the distinction of the laryngeals as the *terminus ante quem.* This should pose no problem if one reasonably assumes that Greek kept the laryngeals apart longer than the other branches, in view of the triple reflex of the syllabic laryngeals ($*CH_iC > *CE_iC$) and in $*C\underset{\circ}{R}H_iC$ ($> *CR\bar{E}_iC$) clusters. In particular, the latter environments are indicative of the capability of laryngeals to individually influence ("color"?) a neighbouring *ə* ($*CR_əH_iC > *CR\bar{E}_iC$) in an *einzelsprachlich* way in Greek (cf. the unvaried outcome *Rā* in Italic and

[117] An overview of the reflexes in individual branches can be found in Beekes 1988: 206-9.
[118] The problem with the Armenian evidence for $*C\underset{\circ}{R}HC$ is not entirely settled, with *CaR* (e.g. *garn* 'lamb' < $*u\underset{\circ}{r}h_1n$-) and *CaRa* (e.g. *čanačʿem* 'know' < $*\acute{g}\underset{\circ}{n}h_3$-*sḱ*-*i̯*-*é/ó*-) being the two relatively secure outcomes. See Clackson 1994: 36-41, Beekes 2003: 194f., Olsen 1999: 775f., Woodhouse 2015a, cf. Mayrhofer 1986: 128.

Celtic).[119] Since the vocalization of the liquid in *$C\underset{\circ}{R}HC$ as *$CR_{\partial}HC$, as suggested above, probably happens at a later stage than that of *$C\underset{\circ}{R}HV$ > *$C_{\partial}RHV$, by extension the loss of the distinction of laryngeals must postdate the advent of *$C_{\partial}RHV$. Thus, there is a conceivable and realistic window for a stage *$C_{\partial}Rh_3V$ to exist and for the proposed change *$C\underset{\circ}{R}h_3V$ > *$C_{\partial}Rh_3V$ > *$CoRV$ to take place.[120]

A major argument against the *o*-timbred outcome for *$C\underset{\circ}{R}h_3V$ clusters in Greek is the lack of defensible parallels where *$C\underset{\circ}{R}h_1V$- becomes *$CeRV$- via a similar "coloring" or "umlauting" process.[121] So states Harðarson (1993: 160): "wenn die Laryngale *h_2 und *h_3 gemäß ihrer Qualität das aus $\underset{\circ}{R}$ entstandene *əR* jeweils zu *aR bzw. *oR umgefärbt hätten, so müßte man davon ausgehen, daß sich in der Gruppe *$C\underset{\circ}{R}h_1$-V in entsprechender Weise eine palatale Resonanz entwickelt hätte, die zu dem Ergebnis *CeR-V geführt hätte". While the structural consistency Harðarson aims at is certainly desirable, one must take into account some particularities of the case at hand. Firstly, the expectation that *$C\underset{\circ}{R}h_1V$- should yield *$CeRV$- as parallel to *$C\underset{\circ}{R}h_3V$ > *$CoRV$- is not warranted *a priori*, as the PIE laryngeals are known to possess idiosyncratic phonetic and articulatory properties, e.g. the voicing effect of *h_3, the aspirating effect of *h_2, etc. With respect to their ability to influence a neighboring vowel, *h_1 for all practical purposes is usually regarded as the non-coloring laryngeal. The absence of coloring, combined with the general lack of other traceable alterations on adjacent sounds, may be taken as an indication of the relatively "weak" coarticulatory effect of *h_1 on

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[119] The "coloring" (or a lack thereof) of the anaptyctic *ə* happens in individual branches and should be kept apart from that of an apophonic *e*, cf. fn.92.

[120] It awaits further investigation whether there are other cases (beyond *$C\underset{\circ}{R}h_3V$) where a *schwa* from a vocalized syllabic liquid is doubly conditioned by a labial and a *h_3.

[121] The *e*-vocalism in ἔτεμον, Arc. ἔζελε, etc. used by Beekes (1969: 221-4) as evidence for *$C\underset{\circ}{R}h_1V$- > *$CeRV$- is permissible (and perhaps better explained) as morphological full-grade *$CeRh_1$-, so with Harðarson (1993: 161), see below 3.3.3, cf. Wackernagel 1916: 14. Against the *e*-timbred reflex of *$C\underset{\circ}{R}h_1V$ speaks ἀρήν 'lamb' < *$\underset{\circ}{u\underset{\circ}{r}}h_1$-*en*- (on the derivation and Myc. *wo-ro-ne-ja* 'wool', see Peters 1993: 390f, fn.74), the *h_1 more directly reflected in πολύρρηνες 'rich in sheep' < *-$\underset{\circ}{u\underset{\circ}{r}}h_1n$- (but cf. Nikolaev 2021: 154f. fn.44, fn.48 for a different analysis of the form).

its neighbours in comparison with *h_2 and *h_3.[122] Secondly, since *$\underset{\circ}{R}$ develops to *oR/Ro* in Aeolic (independent of a surrounding labial consonant or *h_3) and reliable traces of *o*-timbred reflexes besides *aR/Ra* are also found in Arcado-Cyrpiot and already in Mycenaean,[123] it is reasonable to assume that the pronunciation of the original anaptyctic vowel in pre-Greek is perhaps skewed towards the back of the vowel space and/or in some other way prone to be acoustically perceived as the mid back vowel *o*.[124] In other words, the intrinsic phonetic traits of the anaptyctic *ə* render it predisposed to be susceptible to *o*-coloring. The same cannot be said with regard to a development *ə* > *e*, in view of the complete absence of *e*-reflexes from *$\underset{\circ}{R}$ in Greek across the board.[125] If the above analysis holds, that *$C\underset{\circ}{R}h_1V$ > *$CeRV$ fails to operate then becomes understandable, given that the Greek anaptyctic vowel is phonetically distant from *e* to begin with and *h_1 possesses a relatively weak coarticulatory effect on its neighboring sounds.

[122] For more on this idea, see Eichner 1988: 131 for a supposed "Stärkehierarchie" *h_3 -> *h_2 -> *h_1, whereby *h_3 has the strongest "effect" and *h_1 the weakest. According to Eichner, this relative strength of coarticulatory effect can be observed when there is a coloring conflict, e.g. *h_1eh_2, *h_2eh_3, etc, perhaps in cases like Lat. *ōmen* 'omen' < *$h_2$óh$_3$-m$\underset{\circ}{n}$ (but the etymology is entirely uncertain, see EDLIL *s.v.*), cf. Hitt. *ḫā-* 'believe'. Examples are rare and the phenomenon remains a topic to be investigated. The general weakness of *h_1 should not be construed as a suggestion that it has no influence on neighboring sound at all: as indicated by the treatment of *$C\underset{\circ}{R}h_1C$ clusters (> *$CR_əh_1C$ > *$CR\bar{e}C$), *h_1 still "colors" an immediately adjacent *schwa* into *e* in Greek.

[123] Some tantalizing materials suggest possible *u*-reflexes of *$\underset{\circ}{R}$ (e.g. σύρω 'draw' < *t$\underset{\circ}{u\underset{\wedge}{r}}$-$\underset{\wedge}{i}$é/ó-, cf. σαίρω 'sweep'), although the evidence is inconclusive, see van Beek 2021: 21-7, cf. Vine 1999: 570 on the possible effect of Cowgill's Law in *-t$\underset{\wedge}{u}$or- > *-t$\underset{\wedge}{u}$ur-.

[124] Similarly Allen (1959: 249f.): "oρ/ρo could then represent either a development from a more labialized form of the liquid, or simply a means of indicating a rather back mid vowel (e.g. [ʌ]) in dialects where the short *ă* did not have this value".

[125] In presumably palatalizing environments such as σκάλλω 'stir up' < *sk$\underset{\circ}{l}$-$\underset{\wedge}{i}$é/ó-, the neighbouring palatal glide still fails to exert enough influence on the neighbouring *schwa* (†*sk$_ə$l-$\underset{\wedge}{i}$é/ó-* > †*skel-$\underset{\wedge}{i}$é/ó-* > †σκέλλω) in the way that labials do in labializing environments.

2.6 Conclusion

In the preceding sections, I have demonstrated that the metathesis *-eRo- > -oRe-* is generally unfounded in Greek and cross-linguistically untypical. The only supporting evidence (Myc. *re-wo-te-re-jo* vs. Hom. λοετρόν) outside of the materials it was designed to explain is dubious and counterexamples exist. The fact that this process targets only **-eRḫ₃-* and yet is supposed to be a post-Mycenaean development creates a chronologically implausible scenario. The analogy with *CRā-/CaR-* forms may have played a contributing role, but the very marginal presence of the *CRō-* stems in the early materials and the general tolerance of ablaut patterns involving different vowel series in Greek suggests this process alone may not have sufficient analogical pressure to trigger the remodeling of the allegedly inherited *CaR-* forms. It is proposed that the *o*-grade thematic aorists reflect the phonologically regular outcome of the **CR̥h₃V* sequence and the two previously held instances of *CaR*-reflexes have been discussed and shown to be not probative. A more careful examination of the data including possible evidence outside of the group at hand suggests that the change **CR̥h₃V > *CoRV* may be contingent on a preceding labial element.

Given the scarcity of securely reconstructable cases of **CR̥h₃V* in Greek, it is freely acknowledged that our conclusion here may be of a tentative nature. Nevertheless, the solution proposed in this study offers several advantages over the two standing theories. The more significant one is perhaps the ability to dispense with the *ad hoc -eRo- > -oRe-* metathesis, a development hardly tenable in view of the vast number of inherited *-eRo-* that are kept intact with virtually zero positive evidence for the metathesized variants. The sound change supposed here in comparison is phonetically easier to justify and its mechanism is known to Greek. Structurally, with the incorporation of βόλομαι and ὄρομαι as zero-grade formations of the *tudáti*-type this theory may have the advantage of explaining the problematic *o*-vocalism in both the aorist and the present under a single hypothesis.

2.7 Excursus: (*C*)*oRe-s*- forms (ἐλόεσα, ἐκόρεσα, ἐστόρεσα, etc.)

The sigmatic aorists of the shape (*C*)*oRe-s*- are explained as the result of a metathesis from *κεροσ- and *στεροσ- in Ruipérez's original proposal (1950) and this view still prevails as the standard explanation (see, e.g., Harðarson 1993: 222). In my view, however, the apparent existence of a "(*C*)*oRes*"-type aorist is an epiphenomenon.

2.7.1 ἐλόεσα

As was suggested above (cf. 2.2.2), ἐλόεσα can be understood as built directly from the inherited iterative-causative stem λοε- in λοέω (< *$louh_3$-e̯ie/o-, cf. Lat. *lavāre*).[126] There may be two mechanisms through which this development could have happened, in lieu of the expected and productive synchronic pattern pres. -έω : aor. -ησ-. Firstly, besides the iterative-causatives, the *o*-vocalism is also prominent elsewhere in the paradigm of this verb,[127] in particular in the correspondence between Gk. λόω (impf. λόε κ 361) and Lat. *lavere*, both reflecting a root thematic present *$lo\underset{\cdot}{u}h_3$-e/o-.[128] Unless these are to be regarded as independent processes, it is plausible that the *o*-vocalism is generalized and has spread from the original iterative-causative stem early on.[129] Hence, ἐλόεσα may be thought of as a derivation directly based on the *o*-grade root. Whether this happens via *$lo\underset{\cdot}{u}h_3$-s- or *$lo\underset{\cdot}{u}$e-s- (extracted from *$lo\underset{\cdot}{u}(h_3)$-e̯ie/o-) is a matter of chronology. The first scenario is older (and less

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[126] Equally possible is a denominative formation *$louh_3$e-i̯é/ó- or, for Lat. *lavāre*, *$louh_3éh_2$-i̯é/ó- (LIV^2 *s.v.* *$le\underset{\cdot}{u}h_3$-). The difference between a causative-iterative and a denominative derivation does not significantly affects our analysis, but see below.

[127] Additionally, cf. Umbr. imper. *vutu* 'wash' < *$lo\underset{\cdot}{u}(h_3)$-etōd (cf. Meiser 1986: 155, Untermann 2000: 867), Arm. *loganam* 'bathe' (cf. fn.25).

[128] Differently Klingenschmitt (1982: 116f.) takes λόε as a relic form continuing an athematic root aorist *$le\underset{\cdot}{u}o$-t > *$lo\underset{\cdot}{u}$e-t, cf. LIV^2 *s.v.* *$le\underset{\cdot}{u}h_3$- fn.2

[129] Thus in agreement with Woodhouse's suggestion (2014: 200) that the derivation of λοέω "might better begin and end with the causative/iterative *lou-é̯i-e/o-, given that washing, like knocking, frequently requires a series of repeated similar actions." The general idea that the prevailing *o*-vocalism could be inherited is already mentioned in Peters 1987: 290f.

plausible) and must have undergone the loss of the laryngeal via Saussure's Effect (> **lou̯s-*), which would directly produce the commonly assumed "contracted" forms λούσεν, λούσαντο etc. The second scenario happens at a later stage, at which time λοέω essentially coalesces with the patterns exhibited by "real" disyllabic roots such as καλέω : ἐκάλεσα[130] or by denominatives from *s*-stems such as τελέω : ἐτέλεσα'accomplish'. Importantly, the creation of ἐλόεσα would have to happen before the rise in prominence of the -ε-/-η- pattern (e.g. φιλέω : ἐφίλησα) and thus its membership in the -έω/-εσα conjugation as a quasi-disyllabic root is stable.[131] The second mechanism, which only slightly differs from the first, is that λόεω could be (or rather continue to be) interpreted by speakers as a proper denominative/causative-iterative formation[132] of the type *R(o)-éi̯e/o-*, e.g. δοκέω 'think', νοστέω 'return', etc. In this scenario, λοέω/ἐλόεσα joins the rank of κοτέω/ἐκότεσα 'be angry' (Hom.+), ποθέω/ἐπόθεσα 'long for' (Hom.+), and potentially Hsch. o 141 ὄθεσαν· ἐπεστράφησαν 'turned to' : ὄθομαι 'attend, take heed'. It is generally assumed that -εσ- in these formations is the original aorist formant to -έω before the spread of -ης-, following the standard pattern in *-i̯e/o-* denominatives, e.g., φυλάττω : ἐφύλαξα 'guard' < **p*^{h}*ulak-i̯e/o-* : **p*^{h}*ulak-s-* (cf. Tucker 1990: 193f., Willi 2018: 422-5). In this way, ἐλόεσα is special only in that it preserves, along with the other relic forms above, an archaic type of *s*-aorist to a **-i̯e/o-* present in Greek.

2.7.2 ἐκόρεσα

A similar analysis should apply to ἐκόρεσα as well. To begin with, the color of the laryngeal in this verb is highly contested with the root etymology debated between PIE **ḱerh*$_1$- 'grow'

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[130] For a summary of the various types of radical verbs following -έω/-εσα, see Tucker 1990: 19, cf. Jasanoff 2002-3.

[131] The chronology here should not be taken with rigidity, as both patterns (-έω/-ησα and -έω/-εσα) may be productive in tandem at some point. On the contamination of the two patterns, cf. fn.23, Peters 1987: 291. On the origins of the pattern -έω/-ησα from statives based on compounds, see Tucker 1990: 34-72.

[132] The two types are synchronically indistinguishable, cf. Tucker 1990: 162-80

and *ḱerh$_3$- ‘nourish’ (cf. Villanueva Svensson 2012: 239-44, Vine 2016: 134f. for a summary of recent literature). For our purpose, it perhaps is sufficient to highlight that the only reason for the reconstruction of a root-final *h_3 is to accommodate Gk. ἐκόρεσα (via the supposed metathesis *ḱérh$_3$-*s*- > *κεροσ- > κορεσ-), while other evidence all points to and/or fits under *ḱerh$_1$-, e.g. Lat. *crēvī* ‘grew’ (< *ḱreh$_1$-), *Cerēs* ‘Ceres’ (< *ḱerh$_1$-*es*-), Lith. *šèrti* ‘feed’, HLuw. *zīra*/*ilamma*/*i*- ‘fruitful’, etc.[133] The reliance on the Greek data alone for the determination of the color of a laryngeal itself is not uncommon, but it becomes suspicious in the case of ἐκόρεσα, whose stem distribution (available above at 2.1) does not otherwise show signs of *ḱérh$_3$-. In particular, one notices that the perfect κεκόρημαι (firmly established in Homer, n.b. the archaic act. part. with passive meaning in σ 372) stands at odds with the expected †κέκρωμαι (if a primary, deradical perfect), cf. πέπρωμαι, μέμβλωκα, etc.[134] Rather, κεκόρημαι as well as the verbal adjective ἀκόρητος ‘unsated’ point to a base κορέω (a future in Homer). Hence, following Ruijgh 1986 (cf. also García Ramón 2010: 76f.), a better and more economical alternative to reconstructing *ḱerh$_3$- (and potentially a separate *ḱerh$_1$-) with metathesis is to start from an *o*-grade iterative-causative *ḱorh$_1$-éi̯e/o- of *ḱerh$_1$- ‘grow’, which lends itself naturally to the attested meaning in Greek via ‘cause to grow’ => ‘feed, nourish’ => ‘satiate’.[135] The absence of a present κορέω in Homer (or other early materials)[136] does pose a problem, but there are no present forms attested throughout the classical era and κορέω (along with κορέσκω) in Nicander is still the earliest attested present stem of this verb.[137]

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[133] On Luwian *zīra*/*ilamma*/*i*- see Rieken 2003, Mottausch 2011.

[134] The *CRō*- stem is both phonologically regular (< *ke-kr̥h$_3$-*C*-) in the middle forms and the stem shape generalized in the perfect for most **CeRH*- roots, cf. βέβληκα ‘throw’, τέθνηκα ‘die’, τέτληκα ‘suffer’, etc.

[135] Vine (p.c.) brought to my attention the parallelism between ἐπόθεσα : πόθος ‘yearning’, ἐκότεσα : κότος ‘grudge’, and ἐκόρεσα : κόρος ‘satiety’ (all Hom.+). This may suggest that a denominative origin (of the performative type ‘I create satiety; I perform satiety’ => ‘I satiate’ *vel sim.*) is also possible if an unattested present *κορέω is to be assumed.

[136] Whether κορέω ‘sweep, clean’ (υ 149) belongs here and represents a semantically specialized present as per Ruijgh (1986: 392) is difficult to determine. The semantic path (‘feed, satiate, nourish’ => ‘tend, keep in good condition’ => ‘clean, purify’) is not impossible, but the overall connection is weak.

[137] Note also the lack of *CRō*- forms in λοέω and κορέω makes an analogical **CeRo*- =>

2.7.3 ὤλεσα, ἐστόρεσα

Next, the *oRes*- structure in ὤλεσα 'ruin' is readily explicable as **h₃elh₁-s-*, with the initial **h₃* almost guaranteed by the comparison with Hitt. *ḫallanniya-* 'lay waste, ruin' (cf. Puhvel 1991 *s.v.*, EDHIL *s.v.*, LIV² *s.v. *h₃elh₁-*). Its corresponding present ὄλλυμι reflects *ὄλνυμι, remodeled with a secondary *-nu-* from **h₃l̥-ne-h₁-*. This leaves us with ἐστόρεσα and στόρνυμι 'spread, level', which perhaps are the only forms in this group without a compelling explanation. A root **sterh₃-* is secure in view of ἔστρωσα, στρωτός (cf. Lat. *strātus*), and 2sg. Ved. *astarīs* (AV 12.3.38) < **-sterḫ₃-s*. Therefore, the expected forms for the *s*-aorist and the nasal present should be *στεροσ- and *σταρνυ- respectively. While an analogy after ὀλεσ- : ὀλλυ- may seem gratuitous at first sight, in my opinion this still offers the most plausible account.[138] The *o*-grade is probably first introduced to the present where there could have been a systemic pressure to substitute *σταρνυ- with στορνυ- given θόρνυμι, ὀλλυμι, ὀμνυμι, and ὀρνυμι and a general lack of -αRνυ- presents (elsewhere only [?] in ἄρνυμαι 'win, acquire' and πτάρνυμαι 'sneeze'; on θαρνυ-, see 2.4.3.1).[139] The further alteration is effected via a direct proportional analogy with ὤλεσα via ὀλ-νυ- : ὀλεσ- :: στορ-νυ- : X where X is resolved as στορεσ-, resulting in the mirage of a metathesis. The formal comparison with ὤλεσα/ὄλλυμι may be supported by a probable semantic overlap between the two roots, albeit not synchronically in Greek. Kloekhorst (EDHIL *s.v.*) suggests that the Hittite cognate *ḫallanniya-* denotes "the flattening or trampling down of plants and fields"[140] which represents the original meaning of the root and that the sense of 'destroy, ruin, lay waste' is

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**CoRe-* (based on *CRa-* : *CaR-*, cf. 2.3) unlikely as well.

[138] Cf. Harðarson 1993: 222f. The general idea of the analogy is also in Francis (1970: 215-22), although his analysis which seems to be based on a putative thematic aorists *ολε/ο- (and *στορε/ο-) is to me unnecessarily complicated. See also Cowgill (1965: 158) for an assumed vowel assimilation in *στερο- > *στερε- which later became στορε- by analogy with γενε- : γνη-.

[139] The *nu*-present is likely inherited, cf. Ved. *stṛṇómi* 'spread, strew'.

[140] In Bo 3276 Vs. 6 *welku ḫaršani=šši ḫallannianzi* "They *ḫallanniya*-ed the grass on his head". Cf. the gloss 'niederstrecken' in Oettinger 1979: 80.

a secondary development. The semantic drift from ‘flatten, make level’ to ‘demolish, raze, destroy’ is natural and exactly paralleled by Latin *sternō* meaning ‘spread, lay down’ but also ‘overthrow, demolish’ etc. Hence, it may be conceivable that a semantic affinity between *h_3elh_1- and *$sterh_3$- at a time when both roots still possessed the sense ‘flatten, spread’ in pre-Greek may have facilitated the formal contact of the two verbs.[141]

[141] As food for thought, a further source of *o*-vocalism for ἐστόρεσα/στόρνυμι might be the unreduplicated perfect, directly attested in pluperf. ἔστρωτο (K 155). Although Aeol. ἐστόροται reported by grammarians (Ioann. Gr. 245) is too obscure to be used as evidence, purely from a formal point of view, the forms in the singular would be *ἔστορ-α, -ες, -ε and could provide additional support to the creation of a base στορ- in the present and aorist.

Chapter 3
The stem collocation of the type ἔλιπον-λιμπάνω-λείπω

3.1 Preliminaries

Of the various stem-distribution patterns observed with the thematic aorist, the most robustly represented in the Greek data is the one involving coordination with a nasal present. In the Homeric corpus, this pattern appears in 21 of the total 93 verbs that possess a thematic aorist (73 of which have *any* present formations attested in Homer), with an additional 14 showing nasal presents only in post-Homeric materials.

The present chapter focuses on a subtype of the THEM. AOR. + NASAL. PRES. collocation that includes the zero-grade thematic aorist (e.g. ἔλιπον), the "double-nasal" present (e.g. λιμπάνω), and the full-grade thematic root present (e.g. λείπω). The first part (3.2.1-3.2.7) offers a detailed examination of the philological data and etymological background of each verb belonging to this category. The second part (3.3) aims to present a coherent theory on the prehistory of this type and to explore its implication for the development of the thematic aorist in *CeRC*- roots.

3.2 Philological and etymological survey

THEM. AOR.	**ROOT. THEM. PRES.**	**NASAL PRES.**	**ROOT**
ἄδον 'please'	(ἥδομαι)	ἀνδάνω	**su̯eh₂d-*
κυθε 'hide'	κεύθω	(κυνθάνω)	**k/g^{h}eu̯d^{h}-*
ἔλαθον 'escape notice'	λήθω	λανθάνω	**leh₂-d^{h}-*
ἔλιπον 'leave'	λείπω	(λιμπάνω)	**leiˆk^{w}-*
ἐπυθόμην 'learn'	πεύθομαι	πυνθάνομαι	**b^{h}eud^{h}-*
ἔτυχον 'chance upon'	τεύχω	τυγχάνω	**d^{h}eug^{h}-*
ἔφυγον 'flee'	φεύγω	(φυγγάνω)	**b^{h}eug-*

Table 3.1 Homeric examples showing THEM. AOR. + ROOT THEM. PRES. + NASAL PRES.

(post-Homeric forms are in parenthesis)

3.2.1 ἄδον ‘please’

ATTESTATION[x22]: **ἄδε(ν)**[x8] M 80, N 748, ω 465, *h. Ap.* 220, 244, *h. Ven.* 18, 21, [Hes.] fr. 372.3; **εὔαδε(ν)**[x6] Ξ 340, P 647, π 28, *h. Cer.* 205, *h. Ven.* 9, *h.* 14.4[142]; **ἄδον**[x4] *h. Ap.* 22, *h. Ven.* 10, *Th.* 917, *Th.* 926; **ἄδῃ**[x1] *h. Ap.* 75; **ἄδοι**[x2] ζ 245, υ 327; **ἀδεῖν**[x1] Γ 173.

(The corpus used for the analysis below includes Homer [incl. the Homeric Hymns] and Hesiod; the rare data from Mycenaean will be cited when appropriate.)

OTHER STEMS: PRES. **ἀνδάνω** Hom.+; FUT. **ἀδήσω** Hdt. 5.39; PERF. **ἔᾱδώς** Hom.+ (poetic), **ἄδηκα** Hippon. fr. 132.1.

TRANSITIVITY AND EVENT TYPE: INTRANSITIVE STATE[143]

NOTES: the basic meaning of ἀδέ/ό- (ἀνδάνω) is ‘please, delight’, derived from the inherited sense of ‘become sweet’ of PIE **su̯eh₂d-*. Intransitive in origin, ἀδέ/ό- in all cases found in the current corpus takes an object (of the person pleased) in the dative (cf. Schwyzer and Debrunner 1949: 143f.), e.g.

> M 80 ὣς φάτο Πουλυδάμας, ἄδε δ’ Ἕκτορι μῦθος ἀπήμων
> So spoke Polydamas, and this gentle speech pleased Hector.

The augmented form εὔαδε(ν) reflects the geminate treatment of **-su̯-* > **-u̯u̯-* and the subsequent vocalization of the first glide in Aeolic (cf. Chantraine 1948: 184, Janko 1994: 16, van Beek 2021: 371 fn.42). In the present, ἀπο- (x1) and ἐπι- (x1) are occasionally prefixed. The modal forms are only found in the aorist. The verb lacks middle forms altogether.

DISCUSSION: The expected zero-grade thematic stem of the root **su̯eh₂d-* should have

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[142] Possible instance at Hes. fr. 70.3: εὔ]αδεν ἀθανάτ[οισι, ed. Merkelbach and West.

[143] The traditional “Vendlerian” (Vendler 1957) framework of event types consisting of STATES (e.g. *sleep*, *know*), ACTIVITIES (e.g. *run*, *eat*), and ACCOMPLISHMENTS (e.g. *run a mile*, *bury*), ACHIEVEMENTS (e.g. *kill, finish*) is adopted here. On the theoretical backgrounds, see Napoli 2006: 33-44, Hollenbaugh 2021: 60 with references.

lautgesetzlich yielded *υ̯δέ/ό- (< **suh₂d-é-*, cf. the zero-grade in Lith. *sū́dyti* 'season', Ved. *sūdáyati* 'sweeten'[144]) in Greek, and thus ἀδέ/ό- as such must be an analogically remodeled form under the influence of the full grade αὐδ- (cf. Dor. ᾄδομαι, αὐδύς, etc.) and/or the *a*-vocalism in the present ἀνδάνω. There is no compelling evidence to assume that such remodeling happened at an early stage and that ἀδέ/ό- bears testimony to an Indo-European (neo-)root **su̯ad-* (cf. LIV²: 606 fn.1, EWAia *s.v. SVAD*). Ved. *sváda-ti* 'make sweet', though on the surface a simple thematic present compatible with the reconstruction of a root **su̯ad-* (so Gotō 1987: 342f.),[145] has been shown credibly by Lipp (2009a: 69, 2009b: 163-7, followed by LIV² *s.v.* **su̯eh₂d-*, Kümmel 2000: 569 fn.1208, Zasada 2023: 255-7) to be rather the outcome of a thematic nasal infix present **suh₂-n̥-d-e/o-*, with the hiatus caused by the laryngeal still metrically recoverable in RV. 8.50.5 *s(u)vádanti* and 2.1.14 *s(u)vadante*.[146] This interpretation of the Vedic form and the ensuing unitary reconstruction of a single PIE **su̯eh₂d-* has the advantage of economy and is followed here.

Under this scenario, the nasal stem **suh₂-n̥-d-* is comparatively supported by ἀνδάνω, which, according to Lipp (*ibid*., see also LIV² *ibid*.), reflects the regular outcome of a syllabic nasal after a laryngeal in Greek, i.e. **suh₂-n̥-d-* > **suh₂ₐnd-* > **su̯and-* > ἀνδ-,[147] cf. the weak stem of *-nt-* participles θεντ-, σταντ-, δοντ- < **dʰh₁-n̥t-*, **sth₂-n̥t-*, **dh₃-n̥t-*, etc. The comparison with Ved. *svad-* holds in general, but the Greek reflex of **-HR̥C-* sequences in

[144] The causative formation itself may have been backformed from the weak stem of the perfect *suṣūd-*, see Jamison 1983: 99, cf. Kümmel 2000: 569f.

[145] The hypothesis by Lubotsky (1981: 133, 137) that the short *a* in Ved. *svad-* (along with another dozen of examples he provided) is the result of the regular loss of a laryngeal before an unaspirated voiced stop in Indo-Iranian offers an alternative explanation, although this sound change is not universally accepted, cf. Lipp 2009b: 161, Woodhouse 2011. Lipp's analysis of Ved. *svad-* as a nasal-infix present is supported recently by Neri (2017b: 319f.).

[146] The trisyllabic reading in RV is also noted by Gotō (1987: 340 fn.834, 835 with lit.), but he resorts to an unsatisfying solution invoking "Schwebeablaut" for the root, namely **suh₂ad-*/**su̯ah₂d-* (1987: 343 fn.845).

[147] The syllabic reduction in **su-* > **su̯-* may be understood as a type of "reverse of Lindeman's Law" (Vine p.c.) as discussed by Peters regarding σάος 'safe' < **tu̯áu̯o-* < **tuh₂-éu̯o-* (1980a: 290 fn.243), cf. Neri 2017b: 165.

these environments remains unclear and contested. On the one hand, the participial stems quoted by Lipp could contain a full-grade suffix *-*ent*- (generalized from the strong stem) with the preceding laryngeal coloring the apophonic *e rather than an anaptyctic vowel as he seems to suggest.[148] On the other hand, if forms such as ἔφῡν (< *-*b*^{h}*uh*$_{2}$-*n̥t*) and the old 3pl. optative ending -οιατο (< *-*o-ih*$_{1}$-*n̥to*[149]) are informative at all despite the possible obfuscation from paradigm levelling, one might expect *ὐ̃νδ- or *(ὐ)αδ- to be the expected outcomes of **suh*$_{2}$-*n̥-d*-.[150] The latter in particular would make plausible a scenario whereby ἀδέ/ό- (εὐαδέ/ό-) was originally the imperfect of the inherited *ἄδω (< **su̯h*$_{2}$-*n̥-d*-) and was later pushed into the perfective realm after the obscure present stem was recharacterized as ἀνδάνω following the increasingly popular model of the double-nasal type (e.g. λανθάνω, τυγχάνω, etc.; cf. 3.3.3.4).

Another contender for the starting point (or an ancient member) of this root is the full-grade thematic present **su̯éh*$_{2}$*d-e/o*- (cf. Zasada 2023: 256f., LIV^{2} *s.v.* **su̯eh*$_{2}$*d*-), primarily based on the equation between Ved. *svāda-*^{te} ‘become sweet’ and Gk. ἡδομαι ‘enjoy, take pleasure’. The evidence is tantalizing, as Ved. *svādate* is a *hapax* (RV 9.68.2) at the cadence and possibly a one-time poetic form (so Gotō 1987: 342). ἥδομαι itself is not attested until the early iambics (e.g. Semon. fr. 7.90) and Sappho (fr. 95.10),[151] but Mycenaean preserves a proper name *wa-do-me-no* (/wādomenōi/) in PY Vn 130 and the existence of ἥδομαι in the

[148] Moreover, the *e*-, *a*-, *o*- in θεντ-, σταντ-, δοντ-, etc., regardless of what the original outcome should be, could simply be analogically imported from the root vocalism. Another possibility, so Beekes (1988: 73), is that in cases such as διδοντ- (< **di-dh*$_{3}$*-nt*-) it is simply the laryngeal that is vocalized rather than the resonant, although this would not affect the analysis here since the result would be the same, i.e. **suh*$_{2}$*-n-d*- > *su̯and*- > ἀνδ-. For more on the problem of **-IH-(e)nt*-, see Peters 1980a: 192f fn.149, 2004.

[149] On the preservation/restoration of the glide, see the detailed treatment by Jasanoff (2009: 52-60**)**, cf. Rix 1992: 71, 263, Sihler 1995: 597, Willi 2018: 10f.

[150] Nikolaev (p.c.; cf. 2010: 257) points out to me that the underapplication of Rix’s Law in medial positions involving a nasal can be seen in σπάθη ‘blade’ < **sph*$_{2}$*n̥-deh*$_{2}$- (Hamp 1986) if built from σφήν ‘wedge’ (see Vine 2002).

[151] An earlier occurrence is attested in Archil. fr. 328.9, but the fragment is generally regarded as spurious, see Casadio 1996: 78-81.

Homeric language can be inferred from the sigmatic aorist ἥσατο (ι 353). The semantic misalignment between the fientive meaning of Ved. *svādate*, which is generally agreed to reflect the base meaning of the root (LIV^2 *ibid.*), and the underlying factitive semantics of ἡδομαι (cf. the act. tr. ἧσε 'delight' in Anacr. fr. 131 Page, see also DELG *s.v.*) may also need some explanation. It is conceivable that the Vedic and the Greek forms could be independent innovations based on the securely inherited PIE adjective **su̯eh$_2$d-u-*, cf. Ved. *svādú-*, Gk. ἡδύς, Lat. *suāuis*, etc.,[152] although the pattern of a thematic present pairing with a property adjective in the Caland system itself is apparently old (cf. Rau 2009a: 148-56 with more data). If one operates with the assumption that ἥδομαι is inherited, the thematic aorist ἀδέ/ό- could be created following (or using the same mechanism behind) the model of λήθω : ἔλαθον, but this seems unlikely given the diathetic and functional differences between ἀδέ/ό- and ἥδομαι.[153]

For the background of ἀδέ/ό-, it is plausible that the inherited aorist is also a radical **su̯éh$_2$d-/suh$_2$d-́* (Lipp 2009b: 166) that later underwent thematization. The intransitive fientive semantics (preserved in ἀδέ/ό-) of a root formation pairs well with the transitive factitive nasal-infix present (preserved in Ved. *sváda-^{ti}*, predominantly 'make sweet') which conforms to the category's valency-increasing function.[154]

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[152] Notice the close proximity between ἥσατο and ἡδύς in ι 353-4: ὣς ἐφάμην, ὁ δὲ δέκτο καὶ ἔκπιεν. **ἥσατο** δ' αἰνῶς **ἡδὺ** ποτὸν πίνων καὶ ... "So I spoke, and he accepted and drank it, and he enjoyed himself terribly drinking the sweet drink".

[153] A backformed active ἥδω 'please' (cf. Hsch. α 1223 ἀδῶ · ἀρέσκω 'please') may underlie the aorist ἧσε in Anacr. fr. 131.1 but it is otherwise late, cf. Schwyzer and Debrunner 1949: 228. Hsch. τ 2 τὰ ἀδάνοντα · τὰ ἀρέσκοντα 'pleasant (things)' assumes a present ἀδάνω 'please', but the antiquity of this form is uncertain, cf. EDG *s.v.* ἀνδάνω, Baunack 1911.

[154] Pronk (2019: 138) is correct in pointing out the semantic mismatch between Ved. *sváda-^{ti}* and Gk. ἀνδάνω. In the scenario proposed here, the nasal present ἀνδάνω must have acquired its intransitive semantics from the aorist ἀδέ/ό- (cf. LIV^2 607 fn.2b). The transitive factitive meaning is (later) taken up by ἡδύνω (Archil.+) 'season, make pleasant'.

3.2.2 κυθε ‘hide, cover’

ATTESTATION[x1]: **κύθε**[x1] γ 16

OTHER STEMS: PRES. **κεύθω** Hom.+, **κευθάνω** Γ 453, Hsch. κ 4584 **κυνθάνει**· κρύπτει ‘hides’; AOR. **κεκύθωσι** ζ 303, **ἐπικεύσῃς** ο 263; FUT. **κεύσω** Hom.+; PERF. **κέκευθα** Hom.+ (poetic).

TRANSITIVITY AND EVENT TYPE: **TRANSITIVE STATE**

NOTES: κύθε is a *hapax* attested in γ 16

> πατρός, ὅπου κύθε γαῖα καὶ ὅν τινα πότμον ἐπέσπεν
> (in order that you might find out about) your father,
> where the earth hid him and what fate he met.

Aside from a dubious reading in Apollonius,[155] the stem κυθέ/ό- is not otherwise attested in the post-Homeric materials.[156] A poetic word, the verb in general (pres. κεύθω, once κευθάνω in Γ 453), often prefixed with ἐπι-, denotes the action of hiding or concealing, especially in burial contexts (as in γ 16, cf. LSJ *s.v.* κεύθω). It is used mostly transitively, e.g.

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[155] The line in question is 4.59. The transmitted text reads ἦ θαμὰ δὴ καὶ σεῖο, **κύον**, δολίῃσιν ἀοιδαῖς, where the supposed vocative form κύον (“[you] bitch”, accepted as such in Vian Budé ed., Hulse 2015: 64) is variously emended to κύθον (ed. Fränkel, see also Fränkel 1968: 460) and κλύον (see Fantuzzi 2007: 91-3 with lit. and further discussion).

[156] In a 1937 excavation of the city of Kourion in Cyprus, J. F. Daniel found three *pithoi* from structures dated to the Late Cypriot III period (ca. 1200-1050 BC) that bear on the rim the same inscription consisting of two Cypriot or Cypro-Minoan signs that he identified – on sound paleographical grounds – with 𐠊 (*ku*/*gu*/*k*^{h}*u*) and 𐠴 (*te*/*de*/*t*^{h}*e*) in the classical Cypriot syllabary, and he interpreted the word as the imperative form κύθε in the sense of ‘cover (it) up!’ (see Daniel 1939). The vessels are placed underground with just the rim protruding above the floor, and a narrow piece of limestone was found alongside one of the pots that was supposedly used as a lid. As enticing as the evidence it may furnish of both the use of Greek in Late Bronze Age Cyprus and, for the purpose here, a much earlier attestation of the stem κυθε-, this interpretation remains highly speculative given how little was (and still is) known about the writing system, cf. Daniel 1938: 272-5, 1941: 258, Casson 1939: 42-4, Karageorghis and Karageorghis 1956: 355. For a recent treatment of the Cypro-Minoan corpus, see Valério 2016.

I 313 ὅς χ' ἕτερον μὲν κεύθῃ ἐνὶ φρεσίν, ἄλλο δὲ εἴπῃ "…who hides one thing in his mind but speaks forth another" and occasionally takes an accusative of the person something is hidden from, e.g. γ 18 οὐδέ σε κεύσω "I shall not keep (it) secret from you". The intransitive meaning 'lie hidden' appears once in the Homeric corpus (*h. Cer.* 452, ἔκευθε δ' ἄρα κρῖ λευκὸν "the white barley lay hidden") and a few times in tragedies, e.g., S. *OT* 967-8 ὁ δὲ θανὼν κεύθει κάτω δὴ γῆς "but he is dead and already is hidden beneath the earth". Middle forms are rare (x1 in Ψ 244).

DISCUSSION: The aorist system of κεύθω is poorly understood. One finds three different stems in the early materials, all *hapax* attested only in the Odyssey, namely the root thematic κύθε (γ 16), the reduplicated κεκύθωσι (ζ 303), and the sigmatic ἐπικεύσῃς (ο 263). Of these, ἐπικεύσῃς appears to be a clear innovation based on fut. ἐπικεύσω (x10) and imper. ἐπίκευθε (x2). All three occur only at verse-end in a fixed formula with οὐδ(ε) or μηδ(ε) occupying the fifth and sixth feet, e.g.:

Ε 816	τώ τοι προφρονέως ἐρέω ἔπος **οὐδ' ἐπικεύσω**
	So I shall speak to you confidently and hide nothing
π 168	ἤδη νῦν σῷ παιδὶ ἔπος φάο **μηδ' ἐπίκευθε**.
	Now say your word to your son and do not conceal it

ο 263	εἰπέ μοι εἰρομένῳ νημερτέα **μηδ' ἐπικεύσῃς**
	Speak truth to me as I ask you and do not hide it

In composing ο 263, it is conceivable that the bard had this inflectable formula in mind and a sigmatic aorist could be conveniently generated substituting ἐπίκευθε to match the tense of εἰπέ (Beckwith 1996: 54).

The linguistic reality of the subjunctive κεκύθωσι beyond a poetic nonce form is also doubtful.[157] Semantically, there is no discernible difference between this reduplicated form

[157] Notice that the earliest text of this part of the Odyssey transmitted in P. Oxy. XI 1395 (4th cent. CE; see Grenfell and Hunt 1915: 244) reads κεύθωσι here, cf. West 2017: 134 *app.crit.*

and the simple thematic aorist κύθε, in so far as the two isolated attestations can show,[158] and comparatively no cognate formation is identified. As with many other Homeric reduplicated aorists that are (1) not functionally distinct from their non-reduplicated counterpart (cf. λελαθέ/ό- 'cause to forget' : λαθέ/ό- 'escape notice', λαχέ/ό- 'obtain (by lot)' : λελαχέ/ό- 'grant'), (2) do not have robust (or any) comparative support (cf. εἶπον : Ved. *ávocam* 'I spoke'), and (3) are limited to Homeric language, the assumption that it is artificially created in the context of the epic *Kunstsprache* offers the simplest solution. Beckwith (1996: 56 fn.11) speculates that the original narrative with the indicative might be †δόμοι ἔκυθον καὶ αὐλή (*vel. sim.*) or that the reduplicated form could be a misreading of *κε κύθωσι with a "deictic" κε.[159] Similarly, Willi (2018: 72) suggests that with κεκύθωσι "an artificial creation is … even more likely than in the case of πεπύθοιτο".

The thematic aorist κύθε might be a relic form, although the etymological evidence is somewhat tenuous. The standard handbooks (IEW *s.v. (s)keudh-*, DELG, GEW, EDG all *s.v.* κεύθω) connect κεύθω with OE *hȳdan* 'hide' (cf. OFris. *hēda* 'id.', MoE *hide*) under a root *$keu̯d^h$-, with further support potentially coming from Ved. *kuhū́-* 'new moon' (i.e. 'the covered moon'; AV+), Arm. *suzanem* 'sink; hide' (cf. Adams 1994: 310), and Lat. *custōs* 'guardian' (see Nowicki 1978).[160] This root etymon is in principle phonologically unproblematic for the Greek and the Germanic materials, but it is plausible that the yod-present *hȳdan* is of a denominative origin (as suggested in LIV^2: *s.v.* *$keu̯d^h$- fn.1) based on OE *hȳd* 'skin, hide', OHG *hūt* 'id.', etc. The latter group, along with Gk. σκῦτος n., Lat. *cutis*

[158] The fact that κύθε in its only instance does not have an explicit object whereas κεκύθωσι does should not be overinterpreted, since both are clearly used transitively and an object (μιν or the like) can be readily supplied for κύθε in γ 16.

[159] On the deictic particle *ke*, cf. EDLIL *s.v. -ce*, DELG *s.v.* ἐκεῖ, Dunkel 2014: 396f. For a similar case made for κέκλυθι, see Schulze 1892: 391-7, Schwyzer 1939: 84, Willi 2018: 71, Kölligan 2018, cf. Melazzo 2020. For a similar analysis for Arm. *lsem* 'hear' as *$ḱl(e)u̯$ *ke*, see Vine 2017.

[160] Cf. Adams 1994 for further connection of this root with TB *tuk-* 'be hidden' and OE *dēog* 'he concealed himself' via a supposed metathesis *$keu̯d^h$- > *$d^heu̯k$-.

'skin', etc., belongs to a different root **(s)keu̯H-*[161] and is to be kept separate from κεύθω.[162] Ved. *kuhú-* is unclear but Mayrhofer (EWAia *s.v. kúha*) is doubtful on the connection with κεύθω, whereas Nowicki's analysis of Lat. *custōs* as a compound involving **sed-* 'sit', i.e. **kud^{h}-to-sd-* 'der beim Verborgenen sitzt' (1978: 191), is to me semantically far-fetched (cf. EDLIL *s.v.* for further evaluation of this etymology).

A competing theory that goes back to Schmidt (1881: 166f) followed by Szemerényi (1980) and Klingenschmitt (1982: 226) operates with **g^{h}eu̯d^{h}-,* connecting κεύθω with Ved. *gūha-^{ti}* 'hide, cover' (RV+; cf. *gúh-* 'hiding-place').[163] This etymology offers an appealing stem equation between Gk. κύθε and the Vedic thematic aorist *guhas* in RV 8.6.17:

> *támobhir indra táṃ guhaḥ*
> that one, o Indra, you hid with shades of darkness.

and, as a participle, *guhámānas* in RV 4.1.11:

> *apā́d aśīrṣā́ guhámāno ántā*
> *āyóyuvāno vr̥ṣabhásya nīḷé*
> footless, headless, concealing his two ends, continually retracting (his limbs [=flames]), in the nest of the bull.[164]

Contra Cardona's claim that it was originally a *tudáti*-type imperfect (1960: 50), *guhas* is

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[161] See EDLIL *s.v. cutis*, Kroonen 2013 *s.v. *hūdi-*. Alternatively, Vine (2016: 84-6) sets up a "long-diphthong" root **(s)keh$_3$u-*, which provides a good explanation for Myc. *ko-wo* (/kōwos/, cf. κῶας) 'sheepskin' < **keh$_3$u̯-os*.

[162] Further support for **keu̯d^{h}-* may be found in Goth. *huzd* 'treasure' (cf. MoE *hoard*), which has been analyzed as **kud^{h}-to-* containing the same root as κεύθω, see Mayrhofer 1986: 116 fn.78 with lit., Kroonen 2013 *s.v. *huzda-*. This analysis relies crucially on the application of Bartholomae's Law here (**-zd-* < **-d^{h}t-*), but there is very little evidence otherwise that it is ever operative in Germanic, see Woodhouse 1997: 386-8.

[163] The long *ū* is analogical after the participle *gūḍhá-*, possibly under the influence of *-ūha-^{ti}* (cf. *vaha-^{ti}*) 'carry' next to *ūdhá-*, see Hollifield 1977: 68, Gotō 1987: 296 fn.704, EWAia *s.v. GOH*. Jasanoff (1978: 122-5; 2002-3: 144f.) operates with **g^{h}eu̯ǵ^{h}-* and connects Ved. *guh-* with Lith. *gūžėti* 'nestles', cf. below fn.163.

[164] Translation of RV is from Jamison and Brereton 2014 unless otherwise noted.

better situated in the stem distribution of √*guh*- as a relic form, pairing with a secondary *sa*-aorist (*aghukṣat* RV 5.40.8) created out of the need for a more marked aorist next to the present stem *gūha*- (Narten 1964: 109). The pattern is also seen in *aruḳsat* : *aruhat* 'grow', where the thematic aorist is almost certainly old (cf. Hollifield 1977: 68). Another reason against an original *tudáti* present here comes from the clear evidence of an old nasal-infix present illustrated by (1) Gk. κευθάνω (Γ 453) and Hsch. κυνθάνει· κρύπτει, (2) Arm. *suzanem* (see Klingenschmitt 1982: 135, cf. Adams 1994: 310, Beckwith 1996: 54 fn.8 with lit.), and (3) a plethora of nasal formations in Middle Iranian including Sogd. *''-γwnd*- 'cover', Parth. *-gwnd*- 'id.', Khot. *uysgun*- 'reveal' (cf. Gharib 1975, Cheung 2007 *s.v.* **gauz*), etc. Thus, if the etymology should be valid, an inherited pairing of a root thematic **g*^{h}*ud*^{h}*-é*- and a nasal-infix present **g*^{h}*u-n-d*- is plausibly reconstructable based on the comparative evidence.[165]

3.2.3 ἔλαθον 'escape notice'

ATTESTATION^{x46}**: λάθον**^{x2} Ω 331, ν 270; **ἔλαθε(ν)**^{x6} Ξ 1, P 1, P 626, P 676, Y 112, Ω 477;

[165] Some further non-trivial issues with this etymology need consideration. Phonologically, although the reduction of **d*^{h} to *h* is sometimes observed in Indic from the earliest materials, e.g. *hitá*- vs. *-dhita*- 'placed', *róhati* vs. *ródhati* 'climb', 2sg.imper. *-hi* vs. *-dhi*, etc. (cf. Burrow 1955: 69, Kobyashi 2004: 84-91), the almost complete absence of dental reflex in √*guh*- is potentially problematic, with evidence given by Szemerényi (1980: 47f.) including *gudhera*- 'protective' only coming from late, lexicographical materials (cf. EWAia *s.v. GOH*). Moreover, in addition to the dental stems mentioned above (Sogd. *''-γwnd*, etc. < PIr. **gau̯d*-), Iranian also has a slew of verbal and nominal forms with a similar semantics of 'conceal' or 'cover' that formally go back to PIr.**gau̯z*-, e.g. Sogd. *'p-yw'yz*- 'conceal', Parth. *-gwz*- 'id.', etc. In a meticulous examination of the Iranian data, Gharib (1975) concludes that the two roots must be kept apart (*contra* Szemerényi [1980: 44-9], who derives the sibilant stems from the *s*-aorist). Under this scenario, then, it may be more convincing to align Ved. √*guh*- with PIr. **gau̯z*- (< IIr. **g*^{h}*au̯ǵ*^{h}-), considering that it both removes the problem of **d*^{h} > *h* and makes possible the comparison between Ved. *guhas* and YAv. *aguze* 'cover' (impf. of a thematic present according to Kellens 1984: 101, 228), the only Old Iranian form attested of this root. LIV^{2} (*s.v. keu̯d*^{h}- and ?**g*$^{(u)h}$*eu̯ǵ*^{h}-) takes a hybrid approach and connects Gk. κεύθω with Sogd. *''-γwnd*- and Parth. *-gwnd* under **keu̯d*^{h}- but assumes root contamination with **g*^{h}*eu̯ǵ*^{h} in Middle Iranian to account for the initial consonant.

λάθε(ν)[x8] Γ 420, Ο 583, Π 232, Ρ 89, Χ 277, Ψ 388, ι 281, μ 182; **λαθέτην**[x1] χ 179; **λάθῃσι**[x2] Χ 191, μ 220; **λάθοι**[x3] Κ 468, Ω 566, δ 527; **λαθών**[x7] Ι 477, Λ 251, Μ 390, Ο 541, Ω 681, ρ 305, Hes. *Op.* 50; **λάθετο**[x1] Ι 537, **ἐκλάθετο**[x1] κ 557; **λάθοντο**[x3] Ν 835, Ο 322, Π 357; **ἐξελάθοντο**[x1] Π 602; **λάθωμαι**[x3] Ζ 265, Χ 282, *h. Ap.* 1; **λάθηται**[x1] ι 102; **λάθωνται**[x1] Κ 99; **λαθοίμην**[x2] Κ 243, α 65; **λαθοίατο**[x1] κ 236; **λάθεσθε**[x1] Hes. *Op.* 264; **λαθέσθαι**[x2] Μ 235, ι 97

OTHER STEMS: PRES. **λανθάνω** Hom.+, **λήθω** Hom.+, **ληθάνω** η 221; AOR. **λέλαθον** Hom.+ (poetic), **ἐπίλησα** υ 85 (cf. ἔλᾱσα Alc. 377); FUT. **λήσω** Hom.+; PERF. **λέλαθα** (part. λελαθυῖα) Hes. fr. 343.13, **λέληθα** Semon.+, **λέλασμαι** Hom.+ (poetic), **λέλησμαι** Soph.+

TRANSITIVITY AND EVENT TYPE: In the active voice λαθέ/ό- grammatically always takes an object in the accusative, but semantically ('escape notice of', 'remain hidden from') it may be regarded as INTRANSITIVE. The event type is best described as ACHIEVEMENT (cf. Napoli 2006: 225), i.e. a non-durative telic event. Although a STATE-like actionality is often ascribed to the root (cf. LfgrE II 1628 'verborgen bleiben', LIV[2] ?**leh₂d*^{h}- 'id.'), the denotation of λαθέ/ό- in most of its occurrences hinges on an instantaneous change of *(not) being discovered* or *noticed* rather than the state or the durative action of staying hidden or acting stealthily, e.g. ν 269-70 (notice the collocation of λάθον and ἐνόησε):

> νὺξ δὲ μάλα δνοφερὴ κάτεχ' οὐρανόν, οὐδέ τις ἡμέας
> ἀνθρώπων ἐνόησε, λάθον δέ ἑ θυμὸν ἀπούρας.
> A very dark night covered heaven, and no man
> saw us, and I took his life without being noticed.

This event type can be further supported by the following two observations. First, 73% of the instances of λαθέ/ό- in the active voice in Homer (in its finite forms) are negated (cf. a slightly lower but still remarkable 57% in the present stems) and often best translated as 'A (acc.) noticed B (nom.)'.[166] Secondly, in the non-indicative categories (subj., opt., inf., part.), where the choice of aspect is prioritized, λαθέ/ό- outnumbers λήθω by a ratio of 4:1, reflecting the close affinity of an ACHIEVEMENT event with the perfective aspect, which is typologically well observed (Napoli 2006: 131).

[166] The participle λαθών (x7) is curiously never used with negation in Homer.

NOTES: λαθέ/ό- shows a robust pattern of attestation in Homer and is also the standard aorist form to this root in the later language. In the active voice ('escape notice'), the verb consistently takes an accusative of the person an action (expressed by a participle if explicit) is hidden from. The middle meaning 'forget' is ultimately reflexive in origin ('cause/let something to escape oneself') and takes a genitive of the object forgotten.

There is a discernible functional difference between λαθέ/ό- and λελαθέ/ό- whereby the reduplicated aorist (a) in the active (B 600, O 60) has a causative sense of the middle meaning (i.e. 'cause to forget') and (b) in the middle expresses an *intentional* act of forgetting ('make oneself forget', 'remove from memory'; cf. Latacz 1966: 59-61, Bendahman 1993: 64) whereas λαθέ/ό- is always unintentional (with the exception of M 235 for metrical reasons, cf. LfgrE II 1628 with lit.).[167] The causative sense of λελαθέ/ό- is also seen in ληθάνω, a *hapax* and structurally similar to κευθάνω (also a *hapax*, see above), and in the *s*-aorist ἐπέλησεν in υ 85.

The thematic aorist λαθέ/ό- shows a somewhat less agentive/animate profile in comparison to the present λήθω and λανθάνω. Semantically, λαθέ/ό- is never used in the agentive sense of '(actively) escape, elude, lose oneself from' as is sometimes the reading of λήθω, e.g. in the chase scene between Hector and Achilles in X 193:

> ὣς Ἕκτωρ οὐ λῆθε ποδώκεα Πηλεΐωνα.
> so Hector did not lose himself from swift-footed Peleion.

Further, inanimate subjects only appear with λαθέ/ό-, as in μ 182 (νηῦς 'boat'), μ 220 (νηῦς implied), and Ξ 1 (ἰαχή 'cry, shout'). Conversely, subjects in the first and second person, which are generally considered to be much higher in the animacy hierarchy, appear 8x in the present but only 1x in the aorist.

DISCUSSION: The starting point of this family of verb stems including λαθέ/ό- is the full-grade

[167] See Beckwith (1996: 99) for a tabulated synopsis of the semantics of the various stems of λανθάνω.

θ-present λήθω < *leh_2-d^h-*e/o*- (in Homer still more common than λανθάνω with no semantic difference).[168] The dental root-extension added to *leh_2- in this case is a Greek innovation (of a similar type as πλήθω 'be full' : πίμπλημι 'fill', βρίθω 'be heavy' : βριαρός 'heavy, strong', etc.), and traces of the root without the enlargement of *-d^h- can be seen in OCS -*i̯e/o*- present *lajati* 'ambush, lie in wait for' (cf. LIV^2 *leh_2-, Derksen 2008: 268), Lat. *lateō* (from part. *lh_2-*tó*-; cf. Nussbaum 2013 on verbal derivation based on -*tó*- participles), and within Greek in the Hesychian glosses λ 915 λῆτο· ἐπελάθετο 'forgot' (with *i*-extension typical of the "long diphthong" roots)[169] and α 2760 ἀλανές· ἀληθές 'true, unconcealed'.

Of its aorist formations, λαθέ/ό- is demonstrably the oldest. The evidence in support of its antiquity lies not only in its abundant attestations from the earliest materials, against the sparse occurrence of ἐπίλησα (x1) and λέλαθον (x10) in Homer, but also in the fact that λαθέ/ό- alone displays the full functional distribution between the meaning of 'escape notice of' in the active and 'forget, be oblivious of' in the middle, whereas ἐπίλησα and (the active voice of) λέλαθον both became specialized in a causative semantics of the middle use only (i.e. 'cause to forget', never 'cause to elude' *vel. sim.*). In particular, for λέλαθον, Beckwith (1999) made a convincing case that the reduplicated stem was first created in the middle as a metrical substitute for λαθέ/ό- in environments where the flexibility of the augment is not available to the bard, e.g. a non-indicative λάθοιτο following a formula ending in a long syllable.

Given the secondary nature of the dental root itself, the history of λαθέ/ό- can only be established within the Greek materials. If Hsch. λ 915 λῆτο· ἐπελάθετο is to be taken seriously, the earliest formations reconstructable are thus the root aorist *leh_2-*i*- (with *i*-extension) and the d^h-present *leh_2-d^h-*e/o*-. The former ostensibly became obsolete and/or

[168] See DELG, EDG, GEW *s.v.* λανθάνω. The primary reason for assuming λήθω as the oldest form of this family, other than the chronology of the attestation, is that -θω/-d^h- is a well-established *present* formative in Greek (and beyond). For a recent comprehensive treatment of the category, see Rothstein-Dowden 2023.

[169] Cf. Rothstein-Dowden (2023: 21f., 149-52) on the frequent coordination of *d^h- present and **i*-present.

separate from the present early on, which prompted the creation of a new aorist to what had become a standalone present *leh_2-d^h-*e/o*- (common Greek λᾱθε-). It is possible that λαθέ/ό- reflects a genuine zero-grade *$lh̥_2d^h$- following the synchronically established pattern of a full-grade thematic present next to a zero-grade thematic aorist of the type ἔσχον : ἔχω 'hold', ἔτραπον : τρέπω 'turn', etc. (cf. 3.3.2.1, 3.3.3.3, 4.3), assuming the process took place at an archaic stage when the ablaut of *Ceh_2C- : *Ch_2C- was still operative.[170] More plausibly, however, λαθέ/ό- is established following a secondary ablaut pattern, namely *ā* : *a*, observed within the history of Greek (LIV^2 *s.v.* ?*leh_2d^h-). The derivation is probably not directly based on λᾰ́θω/λήθω, since parallel models of a simple thematic present next to a thematic aorist displaying *ā* : *a* (or Att.-Ion. *ē* : *a*) ablaut are all but rare (but cf. ἥδομαι : ἄδον, see above 3.2.1). The default aorist built to λήθω next to the future λήσω should be ἔλησα (cf. πλήθω : ἔπλησα), an extremely marginal form in this verb. This suggests that other tense stems beyond the present are involved in the creation of the aorist stem λαθέ/ό-. A good candidate here is the perfect – although the active λέληθα (in place of the expected *le-loh_2d^h-)[171] does not appear until Semonides (fr. 7.9), the weak stem λελαθ-/λελασ- (< *le-$lh̥_2d^h$-; later superseded by λελησ-) has a firm footprint in the early materials: Homer offers λέλασται^{x1}, λέλασμεθα^{x1}, λελασμένος^{x5}, λελασμένον^{x1}, and the feminine participle λελαθυῖα is attested in Hes. fr. 343.13. If the early existence of a zero-grade perfect can be assumed based on this archaic profile, it may have provided the paradigmatic support for creating λαθέ/ό- via the familiar pattern available in πέπυσμαι : ἐπυθόμην 'learn', πέφυγμαι : ἔφυγον 'flee, escape', ἑαδώς : ἄδον 'please', etc., whereby λέλασμαι >> ἔλαθον.[172]

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[170] The vocalization of the laryngeal in initial *RHC- sequences follows Beekes (1988, esp. 27, 40f.), cf. μακρός 'long' < *$mh̥_2ḱrós$. See Schrijver 1991: 161 for Latin reflexes of *$RH̥C$-.

[171] For further discussion of this type of secondary ablaut in the perfect, see Kimball 1988, cf. above 2.3.1.

[172] The present λανθάνω, as is generally thought, is best explained as an innovation based on λαθέ/ό-, and not the other way round. The derivation of a double-nasal present from a *CāC-e-* present is far less motivated than from a *CaC-e-* thematic aorist. The latter is a common pattern seen in ἔχαδον : χανδάνω, ἔλαχον : λαγχάνω, ἔλαβον : λαμβάνω, etc.; see detailed discussions below in 3.3.

3.2.4 ἔλιπον ‘leave’

ATTESTATION[x156]:

- ACT. IND. **ἔλιπον**[x3] Ε 480, Ω 580, Hes. *Op.* 155; **λίπον**[x10] Β 722, Ε 204, Ι 447, Λ 759, Π 371, etc.; **κάλλιπον**[x3] Ι 364, Χ 156, *h. Cer.* 180; **λίπες**[x1] Κ 406; **κάλλιπες**[x2] Ρ 151, Φ 414; **ἔλιπε**(ν)[x3] Β 106, Ε 696, Ξ 426; **λίπε(ν)**[x36] Δ 292, 364, 470, Κ 287, Λ 99, etc.; **κάλλιπε(ν)**[x7] Ζ 223, Κ 338, Μ 92, α 243, γ 271, λ 279, φ 33.; **λιπέτην**[x3] Κ 272, Ξ 284, γ 485; **λίπομεν**[x1] δ 488;
- ACT. NON-IND. **λίπω**[x1] Ρ 91; **λίπῃ**[x2] Π 453 λ 221; **λίποι**[x3] Ε 685, η 224, σ 91; **λίποιμεν**[x1] Δ 173; **λίποιτε**[x1] Β 176; **λίποιεν**[x3] Β 160, Ρ 667, Hes. *Op.* 440; **λίπε**[x2] κ 266, ο 199; **κάλλιπε**[x1] ε 344; **λιπέτω**[x1] γ 424; **λίπετε**[x1] Κ 443, Τ 403; **λιπεῖν**[x1] Μ 111; **προλιπεῖν**[x1] ν 331; **λιπέειν**[x1] Hes. *Sc.* 332; **καλλιπέειν**[x1] π 296; **λιπών**[x21] Δ 181, Ε 20, Ζ 254, Θ 194, Ν 250, etc.; **προλιπών**[x9] Ρ 275, γ 314, ο 11, ψ 120, Hes. *Th.* 284, etc.; **ἀποπρολιπών**[x1] Hes. fr. 257.3; **ἐκπρολιπών**[x1] θ 515; **λιποῦσα**[x6] Γ 174, Π 857, Χ 363, Ω 144, σ 270, etc.; **προλιποῦσα**[x5] *h. Cer.* 388, *h. Ven.* 66, Hes. *Sc.* 1, fr. 176.3, 195.1[173]
- MID. IND. **λιπόμην**[x4] Λ 693, ι 316, ν 286, ο 481; **λίπετο**[x2] Π 16, Ω 707; **λίποντο**[x4] Μ 14, δ 495, θ 125, υ 67;
- MID. NON-IND. **λίπηται**[x2] Τ 235, δ 710; **λίπησθον**[x1] Ψ 407; **λίπησθε**[x1] Ψ 248; **λίπωνται**[x2] Τ 230, γ 354; **λιποίμην**[x2] Ι 437, φ 116; **λίποιτο**[x1] Γ 160; **λιπέσθαι**[x5] Ε 154, Ξ 485, γ 196, ρ 187, Hes. fr. 43a.154.

OTHER STEMS: PRES. **λείπω** Hom.+ (cf. KN As 1517 *re-qo-me-no* [leik^{w}omenoi]), **κατα-/ἀπυλιμπάνω** Sapph. 94.2, 5+;[174] FUT. **λείψω** Hom.+; PERF. **λέλοιπα** Hom.+, **λέλειμμαι** Hom.+; AOR.PASS. **λείφθην** *h. Mer.* 195+.

TRANSITIVITY AND EVENT TYPE: λιπέ/ό- in the active (‘leave ‘behind’, ‘abandon’, ‘give up’) is consistently TRANSITIVE and the event type is ACHIEVEMENT. The occasional use of the verb in a truly intransitive/anticausative sense of ‘be gone’ or ‘be missing’ (e.g. S. *El.* 514) is absent in the early epics except perhaps in the perfect, e.g. ξ 213 νῦν δ’ ἤδη πάντα λέλοιπεν

[173] λιποῦσα at Hes. fr. 43a.66 and 176.5 with restored preverb προ-, see Merkelbach and West 1967: 29, 85.

[174] The simplex λιμπάνω is attested in Aratus (*Phaen.* 1.128). The *varia lectio* λίνπανε (sic) in Λ 604 for ἔκμολεν of the paradosis (P. Tebt. 2.266 = Mertens-Pack^{3} 883, 2nd cent. CE, cf. Grenfell and Hunt 1907: 7, Kocharov 2019: 195 fn.168) is generally considered to be a Hellenistic verbal form that encroached into the Homeric text, see Rengakos 1993: 26. λιμπάνω (with various preverbs) appears occasionally in inscriptional materials. The earliest instances include ἐνελίμπανεν (Samos) in IG XII 6 1 261.44 and ὑπολι[μπά]νειν (Attica) in IG II² 1361.12, both ca. mid 4th cent. BC.

"and now all is gone" (cf. below in DISCUSSION).

NOTES: Similar to the case of λαθέ/ό- and λήθω (see 3.2.3), the aorist λιπέ/ό- (sometimes prefixed with κατα-/καλ-, προ-, ἐκ-, etc.) appears to show an affinity with lower animacy in comparison with λείπω. Inanimate subjects such as ψυχή (e.g. E 696 τὸν δ' ἔλιπε ψυχή "the life left him") and αἰών (e.g. η 224 ἰδόντα με καὶ λίποι αἰὼν "and may life leave me when I see…") are virtually confined to the aorist (x15 in *Il.* and *Od.*) with only one instance taking the present stem (B 396 τὸν δ' οὔ ποτε κύματα λείπει "the waves never leave him"). The non-indicative forms are predominantly aoristic with 46 instances from λιπέ/ό- and only 5 from λείπε/ο-.[175]

DISCUSSION: λιπέ/ό- goes back to the well-established PIE root *$leiḱ^{w}$-, usually glossed as 'leave behind, abandon' (cf. 'zurücklassen, sich entfernen von' LIV^{2} *s.v.*, 'lassen, zurück-, übriglassen', IEW *s.v.*). Meiser's attempt (1993: 305-10) to assign to the root an original intransitive meaning 'depart, move away from' on the strength of the Iranian data (YAv. *raekah-* 'the act of retreating' [Yt. 17.20, cf. Bartholomae 1904: 1480], MPers. *rēčitan* 'move away', MPers. *gurēxtan* 'flee', etc.) is not supported by the Greek materials.[176] According to Meiser, the oldest construction of *$leiḱ^{w}$- is with an ablative and the transitive use of the root is a secondary development originating from the nasal-infix present due to its valency-increasing function. Empirically, λιπέ/ό- (as with the present λείπω) in the corpus without exception takes the accusative (whether explicit or, rarely, supplied from the preceding clause). The only two instances of the verb used intransitively with a genitive are likely due to the effect of the preverb ἀπο- and ἐν-:

[175] The ratio of all instances of λιπέ/ό- to λείπε/ο- in the current corpus is roughly 3:1. Thus the concentration of non-indicative forms in the aorist is still significant despite the general preference for λιπέ/ό-.

[176] For a full critique of Meiser's proposal with respect to the Indo-Iranian materials, I refer to Kümmel 2000: 425f. and Zasada 2023: 202-5.

η 117-8 τάων οὔ ποτε καρπὸς ἀπόλλυται οὐδ' **ἀπολείπει**
χείματος οὐδὲ θέρευς …
Their fruit never spoils or stays away from winter or summer…

h. Ap. 212-3 ἢ ἅμα Λευκίππῳ καὶ Λευκίπποιο δάμαρτι
πεζός, ὁ δ' ἵπποισιν; οὐ μὴν Τρίοπός γ' **ἐνέλειπεν**
or with Leucippus and Leucippus' wife
you on foot and he on chariot, and he did not fall behind Triops.[177]

The intransitive use of the verb is only tangible in the perfect λέλοιπεν (e.g. ξ 213, cf. above), but even there 2 of the 4 instances (A 235 τομὴν ἐν ὄρεσσι λέλοιπεν "it has left behind the cut stump in the mountains." and β 279 οὐδέ σε πάγχυ γε μῆτις Ὀδυσσῆος προλέλοιπεν "the wisdom of Odysseus has not entirely abandoned you.") are still used transitively.[178]

Formally, the pair ἔλιπον : λιμπάνω is often considered one of the *Paradebeispiele* for the coordination of a zero-grade thematic aorist and a (double-)nasal present in Greek. The nasal stems to this root (**leiḱ*^{w}-) are abundantly attested across Indo-European including Lat.

[177] The lines 212-3 are obscure, but the context points to an allusion to a race between Apollo and Leucippus. The verb probably means 'stay behind', cf. 'hinter jmdm. zurückstehen' LfgrE II 1660. See also Allen, Halliday, and Sike 1934: 232, Richardson 2010: 115.

[178] In ξ 133-4:

τοῦ δ' ἤδη μέλλουσι κύνες ταχέες τ' οἰωνοὶ
ῥινὸν ἀπ' ὀστεόφιν ἐρύσαι, ψυχὴ δὲ λέλοιπεν
By now, dogs and swift birds of prey must have
pulled the skin off his bones, and his soul has left him.

The collocation of ὀστεόφιν and ψυχὴ λέλοιπεν recalls a similar phrase in λ 220-1:

δαμνᾷ, ἐπεί κε πρῶτα λίπῃ λεύκ' ὀστέα θυμός,
ψυχὴ δ' ἠΰτ' ὄνειρος ἀποπταμένη πεπότηται.
as soon as life leaves the white bones,
and the soul, like a dream, flies about and flies away.

If the bones (ὀστέα) are a stock noun in the epic diction that life (θυμός or ψυχή) leaves when death occurs, it is reasonable to assume λέλοιπεν in ξ 134 to have an elliptical object.

linquō ‘leave’ (Naev.+), Ved. *riṇákti* ‘id.’ (cf. Zasada 2023: 202-5), YAv. *irinaxti* ‘releases’,[179] and OIr. *-leíci* ‘let go’ (cf. McCone: 1998, Schumacher 2004: 454f.).[180] The precise mechanism through which the Greek double-nasal stems arose as a type is still under investigation (see below 3.3),[181] but the existence of a nasal-infix present **li-né-k^{w}-/li-n-k^{w}-́* in the proto-language is almost guaranteed given the strong comparative evidence, even if Gk. λιμπάνω as such must have undergone remodeling (e.g. *λιμπ-μι => λιμπάνω, cf. Kuiper 1937: 152f.; Chantraine 1961: 221f.; see below 3.3). The full-grade thematic present λείπω is a likely a Greek innovation but already firmly established since the earliest materials, being the only present of this verb in Homer along with a Mycenaean attestation in Kn As 1517 *re-qo-me-no* (/leik^{w}omenoi/, see Chadwick and Baumbach 1963: 217, Killen 2024: 1025).[182] The comparison with Goth. *leihvan* ‘lend’, OHG *līhan* ‘id.’, etc. is admissible and could in theory point to an old **léi̯k^{w}-e/o-*, but the Germanic forms are generally considered to belong to a group of thematic verbs that continue old aorist subjunctives (see Ringe 2017: 184f., Kim 2019: 98 fn.6 with lit.; cf. Bammesberger 1986: 41, Kroonen 2013 *s.v. *līhwan-*).[183]

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[179] *hapax* in Yt. 10.68, cf. Kellens 1984: 165, 1995: 58, Cheung 2007: 307.

[180] Arm. *lk‘anem* ‘leave’ (< **lik^{w}-ane/o-*) likely belongs here as well and reflects a replacement of the inherited **li-n-k^{w}-* with **lik^{w}-n̥-* (Martirosyan 2010: 310 with lit.), with the infix-less root borrowed from the corresponding aorist *elik‘* (Kocharov 2019: 195f., Jasanoff 2022a: 103f.). Whether the Armenian form (or type) underwent a stage when both the nasal infix and the nasal suffix were present (and thus matching the structure of Gk. λιμπάνω) is uncertain, cf. Clackson 1994: 84f. The Armenian nasal suffix is productive at some point, as it is used to form presents to roots with no pedigree of any nasal stems, e.g. *harc‘anem* ‘ask’ < **pr̥ḱ-* (usually **pr̥ḱ-sḱé/ó-*, cf. Ved. *pr̥ccháti* ‘ask’, Lat. *poscō* ‘id.’), but in the case of **lei̯k^{w}-* there is no reason to consider *lk‘anem* an entirely innovated form as the comparative evidence for an inherited nasal present is overwhelming.

[181] See, among others, Thurneysen 1894, Kuiper 1937: 152-4, Schwyzer 1939: 700f., Willi 2018: 18, Sturm 2021: 197-200, recently Jasanoff 2022a.

[182] The nominal stem λοιπο- is represented in *o-pi-ro-qo* (/opiloik^{w}oi/, cf. ἐπίλοιπος) ‘remaining’ and *pe-ri-ro-qo* (/periloik^{w}oi/, cf. περίλοιπος) ‘id.’

[183] Cf. Kortlandt’s criticism of the subjunctive theory (2011: 144). Bammesberger (*ibid.*) notes that formally Goth. *leihvan* etc. can also be derived from PGmc. **liŋx^{w}- < **link^{w}-* with the loss of the nasal before **x^{w}-*, albeit no other traces of the nasal present exist in Germanic, cf. Fulk 2018: 248.

Also well represented is the root aorist *$leiḱ^w$-/lik^w-. The original ablauting athematic stem can be inferred in Ved. *prá rikthās* (RV 3.6.2)[184] < *lik^w- and Lat. *līquī* < *$leiḱ^w$-.[185] The thematized version is at least seen in Arm. 3sg. aor. *e-lik'* 'left' (cf. imper. *lik'* < *lik^w-e*) and Gk. ἔλιπον. Whether the standard aorist *aricat* in Classical Sanskrit furnishes a *tertium comparationis* is a point of debate. For Cardona (1960: 7f., 64f.), given the Rigvedic attestation of the root aorist, *aricat* is automatically dismissed as a post-Vedic innovation, and thus *lik^w-e-* fails to attain a three-way cognate set to be considered inherited. Willi (2018: 289f.) argues that cases like *ámucat* (RV) and *amok* (AV) 'loose', where the thematized form predates the root formation, indicate that Cardona's assessment – treating the more recently attested amok as morphologically more archaic than *ámucat* – may be affected by an inherent bias.[186] In the case of *$leiḱ^w$-, eliminating *aricat* removes a potentially important parallel to Arm. *elik'* and Gk. ἔλιπε that would all be compatible with a single *lik^w-e-.[187] It remains

[184] Beyond this *hapax*, further evidence for the root aorist in Indic comes from the isolated 3sg. *s*-aorist *á́raik* (x5 in RV). According to Narten (1964: 223f.), *á́raik* belongs to a small group of 3sg. *s*-aorist forms built to roots with a rhyme in *-iC* or *-uC* that are paradigmatically isolated and likely reflects a "clarifying" by-form of an older **árek*, cf. 3pl. root aorist *aśvitan* vs. 3sg. *s*-aorist *aśvait* 'be bright'.

[185] For Lat. *līquī* I follow the interpretation of LIV^2 (*s.v.* *$leiḱ^w$-; no longer held in LIV$^{2\ Add.}$), Ringe (2017: 181 in passing), and Weiss (2020: 438) that it continues a full-grade root stem *$leiḱ^w$-. The older view (LEW *s.v. linquō*, IEW *s.v.* *$leik^w$-, DELG *s.v.* λείπω, still followed in EDG *s.v.* λείπω) that the Latin perfect reflects an unreduplicated *$loiḱ^w$- is inconsistent with Ved. *rireca* and Gk. λέλοιπα, and Goth. pst. *laihu* 'lend' is unprobative. Meiser (2003: 202f.) suggests that *līquī* could be a deredupulicated perfect from *le-$loiḱ^w$-, but even in that scenario one expects *$lilik^w$- > *lik^w- unless the restriction on reduplication of liquid-initial roots (see Meiser 2003: 154f.) is so old that it operated at a time when the full-grade stem was still relevant.

[186] The specific case of *ámucat* vs. *amok*, however, is more nuanced than the simple relative chronology between RV and AV. As Jamison (p.c.) brought to my attention, RV has a number of root aorist forms incl. *ámugdhvam* (5.55.6) and *mucīṣṭa* (7.59.8). Additionally, the zero-grade *s*-aorist *mukṣata* (10.87.19) also suggests that it is a remade root aorist.

[187] Rau's (2024) recent remark that Cardona assumes that "any thematic aorist that can *theoretically* be explained as secondary is secondary and should be dismissed from discussion" is a fairly accurate summary of this approach. A scenario that is central to Willi's broader argument (2018: 296-8) and one that Cardona does not take full account of in his analysis is

possible that besides **li-né-k^{w}-/li-n-k^{w}-́* and **lé̯ik^{w}-/lik^{w}-́*, both securely reconstructable, a thematized **lik^{w}-e-* could have already existed in some capacity in PIE, although ultimately the exact timing of thematization cannot be determined with certainty (cf. the releatively late emergence of λιπέ/ό- etc. below at 3.3.3.2, 3.4). The actual thematization behind ἔλιπον is used by Cardona as a prototype of the analogical process he proposed based on the active participle:

> Now, given that in the flexion of the thematic presents of the type φέρω, the third plural active indicative φέροντι (impfct. ἔφερον) and the participle φεροντ- both had the desinential vowel -o-, it is possible that, in the athematic flexion, the contrast between third plural active **lipen(t)* and active participle *lipónt-* (the former with desinential vowel *-e-*, the latter with *-o-*) was levelled on the model of the thematic flexion, so that **lipen(t)* was replaced by *lipon(t)*.

The general issue regarding the likelihood of such an analogy is discussed in the Introduction (see 1.2.2). For ἔλιπον in particular, in Cardona's favor, participial forms that go back to **lipónt-* or **lipónt-i̯a-* are extremely frequent, occurring 43x in the corpus, accounting for about half of all tokens of the active forms of λιπέ/ό- (x89). There are no concrete traces of the old root aorist in Greek. An unlikely but potentially interesting form is found in Π 507:

> Μυρμιδόνες δ' αὐτοῦ σχέθον ἵππους φυσιόωντας
> ἰεμένους φοβέεσθαι, ἐπεὶ **λίπεν** ἅρματ' ἀνάκτων.

where the Myrmidons after killing Sarpedon restrained (σχέθον) his horses when they (i.e. the horses) *λίπεν* – or *λίπον* – (from) the chariot of their masters. The reading is marred by unresolved textual issues. Most mss. as well as Aristarchus and Scholia D (ed. van Thiel) read λίπεν whereas *h* (archetype of M N P) C R and Zenodotus[188] offer λίπον (cf. ed. West

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that *some* thematized root aorists could coexist with the original athematic ones in PIE as they do sometimes in daughter languages (e.g. ἔκλυον : κλῦθι). Even **u̯id-e-*, one of the two thematic aorists that are generally agreed to be of PIE antiquity, probably had a root aorist at some point as informed by Lat. *vīdī*, cf. Meiser 2003: 115.

[188] Cf. Scholia A ὅτι Ζηνόδοτος γράφει "ἐπεὶ λίπον", ἀγνοῶν ὅτι τὸ λίπεν νῦν οὐκ ἔστιν

app. crit.), and modern authorities still disagree on the choice between the two variants.[189] The main argument in favor of λίπον against what seems to be better represented in the transmitted texts and supported by ancient scholia is that the same phrase appears early in Π 371 in a similar battle scene involving horses and chariots:

> πολλοὶ δ' ἐν τάφρῳ ἐρυσάρματες ὠκέες ἵπποι
> ἄξαντ' ἐν πρώτῳ ῥυμῷ **λίπον ἅρματ' ἀνάκτων**
> and in the ditch many fast chariot-pulling horses,
> broken short at the joining of the pole, left their master's chariots.

Π 507, the theory goes, is nothing more than a slightly inadvertent (and inaccurate) use of that formula, since here the horses, presumably still yoked to the chariot, could not have left the vehicle unlike in Π 371 where the chariot-pole is explicitly said to have snapped (see van der Valk 1964: 74f., Janko 1994: 381f., Brügger 2016: 228). But this would assume an almost inexplicable correction from λίπον to λίπεν – the putative 3pl. of a passive aorist ἐλίπην, which is never attested in Homer or elsewhere – based purely on semantic considerations by Aristarchus and other ancient sources. As van der Valk (*ibid.*) suggests, a corruption from λίπεν into λίπον is much more trivial. λίπεν, therefore, must be the authentic form. Aristarchus's own interpretation of λίπεν as a 3pl. aorist passive (as reported in the scholia; cf. Leaf 1902: 192) with the chariots being the subject, i.e. "since the chariots of their lords were abandoned", does not seem unreasonable on its own but is still problematic since the syntax of this reading diverges significantly from that of the almost identical phrase in Π 371, with which it clearly forms a shared formulaic expression.[190] A very tentative third possibility is that one is dealing with a real active aorist λίπεν, with what is essentially the

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ἑνικόν, ἀλλὰ ἀνάλογον τῷ ἐλείφθησαν τὰ ἅρματα, ὡσπερ "κόσμηθεν" (Γ 1) καὶ ... See Schol. A ad Π 507 (ed. Erbse).

[189] For λίπεν: Leaf 1902: 192, van Thiel 2010: 317, cf. LIV^2 *s.v.* **leiḱ^w*-; for λίπον: Allen (1931: 119), West *ibid.*

[190] Formally a "passive" aorist λιπη- with an archaic anticausative semantics 'be free, get loosened' (cf. καη- 'burn', ἀγη- 'break') is not implausible, cf. Romagno 2014: 179. Chantraine (1948: 401) is agnostic on the passive reading of λίπεν.

athematic 3pl. ending -εν (cf. opt. -οι-εν), continuing an old **lik^{w}-ént* before the substitution of **-ent* by *-ont* took place envisaged by Meillet and Vendryes (1948: 204f.), Cardona (1960: 66-9), and others (see 1.2.1, 1.2.2). Under this hypothesis, λίπεν would be comparable to the Pindaric anomaly δρακείς (< **dr̥ḱ-ent-*; cf. Forssman 1964) in *P*. 2.20 and *N*. 7.3. next to the expected thematic δρακών (E. *Her.* 951, etc.) as a pre-thematization archaism.[191] The isolated preservation of λίπεν in Π 507 may precisely be due to the semantic inadvertency conjectured by the modern commentators. In other words, if at some point in the early stage of the epic oral traditions the old athematic λίπεν was replaced by λίπον *en masse*, Π 507 may have escaped that process because the semantic ambiguity of the verse created some confusion for the bards and allowed for a mis-/reanalysis of the form as an aorist in -η. The hypothesis advanced here has the advantage of (1) respecting the authority of λίπεν which is no doubt the *lectio difficilior* in this case, (2) maintaining formulaic parallelism with Π 371, and (3) identifying λίπεν as a form predicted in the diachrony of the category. That said, at least two objections may be made against treating λίπεν as anything archaic. First, the antiquity of Pi. δρακείς is questionable (see fn.189) and the comparison therewith may not hold much value. Secondly, ἅρματ' ἀνάκτων shows the absence of the effect of the digamma, which is relatively rare for ἄναξ (34 out of 304 instances according to LfgrE I 782) and could suggest that this formula specifically is relatively young.

Overall, the Greek evidence points to a system in which the nasal-infix present was inherited and preserved in a remodeled form, while the full-grade thematic present λείπω represents an early Greek innovation that quickly became dominant. The aorist appears to have undergone early thematization, with ἔλιπον replacing any older root-aorist reflexes; nevertheless, the problematic Homeric λίπεν of Π 507 may preserve an archaism that briefly reveals the pre-thematized 3pl. before analogy took full effect.

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[191] Caution must be taken as this form has been dismissed as an archaic innovation with the vocalism borrowed from athem. 3pl. *-ent(i)* replacing the original form in -οντ- as preserved in the nominal δράκων 'snake', see Peters 2004, Barnes 2013, more recently van Beek 2021: 375-7, cf. also Morpurgo Davies 1978 on Thess. ἐντ-, Harđarson 1993: 44f.

3.2.5 ἐπυθόμην ‘hear, learn’

ATTESTATION[x50]: **πυθόμην**[x5] δ 732, ξ 321, ψ 40, *h. Mer.* 263, 363; **πυθέσθην**[x1] P 427; **ἐπύθοντο**[x4] E 702, O 224, 379, Ξ 530; **πύθηαι**[x5] E 351, P 685, Ω 592, γ 15, δ 494; **πύθηται**[x3] T 337, δ 713, τ 486; **πύθησθε**[x1] θ 12; **πύθωνται**[x1] H 195; **πυθοίμην**[x4] P 102, T 322, β 43, κ 147; **πύθοιτο**[x4] K 207, 211, β 31, *h. Mer.* 269; **πυθοίατο**[x1] A 257; **πυθέσθαι**[x21] B 119, Z 465, K 308, 320, 395, etc.

OTHER STEMS: PRES. **πεύθομαι** Hom.+, **πυνθάνομαι** Hom.+; AOR. **πεπυθόμην** 50, K 381, Λ 135 (poetic), FUT. **πεύσομαι** Hom.+; PERF. **πέπυσμαι** Hom.+.

TRANSITIVITY AND EVENT TYPE: πυθέ/ό- is predominantly TRANSITIVE, taking an object in the accusative when acoustic perception is expressed (e.g. O 379 ὡς ἐπύθοντο Διὸς κτύπον “as they heard the thunderstroke of Zeus”) or genitive in the sense of ‘learn that…’ (e.g. P 427 πυθέσθην ἡνιόχοιο κονίῃσι πεσόντος “they learned that their charioteer had fallen in the dust”). The event type is ACHIEVEMENT.

DISCUSSION: πυθέ/ό- is another potentially important form (or represents an important type) in our quest for patterns of the early spread of the thematic aorist. The comparison of ἐπύθοντο with Ved. 3pl. inj. *budhánta* ‘woke up’ (otherwise athematically inflected in the aorist, cf. *abudhran*, part. *budhāna-*, etc.) has led to much suspicion that *b^hud^h-*é*/*ó*- (or at least the 3pl.) is another inherited thematic aorist (cf. Burrow 1955: 334, Strunk 1967: 98, Willi 2018: 64). In particular, Grestenberger 2019 (cf. Grestenberger 2025) frames this problem under the *h_2e-conjugation / “proto-middle” theory and proposes that *-*ónto* (whatever its origin) replaced *-*r(ó)* in PIE in the 3pl. middle of older “stative-intransitive” *h_2e-aorists such as *budhánta*/ἐπύθοντο, (*á*)*juṣanta* ‘were pleased’, *mṛṣanta* ‘neglected’, and a few other less secure cases. Without getting into the details of her claims related to the *h_2e-conjugation, the supposed PIE/ “Graeco-Aryan” mid. 3pl. *-*onto* to me does not gain us much. Firstly, the only convincing pair is ἐπύθοντο and *budhánta,* the other cases all lacking Greek counterparts. But even for the one good example, the significance of this coordination is unclear because for all we know the 3pl. -οντο in Greek from its earliest attestation is part of

a regular thematic paradigm (-ομην, -εσθε, -οντο, etc.). Hence, there is nothing extraordinary about the Greek form (as there would be, for instance, if πυθ- otherwise inflects athematically in the aorist like *budh*- or if certain isolated 3pl. -οντο forms are attested), as ἐπύθοντο is exactly what one would expect in a productive paradigm. Therefore, the "odd" one, so to speak, in the pair is *budhánta* – together perhaps with *mṛṣanta* and (*á*)*juṣanta* – which stands out because of its seemingly thematic form in an otherwise athematic aorist paradigm, essentially replacing/coordinating with *abudh-ran*. The relation between IIr. mid-pass. *-*ran/-ram/-anta/-ata* is a complicated problem in its own (especially on the relative chronology between Ved. -*anta* and -*ata*, see Narten 1964: 24-5, Watkins 1969: 37, Gotō 2013: 90, 109, 119), but whatever one's story on the chronology or the distribution (e.g. ind. -*ata* vs. inj. -*anta*, per Watkins *ibid.*), the paradigmatic pressure to get rid of the ostensibly older *r*-ending (as in Grestenberger's *h_2e-theory framework, but also see Rix 1992: 249) – or, failing that, create a more regular looking alternative – within Indo-Iranian must be significant if one starts with a mid-pass. paradigm that looked like:

	Primary	Secondary
3sg.	*-*tai*	*-*ta*
1pl.	*-*mad^{h}ai*	*-*mad^{h}a*
2pl.	*-*d^{h}vai*	*-*d^{h}vam*
3pl.	*-*ntai*	***-ran/ram***

Table 3.2 Indo-Iranian mid.-pass. paradigm

Hence, projecting Ved. -*anta* to an inherited *-*onto* appears to have no advantage over the classic treatment since Brugmann 1904: 596 (followed by Cardona 1960: 27) that assigns the origin of athematic -*anta* to an analogy with thematic stems such as *abhavan* : *abhavanta* (:: *akran* : X where X is resolved as *akranta*). Once this ending is accepted as an alternative to -*ran*, it could easily have spread in Vedic to other athematic aorist stems including the *media tantum* or the semantically "proto-middle" ones that Grestenberger discussed. Narten's (*ibid.*) view that the -*ata* (< *-*n̥ta*) reflects the older, phonetically regular shape of the ending poses

no difficulty here as *-ata* could also be an analogical creation based on the model set in the singular or simply inherited (< *-*n̥to*). Secondly, even if PIE (or "Graeco-Aryan") does have an athem. 3pl. mid-pass. ending in the shape of *-*Vnto*, one would expect it to be *-*ento* (cf. Gotō 2013: 89), as in, e.g., ἐπρίαντο (with coloring from *$k^{w}rih_{2}$-*énto*), thus structurally conforming with the athem. act. 3pl. ending *-*ent*. In other words, although operating with *-*onto* for *budhánta*/ἐπύθοντο potentially makes it easier to explain the Greek vocalism, (1) it is not required as Greek -οντο is simply in a productive paradigm and never appears in a paradigmatically unexpected position and (2) establishing *-*onto* for the ancestor of Greek and Indo-Iranian still leaves the substitution of *-*ento* with *-*onto* unexplained. Lastly, and more relevant to our purpose, even under the assumption that we do have *-*onto* in the ancestor of Greek and Indic, it is unlikely that this is the starting point of the whole thematic aorist category, since *-*onto* being a *plural* and *medio-passive* ending should not have much analogical influence (as is typologically supported, cf. Albright 2008: 145f.). Indeed, the fact that the athem. 3pl. mid-pass. ending itself in Indo-Iranian has several variations suggests that it is a slot that often has analogical patterns elsewhere imposed on itself and not the other way round. Thus, it is the view here that the equation between *budhánta* and ἐπύθοντο alone does not prove the inherited status of this thematic aorist nor is the evidence sufficient to support the proposed significance of the 3pl. mid-pass. in the genesis of the category.

The nasal present πυνθάνομαι is echoed by OIr. *ad-bond-* 'announce' and Lith. *at-buñda* 'wake up'. While Beekes (EDG *s.v.*) and LIV^{2} (*s.v.* *$b^{h}eu̯d^{h}$-) quite reasonably flag them as possible innovations given the productivity of these formations, independent attestation in Greek, Celtic, and Balto-Slavic should suggest a strong likelihood of a nasal-infix present to this root in PIE.[192] As will be discussed in 3.9, the Greek double-nasal formation itself is

[192] It is worth noting that Gorbachov (2007 *passim*, esp. 24, 156-8, 165f., 203) considers Lith. *at-buñda* 'wake up' and OCS *-bŭ[d]netŭ* 'id.' to reflect one of the inherited *thematic* nasal inchoative formations (*b^{h}und-é/ó-), which are purported to be originally an *$h_{2}e$-conjugation category (215-9). Since this type of intransitive inchoative nasal present is limited to Germanic and Balto-Slavic (cf. Rau 2024: 2), the Celtic and Greek materials still point to a standard athematic nasal present *b^{h}u-n(e)-d^{h}-.

likely of a secondary origin, remade from a nasal infix present *$b^hun(e)d^h$-.[193] The simple thematic πευθόμαι could be equated with Goth. *-biudan* 'order', OCS *bljudǫ* 'pay attention, observe', and Ved. *bódhati*. Note, however, that compared with πεύθομαι, the Vedic forms is an *activum tantum* meaning 'perceive, understand' and later largely superseded by the *medium tantum búdhya-* 'wake up'. Despite the comparative parallels, many authorities have plausibly suggested that the thematic present in Vedic is rather to be taken as aorist subjunctive (see Hoffmann 1967: 231f, Hollifield 1977: 179, cf. Gotō 1987: 217 fn.451, Beckwith 1996: 49f.), and that πεύθομαι and Goth. *-biudan* are also likely to be independently innovated (Watkins 1987: 310f.).

3.2.6 ἔτυχον 'meet, chance upon'

ATTESTATION^{x22}: **ἔτυχες**^{x1} E 287; **τύχε**^{x2} E 587, Λ 684; **τύχωμι**^{x3} E 279, H 243, χ 7; **τύχῃς**^{x1} ζ 290; **τύχῃ**^{x2} Θ 430, Ψ 857; **τύχῃσι**^{x2} Λ 116,[194] *h. Mer.* 566; **τύχοις**^{x1} μ 106; **τυχών**^{x10} E 98, 582, 858, N 371, 397, etc.

OTHER STEMS: PRES. **τυγχάνω** Hom.+; AOR. **τύχησα** only Hom.; FUT. **τεύξομαι** Hom.+; PERF. **τετύχηκα** Hom.+; **τέτευχα** (Hdt. 3.14)

DISCUSSION: The thematic aorist τυχέ/ό- stands as the most established stem of this verb in the early epics, accounting for the majority of the attestations (22 out of 36) as well as exhibiting the more archaic-looking usage of the root in the sense of *hitting* (a target), e.g. Ψ 857 ὃς δέ κε μηρίνθοιο τύχῃ ὄρνιθος ἁμαρτών "he who hits the string but misses the bird"

[193] Recently Nikolaev (2023: 865) notes that πυνθάνομαι fails to exhibit the effect of Miller's Law whereby a voiced aspirated stop would be deaspirated after a nasal (863 fn.22 with lit.), thus not yielding the expected *φυνδ- (without Grassmann's Law affecting the initial *b^h-). But this may be due to the fact that, pending the relative chronology of the accent retraction in Greek, the original sequence *-nd^h- may never have been in a position where it followed an accented vowel (cf. *b^hu-*né*-d^h : b^hu-*n*-d^h-́), which is a condition of Miller's Law (863).

[194] The line reads ἢ δ' εἴ πέρ τε τύχῃσι μάλα σχεδόν, οὐ δύναταί σφι / χραισμεῖν… "and even if (a doe) happens to be near, she is not able to be useful to them…" Aristarchus acknowledges both the reading τύχῃσι and τετύχῃσι, cf. Erbse 1974: 148, Beckwith 1996: 59 fn.16. The reduplicated aorist is otherwise a late and artificial form.

and notably in the formula αἴ κε τύχωμι "if I might hit…" (E 279, H 243, χ 7). In Homer, the present τυγχάνω is still exceedingly rare (1x τύγχανε ξ 231, 1x παρετύγχανε Λ 74) and entirely lacks comparative evidence of any nasal present elsewhere, strongly suggesting it is a relatively late creation (based on τυχέ/ό- with ample models of the type to follow, cf. 3.3.4). The thematic aorist is accompanied by a less frequent *s*-aorist ἐτύχησα. The form is confined to the epic language (7x in *Il.* and *Od*., Hes. fr. 35.8) and often considered an artificial form along with other isolated ησ-aorist.[195] According to Risch (1974: 300, 308), the coordination of ησ-aorist and thematic aorist – or rather the creation of the former – seen also in ἐπίθησα : ἐπιθόμην 'convince; trust', is effected via the reanalysis of the inf. -εῖν, perhaps through an analogy φιλεῖν : ἐφίλησα :: τυχεῖν : X where X is resolved as ἐτύχησα. While plausible in theory, this scenario presupposes forms that either do not exist (†πιθεῖν, which is especially fatal to this explanation being applied to ἐπίθησα, cf. Peters 2007: 263) or, in the case at hand, are not attested in the early materials (τυχεῖν first appearing in the Theognidea). Metrically, all 5 instances of the participial form τυχήσας (out of the 8 occurrences overall of this stem) appear at verse-end:

Δ 106	ἀγρίου, ὅν ῥά ποτ' αὐτὸς ὑπὸ στέρνοιο **τυχήσας**
E 579	ἑσταότ' ἔγχεϊ νύξε κατὰ κληῖδα **τυχήσας**
M 189	Ἱππόμαχον βάλε δουρὶ κατὰ ζωστῆρα **τυχήσας**.
M 394	ἀλλ' ὅ γε Θεστορίδην Ἀλκμάονα δουρὶ **τυχήσας**
Φ 13	δῶρα τά οἱ ξεῖνος Λακεδαίμονι δῶκε **τυχήσας**

Thus, it is rather likely that the ησ-aorist may have started as a metrical substitution of τυχών in the cadence (so briefly Chantraine 1948: 415f),[196] which then prompted the backformation of the (only) finite form ἐτύχησε(ν), whose usage exhibits the more familiar constructions common in the later language such as 'succeed in…' (Ψ 466 καὶ οὐκ ἐτύχησεν

[195] There are roughly 20 such ησ-aorists that are not based on an έω-present or otherwise paradigmatically supported, cf. van de Laar 2000: 411 fn.15.

[196] Note ἐπίθησα is also almost exclusively attested in its participial form πιθήσας at verse-end.

ἑλίξας "…and he did not succeed in turning [during a chariot chase]") and 'happen to…' (Hes. fr. 35.7-8 …ἱππότα Νέστωρ | ξεῖνος ἐὼν ἐτύχησε… "…the horseman Nestor happened to be a guest friend…"). Additionally, as Meister (1921: 119 fn.2) comments in passing, ἐτύχησα (as well as perf. τετύχηκα) never takes an object either in the accusative or in the genitive. Such transitive use of the verb is limited to the thematic aorist, as seen above in Ψ 857 with gen. μηρίνθοιο and further with an accusative object in, e.g., E 581-2 ὃ δ' ὑπέστρεφε μώνυχας ἵππους | χερμαδίῳ ἀγκῶνα τυχὼν μέσον "he turned the single-foot horses and hit (him) with a stone right upon the elbow".[197] Thus, as far as the system of τυγχάνω is concerned, the only form with a secure claim of antiquity within Greek is the thematic aorist τυχέ/ό-, with the stem τυχη- being a possible metrical accommodation (but cf. fn.197) and τυγχάνω a late innovation.

Cross-linguistically, one finds a formal word equation between Gk. ἔτυχε and Vedic class VI imperfect *aduhat* in RV X 61.19:

> *dvijā́ áha prathamajā́ ṛtásyedáṃ dhenúr aduhaj jā́yamānā*
> having two births, first born of truth. This (earth?) here did the cow give as her milk as she was being born.

[197] Given the functional difference, there is the alternative possibility that τύχησα may be a genuine form, based on an old stem *τυχη- (listed as 'fientiv' in LIV^2 *s.v.* *$d^heu̯g^h$- fn.6), which can be variously explained as denominative (from τύχη 'coincidence, luck'), deadjectival (from εὐτυχής, δυστυχής, etc., so Tucker 1990: 68f.), or deverbal (from a stative stem *d^hug^h-eh_1-, ultimately a denominative type according to Jasanoff [1978: 124f, 2003], cf. Willi 2018: 275f). The deverbal explanation, in particular, may find distant support in the general 'stative'/middle inclination of the root *$d^heu̯g^h$- as evidenced by the archaic Ved. 3sg. *duhé* (for *-té*) and 3pl. *duhré* (for *duhaté*) 'give milk' (cf. Kümmel 1996: 53-64), although the semantic gap in this case is considerable (cf. below on the semantics) and the paradigmatic and functional connection (if any) between the 'stative' suffix *-eh_1- (see, among others, Watkins 1971, Jasanoff 2002-3, Yakubovich 2014b) and the purported diathesis 'stative' (see, e.g., Oettinger 1976, Rix 1988, cf. Harðarson 1988) is still unclear. A further trace of τύχησα being potentially more than a poetic creation lies in the perfect τετύχηκα (Hom. but also standard in Attic prose), where the otherwise elusive insertion of -η- could reflect the influence of the ησ-aorist (cf. Nikolaev 2010: 83 fn.106 with lit.).

A direct comparison, however, remains difficult, primarily because *aduhat* is almost certainly the middle 3sg. *aduha* (cf. Wackernagel 1907: 309f.) with a secondary, 'clarifying' addition of the *mi*-conjugation *-t*, based both on semantic considerations (here 'gave milk' rather than 'milked') and the fact that the archaic, unextended middle 3sg. *aduha* is still attested in MS (III 3.4: 36.4, etc.). Therefore, a connection between ἔτυχε and *aduhat* – if any – can only be sought at a depth of reconstruction where Jasanoff's proposal (2017) that the thematic conjugation started off from the 3sg. of the **h₂e*-conjugation (e.g. *u̯idé* => *u̯idé-t*) is relevant and operative (cf. above 1.2.4).

The synchronically separate τεύχω 'make, produce; prepare' (aor. τεῦξα, τέτυκον, fut. τεύξω, all Hom.+) can on formal and etymological grounds be related to τυχέ/ό-, in a similar way as λείπω to λιπέ/ό-. The separate semantic development can already be seen in Myc. *te-tu-ko-wo-a* (KN L 871; perf. part, cf. Hom. τετευχώς), a word describing clothes (and once chariot in PY Sa 682, spelled with a final *a₂*) and usually glossed as 'fully worked, finished' (Killen 2024: 1038, cf. DMic *s.v.*). The thematic present itself is unlikely to be the original formation in PIE, in view of the strong presence of a root present in Vedic (3sg. *ádhok*, 3pl. *duhaté*, part. *duhāná-*, etc.). Kuiper (1937: 156) postulates an origin in the subjunctive for τεύχω, similar to λείπω/Goth. *leihvan* discussed above (3.2.4), although this remains speculative. The semantic difference between τυγχάνω and τεύχω is usually ascribed partially to a prehistoric root contamination with **teu̯k-* 'hammer, strike' (=> 'forge, make'), which arguably still underlies the reduplicated aorist τέτυκον (LIV^2 *s.v.* 1.**teu̯k-*, cf. Willi 2018: 64f.), and partially to the factitive function of the active thematic root present itself ("faktitives Aktiv" LIV^2 *s.v.* **dʰeu̯gʰ-* fn.3). The factitivity opposition between τυχ- and τευχ- can be made apparent if one considers the Germanic reflexes of the root as in Goth. *dugan* 'be useful, of value; be available' to preserve the old intransitive meaning which is still relatable to Gk. τυχ- in its sense of 'succeed, hit upon; happen to be', whence a factitive meaning 'make useful, make ready; put together' => 'make, forge, arrange' can easily

emerge.[198]

In light of the fact that (1) the nasal present τυγχάνω is clearly of a secondary nature, whereas τεύχω is more firmly established since the earliest materials and (2) τύχον and τεύχω form a somewhat contrastive pair of intransitive THEM. AOR. ('succeed, happen to be') and transitive ROOT THEM. PRES. ('make, forge, prepare'), this puts τυχέ/ό- in a pattern that is shared with several other thematic aorists that are used intransitively (in the active) in verbs that are otherwise active, e.g. τράφον 'grow up' – the active voice of which in Homer is always used in the middle sense – next to the factitive τρέφω 'nurture, thicken', and τράπον 'turn' which is usually transitive but is used intransitively in κ 469 and Π 657, next to τρέπω 'turn (tr.)'. In this way, τύχον and τεύχω could furnish another example of this potentially archaic pattern that has been argued to ultimately reflect a connection between the thematic inflection and the perfect and mediopassive endings (cf. Watkins 1969: 100f., Jasanoff 2017, Rau 2024; cf. 1.2.4, 4.3).

3.2.7 ἔφυγον 'flee'

ATTESTATION^{x119}:

- ACT. IND. **φύγον**^{x8} Γ 4, Θ 137, Λ 128, Ψ 465, α 11, δ 192, ζ 170, Hes. fr. 302.18; **ὑπέκφυγον**^{x2} ι 286, μ 446; **ὑπεξέφυγον**^{x1} Λ 383; **ἐξέφυγον**^{x1} ψ 236; **ἔφυγες**^{x2} Λ 362, Υ 449; **ὑπέκφυγες**^{x1} Υ 191; **φύγε(ν)**^{x15} Δ 350, Ξ 83, Σ 117, Υ 350, Φ 493, etc.; **ἔκφυγε(ν)**^{x11} Ε 18, Ι 355, Λ 376, 380, Ξ 407, etc.; **ἐξέφυγε(ν)**^{x2} Hes. fr. 35.9, 76.22; **ὑπέκφυγε(ν)**^{x3} Ε 22, Π 687, Ψ 320; **ὑπεξέφυγε(ν)**^{x2} Θ 369, Χ 202; **φύγομεν**^{x3} ι 61, 467, μ 260; **ἐκφύγομεν**^{x1} μ 212
- ACT. NON-IND. **φύγῃσι(ν)**^{x5} Ε 258, Η 118, 173, Τ 72, Φ 296; **παρεκπροφύγῃσιν**^{x1} Ψ 314; **φύγῃ**^{x4} Σ 271, Φ 103, Χ 487, ξ 184; **προφύγῃ**^{x1} Ξ 81; **ὑπεκπροφύγῃ**^{x2} Hes. *Sc.* 42, fr. 195.42; **φύγωμεν**^{x1} Ρ 714; **φύγοιμι**^{x1} Ξ 312; **ὑπεκπροφύγοιμι**^{x2} μ 113, υ 43; **φύγοις**^{x1} Hes. *Op.* 684; **φύγοι**^{x7} Ζ 59, Ξ 507, Π 98, 283, δ 789, ο 300, χ 43; **ὑπεκφύγοι**^{x4} Ζ 57, μ 287, π 372, υ 368; **φύγοιμεν**^{x5} Α 60, γ 175, ι 489, κ 129, μ 157; **φυγέειν**^{x5} Β 393, Ν 436, Ξ 80, δ 504, μ 120; **ἐκφυγέειν**^{x6} Φ 66, ε 289, 414, τ 157, 231, Hes. fr. 150.29; **ὑπεκφυγέειν**^{x2} Θ 243, μ 216; **φυγεῖν**^{x3} Β 401, Μ 327, Hes. fr. 76.8; **προφυγεῖν**^{x2} Λ 340, Hes. *Op.* 647; **ὑπεκπροφυγών**^{x2} Υ 147, Φ 44; **φυγών**^{x7} Μ 322,

[198] Jasanoff (1978: 74 fn.30) speaks of a two-way "reciprocal" sense, i.e. 'produce' as well as 'receive (as product)', inherent to the root *$d^heu̯g^h$-.

Φ 57, ι 17, ο 277, π 21, ρ 47, ψ 238; **προφυγών**[x4] Z 502, H 309, λ 107 *h. Mar.* 17

- MID. NON-IND. **προφύγοισθα**[x1] χ 325

OTHER STEMS: PRES. **φεύγω** Hom.+, **φυγγάνω** (A.+; in prose with ἀπο-, δια-, ἐκ-, κατα-, etc.); FUT. **φεύξομαι** Hom.+, **φευξοῦμαι** E.+; PERF. **πέφευγα** Hom.+ (Φ 609, α 12), **πέφυγμαι** Hom. (only part.)[199]

TRANSITIVITY AND EVENT TYPE: φυγέ/ό- is mostly TRANSITIVE, often taking up an accusative object denoting death or disaster (cf. fn.207; e.g. Y 449 ἐξ αὖ νῦν ἔφυγες θάνατον "now you escaped death once more", α 11 ὅσοι φύγον αἰπὺν ὄλεθρον "who escaped sheer destruction"), rarely with an object of the person escaped from (e.g. Θ 137 Νέστορα δ' ἐκ χειρῶν φύγον ἡνία σιγαλόεντα "the glittering reins escaped Nestor from his hands") or used intransitively (e.g. Φ 496 ὣς ἣ δακρυόεσσα φύγεν "as she fled weeping"). The verb is virtually *activum tantum* with one isolated middle form attested (προφύγοισθα in χ 325; possibly metrically motivated in replacement of †προφύγοις at verse-end). The event type is ACHIEVEMENT.

DISCUSSION: The PIE root underlying φυγέ/ό-, namely *$b^heu̯g$- 'flee, be released' (LIV^2 *s.v.* 1.*$b^heu̯g$- 'enfliehen, freikommen'), shows a bewildering pattern of homonymy and possible semantic connection with *$b^heu̯g$- 'benefit, be useful' (cf. Ved. *bhunákti* 'id', Lat. *fungor* 'enjoy, act', etc.) and *$b^heu̯g^{(h)}$- 'to bend' (cf. Ved. *paribhujáṭ* 'wraps around', Goth. *biugan* 'bend'; LIV^2 *s.vv.*).[200] The only comparison that is relatively secure both formally and semantically is that between Gk. φυγέ/ό- and Lat. *fūgī* 'escaped, fled'. The latter, as Meiser noted (2003: 201) following earlier observations by Watkins (1969: 149) and Cardona (1960:

[199] Also found in Homer is πεφυζότες (Φ 6, 528, 532, X 1), a perfect stem of uncertain formation. Schwyzer (1939: 771) suggests an underlying present *φύζω in parallel with Lat. *fugiō*, whereas Hackstein (2002: 194-6) considers πεφυζότες a metrical substitution of *πεφυζηότες, the latter supposedly a denominative from the noun φύζα 'flight', cf. also Coray and Krieter-Spiro 2021: 18.

[200] The root *$b^heu̯g$- 'be useful', 'make use of' might have traces of it preserved in ἀφυσγετός 'drift wood; debris' (Λ 495), if the etymology suggested by Nikolaev (2014: 710) deriving the word from *$n̥$-b^hug-$sḱe/o$- with a meaning 'unusable' => 'useless stuff, garbage' holds.

108), continues a root aorist PIt. **φeu̯g-*.[201] The alternation between the full-grade root in *fūgī* (< **b*^{h}*eu̯g-*) and the zero-grade in φυγέ/ό- (< **b*^{h}*ug-*) points to an original ablauting root paradigm.

Whether thematization in this case already took place in PIE is difficult to assess. Rix (1992: 215), in fact, tentatively lists φυγέ/ό- – next to (Ϝ)ιδε- and ἐλυθε- – as one of the inherited thematic aorists based on the equation with OIr. *-buich*, *-buig*, the unreduplicated preterite to the nasal present *-bongaid* 'break'. The idea can be traced back to Thurneysen (1917: 65f.), who suggests that *-buich* is an "altertümliche oder altertümelnde Nebenform" of the reduplicated *-bobaig*, explicitly comparing its archaic status to *luid* 'went' along with Hom. ἤλυθε.[202] The antiquity of the preterite form *-buich*, however, is debatable, not only due to the lateness and rarity of its attestation in the manuscripts (see Schumacher 2004: 239f. for details), but also because the preservation of both inherited aorist and perfect formations is highly unusual in Old Irish. In general, the two categories converged into the suffixless preterite with the old aorist surviving in weak verbs and the old perfect regularly appearing with strong verbs (McCone 1991: 126, 131, Charles-Edwards 2021: 271f.). Semantically, the gap between OIr. *-buich* 'break, strike' and Gk. φυγέ/ό- 'flee' is significant, the former arguably more relatable to the root **b*^{h}*eu̯g*^{h}- 'bend' instead of **b*^{h}*eu̯g-* 'flee'.[203] Thus, following McCone (*ibid*.) and Schumacher (*ibid.*), the Irish form may find a better explanation as an innovatory *s*-preterite developed within the language, although the case remains unclear.

Two formal equations of φυγέ/ό- are found in Vedic. First, the root *bhuj-* 'bend' produces

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[201] The existence of a root aorist in the case of *fūgī* may be further supported by the **-i̯e/o-* present *fugiō*, which according to Meiser (2003: 196 fn.6) almost always pairs with a root aorist, cf. *capiō* : *cēpī*, *faciō : fēcī*, *iaciō : iēcī*, etc.

[202] Cf. also Thurneysen (1946: 431) referring to both *luid* and *buich* as "evidently old aorists like Gk. ἤλυθον and ἔφυγον". Pedersen (1913: 477) laconically comments that *-buich* is the preterite of an unattested present **buig-*, comparing it to *dluig-* next to *dlongaid* 'split'.

[203] On the possible semantic path from 'break, divide' (OIr. *-bongaid*) to 'enjoy, use' (Ved. *bhunákti* and Lat. *fungor*), see McCone 1991: 43.

a class VI present *pari-bhuját* ‘wrap around’ attested in 1.100.14 (and standard RV+):

> *yásyā́jasraṃ śávasā mā́nam ukthám*
> *paribhujád ródasī viśvátaḥ sīm*
> Whose inexhaustible measure in its vastness, (like our) hymn,
> will coil around both worlds on all sides.

The semantic issue notwithstanding, the comparison here is challenged by the fact that the Germanic cognates in this family of words including Goth. *biugan*, ‘bend’, OHG *biogan*, ‘id.’, OE *būgan* ‘bend, bow; flee’ and so on unambiguously point to a PIE **b^heu̯g^h-*. If IIr. **b^hau̯g-* ‘bend’ (cf. Khot. *haṃ-buśdä* ‘bow down’) is to be connected, as seems preferable on semantic grounds (so EWAia *s.v. BHOJ*[1]), the final unaspirated **-g-* must reflect a prehistoric influence from **b^heu̯g-* ‘flee’ following LIV[2] (*s.v. *b^heu̯g^h-* fn.3). The ‘flee’ root does not leave traces in Sanskrit but survives – in its causative/factitive sense – in Iranian in YAv. *buṇjaiṇti* ‘free’ and MPers. *bōxtan* ‘id.’ (more on the homonymy with **b^hau̯g-* ‘bend’, see below).[204] The exact semantic motivation is unclear, but a drift from ‘bend oneself (away), turn away’ to ‘flee’ as suggested by EWAia and also Kluge and Seebold (1989: 83) does not seem impossible, assuming both roots were at some point present in Indo-Iranian. In this scenario, Gk. φυγέ/ό- must be kept separate from Ved. *(-)bhuja-* ‘bend’ despite the formal equivalence.

The other Vedic form of interest is the optative 1pl. *bhujema* based on what appears to be a thematic aorist *bhuja-* from the root *bhuj-* ‘benefit, be useful’. All 5 occurrences of the form appear in the grammatical solecism of *mā* followed by an optative and are used in a quasi-formulaic expression of ‘pay for the sin done by another’, e.g.:

> 6.51.7 *mā́ va éno anyákr̥tam bhujema*
> Don’t let us pay for the offense done to you by another.
> 7.52.2 *mā́ vo bhujemānyájātam énas*
> Let us not suffer the guilt toward you that is born from another.

[204] On *bhujyú-* denoting ‘gazelle’ (<= ‘fleeing one’) in RV 10.95.8 and possibly to be connected with **b^heu̯g-* ‘flee’, see EWAia *s.v.*

7.88.6 *mā́ ta énasvanto yakṣin bhujema*
may we, burdened with guilt, not pay for (the transgression) against you.

In principle, the isolated thematic optative as such – next to an otherwise athematic root paradigm (cf. *mā bhojam* 2.28.9, mid. *bhójate* 1.72.8) – does not normally have a claim to be archaic. Hoffmann (1967: 95-7; cf. Jamison 2024: 2.1.4) suggests the form originated from the misanalysis of a phrase involving the infinitive *bhujé* and the first-person pronoun, i.e *bhujé me* or *bhujé naḥ* 'ist mir/uns zur Buße', which gave rise to *bhujema* 'let us pay' following the 'akustische Muster' provided by models such as *huve* : *huvema* 'call'. Cardona (1960: 30-2) resorts, again, to the influence of the 3pl. **bhuján* (of the root aorist, < **b*^{h}*ug-ént*), which he opines to have been interpreted as a thematic **bhuja-n* whence the thematic optative, with additional motivation from the metrical pressure to substitute a putative athem. **bhujyāma* in the cadence of a triṣṭubh. Neither solution seems conclusive to me. The fact that *bhujema* is preserved in the synchronically inadmissible construction of *mā* + opt. and employed in an expression that is demonstrably formulaic ([PAY (for)$^{*bhaug-}$ – SIN$^{*aj\text{ɲ}as-}$ – (OF ANOTHER)] *vel. sim.* with an Iranian counterpart *aēnaŋhō… būjəm* in Y. 31.13 (cf. EWAia *s.v. BHOJ*^{2}, Hoffmann 1967: 96):

yə́ vā kasāuš ***aēnaŋhō*** *ā mazištąm (a)yamaitē* ***būjəm***
or when one for (just) a small sin incurs a very great atonement.[205]

constitutes a strong piece of evidence that, rather than a late artificial creation, the thematized aorist may have been a relic form inherited into the (pre-)Vedic diction – side by side with the original root aorist – from Proto-Indo-Iranian and possibly beyond if an ultimate connection with φυγέ/ό- could be made. What makes such a connection potentially viable,

[205] The translation follows Humbach and Faiss (2010: 87). This reading that takes the genitive phrase *kasāuš aēnaŋhō* as dependent on the verb ('[incur an atonement] for a small sin') aligns with that of Kellens (1974: 56), Hoffmann (*ibid.*), and EWAia (*ibid.*). Insler's translation (1975: 186f., following Hinz 1960: 85f.) "either the one who is guilty of a small offense, (or the one who) shall receive a very great punishment" requiring a change of subject and supplying of *astī* is less preferable.

despite the fact that most authorities assign the two verbs into separate (homonymous) roots (cf. LIV^2 *s.v.* 2.*$b^heu̯g$-, IEW *s.v.* 4. *bheug*-), is the semantic peculiarity of the aorist of Ved. *bhuj*-. The active aorist forms, i.e. *bhojam* (2.28.9) and *bhujema* (x5 in RV), as Zasada (2023: 169) notes following Hoffmann (*ibid*.), all mean 'pay (for), atone (for)', 'büßen' (+ acc. rei),[206] a usage that stands out from the other tense stems (pres. *bhunákti*; perf. *bubhója*) of the verb which denotes 'benefit, help, be useful' (+ acc. pers.) in the active and 'enjoy, use' (+ instr. rei) in the middle (see Cardona 1987). The (non-aorist) active is not attested in finite forms until the brāhmaṇas, and even as participles the active voice is rare in RV (1.48.5, 10.89.17). Further, the negative connotation of the objects that the active aorist is construed with – 3x *énas* 'sin, evil', 1x *yakṣá* 'phantasm, ghost', 1x *dákṣam* '(cunning) skill' – marks a clear distinction from the arguments that the verb takes otherwise, i.e. persons in the accusative (e.g. TB 1.6.2.1 *tásmād yáṃ ca prajā́ bhunákti* 'therefore, he whom posterity serves…') or words associated with well-being or prosperity in the instrumental (e.g. *ukthá* 'wealth' 5.42.9, *ūtí* 'help, protection' 8.67.16, *gó* [*...jīvā́*] 'cow [alive]' 10.19.6). The semantic realm of the active aorist of *bhuj*-, in fact, echoes the use of φυγέ/ό-, which in Homer is prominently associated with concepts (acc.) referring to suffering or adversity (κακόν, κακότης, πήμα, μόρος, κῆρ, θάνατος, ὄλεθρος, etc.), e.g.:[207]

ι 488-9 ὦσα παρέξ, ἑτάροισι δ' ἐποτρύνας ἐκέλευσα
ἐμβαλέειν κώπῃς, ἵν' ὑπὲκ **κακότητα φύγοιμεν**
I pushed off and along, then, by nodding my head to my comrades
urged and bid them lay on the oar handles, so we'd **get out of danger.**

[206] Cf. Grassmann (1873 *s.v.* 2.√*bhuj*) 'etwas [A., selten I.] büssen, für eine Uebelthat [A.] von jemand [G.] Strafe erleiden'.

[207] The unsurprising association of φυγέ/ό- and negative connotations may find support already in the Mycenaean materials – recently García Ramón (2009) analyzes the Mycenaean personal name *pu$_2$-ke-qi-ri* as a compound consisting of φυγέ/ό- and βρι-, denoting someone who escapes something "heavy" (e.g. enemy, spears, misfortunes, etc.). Morphologically, García Ramón interprets the first member Myc. /p^{h}uge-/ as reflecting (or based on) ἔφυγε, in which case the thematic aorist may have already existed in Mycenaean Greek.

Ξ 312 ἱστὸν ἀμαιμάκετον νηὸς κυανοπρώροιο
ἐν χείρεσσιν ἔθηκεν, ὅπως ἔτι **πῆμα φύγοιμι**
(Zeus) put the huge mast of the dark-prowed ship
in my hands, so I'd still **escape misery**.

The semantic specialness of *bhujema* and *bhojam* within Vedic and its functional affinity with φυγέ/ό- allows for the conjecture that both verbs are derived from a single root *$b^heu̯g$- with an original meaning '(intr.) be free, get out, release oneself' and '(tr.) free, release' (from an adverse event or state of affairs). The development towards φυγέ/ό- (and Lat. *fugiō*) 'flee, escape' is self-explanatory. As for Ved. *bhuj*-, the sense of 'pay (for), atone (for)' can be reached through well-attested semantic pathways that connect ideas of departure, release, separation, and compensation, whereby paying/atoning is cognitively framed as removing a moral or financial burden or getting free from an obligation (cf. Lat. *solvō* 'release, loosen, free' => 'pay up', MoE *acquit*, French *s'acquitter*, etc.). Thus, the use of *bhujema* in, e.g., 6.51.7 *mā́ va éno anyákr̥tam bhujema*, could then be understood as "Let us not (have to) release/redeem ourselves from the sin done by another". Under this hypothesis, the active aorist of *bhuj*- shows semantic archaism that aligns itself closely with φυγέ/ό-, complementing the formal equivalence between the two thematic aorists *bhujema* and

φυγέ/ό-.[208, 209]

Regarding the present stems, the thematic φεύγω is the earliest and the most established

[208] The semantic path suggested here has the advantage over that of Hoffmann's (*ibid.* 'jemandem (Akk.) Nutzen schaffen' zu 'Genugtuung leisten') in that, as shown above, the active voice and the associated meaning 'be useful, benefit; grant pleasure' with an *accusativus personae* appears empirically to be later than (finite forms ŚB+) and secondary to the middle 'enjoy, relish, help oneself' with an instrumental (RV+), which renders a development from 'be useful, benefit' => 'provide satisfaction' => 'pay, atone' anachronistic in Vedic. The use of *bhuj-* in the middle is reconcilable with the sense '(cause to) be free, escape' if one considers the state of being free from an adversity, escaping danger, or being released from constraints as semantically approximating the idea of experiencing ease and content, whence the sense 'enjoy, help oneself'. A direct cross-linguistic example of this particular semantic development can be found in Chinese 逸 (< **lit*, cf. Baxter and Sagart 2014: 137) originally meaning 'flee, escape' but also used to denote 'leisurely, at ease' and further 'indulge, be dissolute'. A trace of the older semantics of *bhuj-* arguably is preserved in 8.68.14-16:

> *té na āsnó vr̥kāṇām ā́dityāso mumócata | stenám baddhám ivādite*
> Being such, free us from the mouth of wolves, o Ādityas,
> like a thief who's been bound, o Aditi
>
> *ápo ṣú ṇa iyáṃ śárur ā́dityā ápa durmatíḥ | asmád etv ájaghnuṣī*
> Let this arrow go away from us, o Ādityas,
> away malevolence, without striking us
>
> *śáśvad dhí vaḥ sudānava ā́dityā ūtíbhir vayám | purā́ nūnám bubhujmáhe*
> Because, o Ādityas of good gifts, over and over,
> previously and now, we have benefited by your help

As the mid. perf. *bubhujmáhe* directly responds to the earlier requests of 'free us from…' (*mumócata*) and 'let this arrow go away…without striking us' (*ápa… etv ájaghnuṣī*), it is not a significant leap to assume the form may convey some sense of 'have escaped, fled', 'have survived' ([from] the 'mouth of the wolves', the 'arrow', and other calamities). The typical use of the middle with the instrumental in the sense of 'enjoy' may develop from a semantic drift from 'be free from, escape (an adversity by means of [instr.])' => 'survive, be at ease (thanks to [instr.])' => 'benefit (oneself), enjoy (sth. [instr.])'.

[209] Two further cases in Indo-Iranian that formally reflect *b*^{h}*au̯g-a-* are not considered here: (1) YAv. *būjaṯ* 'save' (cf. *būjaiia-*, *buŋj-* 'id.') is attested once in the corrupt Yt. 4.3 – both LIV^{2} (*s.v.* 1.**b*^{h}*eu̯g-* fn.2) and Kellens (1984: 102, 135 fn.4) doubt its authenticity, (2) Ved. *sambhújam* in RV 2.1.4 is taken as a 1sg. subjunctive/injunctive based on a thematic aorist of *bhuj-* by Grassmann (1873 *s.v.* √*bhuj-*) but a nominal reading 'a common meal' is generally accepted now, see Jamison 2024 *ad. loc.* for detailed discussion with lit.

present Greek knows since Homer (x94, cf. x119 in aor.). The form has no extra Greek support and is taken as an innovation there by LIV^2 (*s.v.* 1.*$b^heu̯g$-). EDG (*s.v.*) states that φεύγω is "built on a PIE aor. continued in φυγεῖν", albeit unclear as to whether it is created via the more ancient process involving the aorist subjunctive (cf. Goth. *leihvan* above) or a more recent process Greek-internally based on the ample models of the type such as ἐσπόμην : ἕπομαι 'follow', ἔπλον : πέλω 'become', etc. The nasal present φυγγάνω has a sparse but secure presence in the post-Homeric materials, appearing three times in tragedy ([A.] *Pr.* 513, S. *El.* 132, with ἐκ- A. *Pr.* 526) and used, with prefixes, sporadically by prose authors. Indirectly, Alcaeus is said to use a perfect participle πεφύγγων (fr. 421 Voigt), ostensibly derived from φυγγάνω or even *φύγγω(?). [210] Outside of Greek, nasal presents are abundantly represented for *$b^heu̯g$- 'flee' as well as the homonymous *$b^heu̯g$- 'benefit, be useful'. YAv. *buŋjaiṇti* 'save, free' (cf. Bartholomae 1904 *s.v. baog*- 'lösen' 'befreien, retten', Kellens 1984: 135f., 165) and Pāli *pari-bhuñjati* 'purifies' both fit readily under the semantic coverage of the proposed unified root *$b^heu̯g$- 'free (from), get out, release' as forms originally causative in nature (cf. LIV^2 *s.v.* 1.*$b^heu̯g$- fn.1). Ved. *bhuñjáte* 'enjoy, help oneself' and *bhunákti* 'benefit, be useful, serve' as discussed above (cf. fn. 208) follow a similar semantic trajectory from 'be free, escape, release oneself' (from danger) to 'be at ease, feel relief' => 'enjoy' (mid.) and 'cause to be safe, make…free from' => 'benefit, serve, help' (act.). Further, Lat. *fungor* 'perform; enjoy; survive' – assigned to the root *$b^heu̯g$- 'be useful' in LIV^2 – provides another example of this development. Osthoff in 1895 (293f.) makes an insightful observation on the semantic core of Lat. *fungor* which is notoriously versatile in meaning:

> Man muss für *fungor* von dem Grundbegriff 'sich von etwas frei machen, sich losmachen', 'sich einer Sache entledigen', franz. 's'acquitter de' ausgehen …. daraus entwickeln sich die übrigen Anwendungsweisen

[210] On perf. part. -ων in Aeol., see recently Scarborough 2023: 117-20 with lit. The form is unclear – Voigt cites Epim. AN. Ox 1.366, 22 and 29 (Cramer) where πεφύγγων is understood by grammarians as a type of consonant gemination peculiar to Aeolic.

> ‘mit etwas fertig werden’, ‘handelnd abmachen, vollbringen, vollziehen, verrichten, leisten’, ‘leidend durchmachen, erleiden, etwas überstehen’ usw.

Based on this, Ostoff connects *fungor* with YAv. *buṇjaiṇti* ‘free’, *baoxtar-* ‘liberator’, etc., although for the same semantic consideration he rejects the then already prevalent opinion that Ved. *bhuñjáte*/*bhunákti* should be related, seemingly not aware of the special usage of the active aorist *bhojam* and *bhujema* ‘pay (for)’ in Vedic as examined above. Our discussion has shown that Ved. *bhuj-*, in fact, can be traced back to the same semantic origin that Osthoff identified.[211] The combined evidence presented here points to a few conclusions. The thematic aorist φυγέ/ό- is underscored by a root aorist reflected in Lat. *fūgī* and more directly in Ved. *bhojam* and *bhójate*. A direct comparison between φυγέ/ό- and Ved. *bhujema* is tentatively established based on the semantic affinity. The present φεύγω is standard in Greek but has no parallel elsewhere, while φυγγάνω is relatively marginal, but nasal presents to the root are highly prolific outside Greek. It is suggested that the avatars of the two separate **b*^{h}*e̯u̯g-* in LIV^{2} (‘enfliehen, freikommen’ and ‘jmdm. nützen, Nutzen bringen’) may be unified into one with a meaning ‘(cause to) be free, release (oneself)’. The original, semantically intransitive/reflexive usage is mostly preserved in Gk. φυγέ/ό- ‘flee, escape’, Lat. *fūgī* ‘id.’, Ved. *bhojam* and *bhujema* ‘pay (for), atone’, with the nasal presents in Indo-Iranian YAv. *buṇjaiṇti* ‘free’, Pal. *paribhuñjati* ‘purify’, Ved. *bhunákti* ‘benefit, help’ exhibiting a causative function of the root.

3.3 The origin and diachrony of the type ἔλιπον-λιμπάνω-λείπω

With the results from the survey above as our empirical basis, the following sections will attempt to establish a diachronic account for the stem collocation pattern of the type ἔλιπον-

[211] Notice also that the oldest usage of Lat. *fungor* as observed consistently in Plautus and Terence is with an *accusativus rei* (often *munus*, *officium*, etc.), whereas the use with the ablative is a later development. The case syntax aligns with that of Gk. φεύγω and Ved. *bhujema*.

λιμπάνω-λείπω, in particular regarding the polymorphic juxtaposition of the *-N-C-ane/o-* present and the thematic root present and its implication for the early history of the stem building tendencies of **CeRC-* roots in Greek.

3.3.1 The thematic aorist

To start with a relatively straightforward observation, for all members of this group – with the possible exception of λαθέ/ό- (cf. above 3.2.3) – the thematic aorist clearly forms (one of) the starting point(s) of their individual paradigm. Empirically, all (except the *hapax* κύθε) are well attested in the early epics and often significantly better represented than forms built on the present stem(s) (cf. τυχέ/ό-: 22 vs. 2, λιπέ/ό-: 156 vs. 54, πυθέ/ό-: 50 vs. 19, λαθέ/ό-: 46 vs. 25; but ἀδέ/ό-: 9 vs. 25). Within Greek, the thematic aorist is the only historically authentic aorist known to this type. The concurrent *s*-aorists are either artificial (ἐτύχησα, cf. 3.2.6) or late (ἔλειψα: once as part. λείψας in Ar. fr. 965, otherwise not attested until Polybius, cf. DELG *s.v.*), and the reduplicated aorists – πεπύθοιτο (formulaic in Z 50 = K 381, Λ 135; cf. Bendahman 1993: 106, Beckwith 1996: 51), λέλαθον (Beckwith 1999), and the *hapax* κεκύθωσι – have been previously argued correctly to be poetic creations. Most thematic aorists found in the group have good comparative support from root aorists elsewhere in Indo-European (whether already thematized or still radical), including λιπέ/ό-: Ved. *rikthās*/*aricat*, Arm. *elik'*, κυθέ/ό-: Ved. *guhámānas* (cf. above *s.v.*), πυθέ/ό-: Ved. *budhánta*, and φυγέ/ό-: Ved. *bhujema*, *bhojam*, Lat. *fūgī*, etc., although some are admittedly isolated (ἀδέ/ό-, τυχέ/ό-) or a possible Greek innovation (λαθέ/ό-). Recoverable traces of the underlying athematic root aorists within Greek include Hsch. λῆτο· ἐπελάθετο (for **leh₂-* before the extension) and tentatively λίπεν (Π 507; cf. 3.2.4).[212]

[212] Cf. 3.2.3 for the creation of λαθέ/ό- based on λήθω or the weak perfect stem λελαθ-. Alternatively, if the root aorist in the Hesychian gloss (< **leh₂i-*) is considered the basis, one postulates that a thematic form *λαε- (< *λαι̯ε- < **lh̥₂i̯-e-*) existed at some point, which is remade into λαθέ/ό- and absorbed into the paradigm of λήθω.

3.3.2 The present

In the realm of the present stems, the materials examined picture a puzzling discrepancy between the relative chronology of the nasal and the thematic root presents exhibited in Greek and that implied by the comparative data. On the one hand, the double-nasal forms, as such, are either marginal in the early materials *vis-à-vis* the thematic present (πυνθάνομαι 2x vs. πεύθομαι 17x, τυγχάνω 2x vs. τεύχω 23x, λανθάνω 5x vs. λήθω 18x) or post-Homeric/dialectal as in κυνθάνω (Hesychian), φυγγάνω (A.+ and sporadic), and λιμπάνω (Aeol.). Their extra Greek nasal comparanda, however, as shown in the discussions above, are amply attested for at least πυνθάνομαι (cf. OIr. *ad-bond-*, Lith. *bundù*), φυγγάνω (cf. Ved. *bhunákti*, Lat. *fungor*, etc.), and λιμπάνω (cf. Ved. *riṇákti*, Lat. *linquō*, etc.), and possibly κυνθάνω (cf. Arm. *suzanem*, PIr. **gau̯d-*; see 3.2.2 fn.163 on the etymological complication) and ἀνδάνω (cf. Ved. *svāda-*^{ti}). The evidence for inherited nasal-infix present underlying the Greek double nasal type is thus more prevalent than previously supposed.[213] On the other hand, the thematic presents are clearly the older formation within Greek, all well established and the dominant variant in Homer (for the exception ἥδομαι, cf. 3.2.1) with some found already in Mycenaean (ἥδομαι: *wa-do-me-no*, λείπω: *re-qo-me-no*). Comparatively, πεύθομαι offers the only case with extra-Greek parallels in view of Ved. *bódhati*, Goth. *-biudan*, etc. < **b^{h}éu̯d^{h}-e/o-*, but even for this item the alternative possibility that the thematic present is an offshoot of the old root aorist subjunctive is often suspected (see 3.2.5). The remaining members of the group have no good external support. The contrasting patterns above lead to the questions (1) which of the two present formations originally accompanied the thematic aorist and (2) how the polymorphic presents came into being with the chronology they display and what the role the thematic aorist might have played in such a development.[214]

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[213] Risch (1974: 272) and Jasanoff (2022a: 90) only recognize λιμπάνω and πυνθάνομαι as inherited, whereas Thurneysen (1894) largely rejects the connection of the Greek double-nasal type with the inherited nasal-infix present. See Sturm (2021:200-3) for brief etymological comments on the Greek materials.

[214] Polymorphism is commonplace among Greek verb stems including the present, but the

3.3.2.1 Hypothesis I: original THEM. AOR. + ROOT THEM. PRES.

The null hypothesis would be that early Greek started with THEM. AOR. + ROOT THEM. PRES., directly aligning with what is reflected in the philological evidence. This is essentially the view of Thurneysen (1894), who opines that the type started from the pair ἔχαδον : *χένδω 'hold', with the latter remodeled at some point into *χενδάνω with an empty suffix -άνω and subsequently borrowing the vocalism from ἔχαδον, resulting in χανδάνω. This pattern, according to Thurneysen, spread by analogy to ἔλιπον => λιμπάνω, ἔτυχον => τυγχάνω, etc., without any connection between the double-nasal present and the inherited nasal-infix present (cf. 1894: 82 "Somit scheint mir λιμπάνω von lat. *linquō* absolut zu trennen."). The issue with creating a category *ex nihilo* purely analogically notwithstanding, the assumption that the nasal present is an entirely new member of this class is problematic not the least because it goes against the comparative profile of many of the verbs of the group as discussed above and would require a scenario where comparatively well supported λιμπάνω, πυνθάνομαι, and φυγγάνω must be regarded as having developed independently from their apparent IE cognates. More importantly, if the pristine stem pairing was ἔλιπον : λείπω, ἔφυγον : φεύγω, etc., the creation of extra polymorphic double-nasal presents lacks any motivation. The combination of the zero-grade thematic aorist and the full-grade thematic present is a highly stable and prolific class in Greek. Some of the better represented instances in Homer include:

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coexistence of a nasal present (of any kind) with a thematic full-grade present is, in fact, not frequent. The only other case among verbs involving the thematic aorist is ἀπεχθάνομαι : ἔχθομαι 'be hateful', which is a denominative from ἔχθος (EDG *s.v.*). Elsewhere in the Homeric corpus, one finds ὀρέγνυμι : ὀρέγω 'stretch' (the νυ-present is only attested as ὀρεγνύς 2x at verse-end in Hom.; ὤρεξα extremely frequent, cf. Lat. *rēxī*), ἔργνυμι (*hapax* κ 238; verse-end) : (ἐ)έργω 'confine' (ἔρξα common), and more distantly πλύνω 'wash' : πλέω 'sail' (< *pleu̯-).

ἔδρακον	:	δέρκομαι	‘look’
ἤριπον	:	ἐρείπω	‘ruin’
ἤρυγον	:	ἐρεύγω	‘belch’
ἐπιθόμην	:	πείθομαι	‘trust’
ἔπραθον	:	πέρθω	‘destroy’
ἔτραφον	:	τρέφω	‘nurture’
ἔτραπον	:	τρέπω	‘turn’
ἐσπόμην	:	ἕπομαι	‘follow’
ἔσχον	:	ἔχω	‘hold’

Table 3.3 Selected examples of THEM. AOR. + ROOT THEM. PRES. in Homer

Pairs such as ἔλιπον : λείπω would have naturally settled into the pattern, especially considering that some members of this group share a (synchronically) similar root shape *-*CeC*-*IC*- (ἤρυγον, ἐπιθόμην). Another observation on the synchronic Greek data that potentially sheds light on the chronology of the λείπω-type presents of this group is that while all double-nasal presents discussed so far are accompanied with a full-grade thematic present, the remaining two Homeric examples, namely χανδάνω (: ἔχαδον) and λαγχάνω (: ἔλαχον), appear without †χένδω or †λέγχω.[215] To these one may add two post-Homeric cases, μανθάνω (Pi.+, next to ἔμαθον Hom.+) and λαμβάνω (Arch.+, next to ἔλαβον Hom.+), without concurrent †μένθω or †λήβω.[216] Of these λαμβάνω can be safely dismissed as it has an old -*i̯é/ó*- present λάζομαι (*Il*.+; <= *$s_{\text{l̥}}h_2g^w$-*i̯é/ó*-; on the short -α- see Bendahman 1993: 107 fn.86)

[215] PIE *g^hed-, originally nasal-less (cf. Lat. *praeda* ‘booty’, Goth. -*gitan* ‘get’), historically makes a nasal-infix present (more below), but the nasal is incorporated into the root early and widely enough that a neo-root *g^hend- can be observed in several branches (e.g. Lat. *prehendō*). In Greek, the synchronic root is most straightforwardly understood as χενδ- with full-grade in fut. χείσομαι (σ 17, *h. Ven.* 252) < *χενδ-σ- and *o*-grade in pluperf. κεχόνδει (*v.l.* of κεχάνδει Ω 192, cf. Chantraine 1948: 427).

[216] The root only appears as μαθ- in the verbal paradigm (ἔμαθον, μαθήσομαι, μεμάθηκα, etc.) and its underlying structure is debated. The traditional interpretation that it reflects a secondary root *$men(s)$-d^h- (EDG *s.v.* μανθάνω, NIL *s.v.* $mn̥(s)d^h(e)h_1$- but cf. fn.1, fn.13 there, Chantraine 1961: 222) connects it with Hsch. μ 849 μενθήρη · φροντίς ‘thought’, but this has been challenged by Kümmel (2005: 324-6), who advocates for a connection with προμηθής ‘forethinking’ going back to a root *$(s)meh_2d^h$-, and more recently by Vine (fthc.), cf. Rothstein-Dowden (2023: 59f.),

with an attractive Germanic isogloss OE *læccan* 'grasp' (< PGmc. **lak-i̯an-*; Kroonen 2013 *s.v.*, LIV^2 *s.v.* **sleh$_2$g^w-* fn.3). The three instances missing from the otherwise surprisingly regular triadic pattern of ἔλιπον-λιμπάνω-λείπω are thus †χένδω, †λέγχω, and possibly †μένθω, all of the shape *CeNC-*. The possibility of a mere historical coincidence cannot be ruled out, but diachronically these exceptions might be explicable if one assumes that the full-grade thematic presents of this group are secondarily created at a time when the mechanisms for generating full-grade present to a base *CVC-* (as exhibited by all verbs here in the thematic aorist), are still functional for λείπω, φεύγω, etc. but failed in ἔχαδον >> †χένδω, ἔλαχον >> †λέγχω, and ἔμαθον >> †μένθω.[217] Such underapplication of a process *CaC >> CeNC-* is understandable given that the "correct" ablaut relationship between zero-grade *-a-* and full-grade *-en-* is probably less tangible in Greek than *u* : *eu*, *i* : *ei*, or *Ø* : *e* (due to the inherent ambiguity of *-a-*) and likely becomes somewhat inactive in the grammar at a fairly early point. It is perhaps not a coincidence that no present forms in *-éNC-e/o-* in Greek have the zero-grade root *CaC-* attested in their paradigms:

- ἐλέγχω 'revile' (Hom.+) < **h$_1$leng^h-* (cf. Hitt. *linkanzi* 'swear', EDHIL *s.v. li(n)k-^{zi}*): full-grade ἐλεγχ- only incl. fut. ἐλέγξω (E.+), aor. ἔλεγξα (Hom.+), etc.
- μέμφομαι 'blame' (Hom. [ἐπι-]+) < ?: full-grade μεμφ- only incl. fut. μέμψομαι (Hes.+), aor. μεμψάμην (A.+), μέμφθην (Pi.+), etc.
- πέμπω 'send' (Hom.+) < **pemp-* (substrate word per EDG): full-grade πεμπ- in fut. πέμψω, aor. πέμψα (Hom.+), etc.; *o*-grade in πέπομφα (Hdt., Th.+).
- ῥέγκω 'snore' (A.+) < ? (possibly onomatopoeic, cf. DELG, EDG *s.v.*): only attested in the present.
- ῥέμβομαι 'wander' (Men.+) < ?: late and only attested in the present.
- σπένδω 'pour libation' (Hom.+) < **spend-* (cf. Hitt. *ispanti*): full-grade σπενδ- only

[217] It is assumed that if such presents were to exist, the derivational basis should be the aorist, as is most common in Greek. In theory, a speaker could infer the correct full-grade forms, in the case of †χένδω and †λέγχω, from the perfect κέχονδα and λέλογχα. But both perfects are peripheral in their paradigms and at least for κέχονδα (only as *v.l.* once in Iliad) the *o*-grade perfect is already being phased out in Homer (cf. κεχανδότα, κεχάνδει). The future χείσομαι (< **g^{h}end-s-*), which could be an inherited desiderative (cf. LIV^2 *s.v.* **g^{h}ed-*), likely also did not provide enough cues to generate †χένδω.

incl. aor. σπεῖσα (Hom.+), fut. σπεῖσω (Hdt.+), perf. ἔσπεικα (Plut.), etc.

- τέγγω ‘wet, moisten’ (Alc. [fr. 347. 1 Voigt]+) < **teng-* (exact equivalent in Lat. *tingō*): full-grade τεγγ- only incl. fut. τέγξω (Pi.+), aor. τέγξα (A.+), τέγχθην (Pl.).
- τένδω ‘gnaw’ (Hes. *Op.* 524, Hsch. τ 472 τένδει· ἐσθίει ‘eats’, ἢ λιχνεύει ‘gormandize’) < **tend-* (cf. Lat. *tondeō*, OIr. *tenn-* ‘cut’): only attested in the present.
- φθέγγομαι ‘sound’ (Hom.+) < ?: full-grade φθεγγ- only incl. fut. φθέγξομαι, aor. φθεγξάμην (Hom.+), perf. ἔφθέγμαι (Pl.+).

In the data involving the thematic aorist, ἔπαθον ‘suffer’ (< **b^{h}end^h-*) and ἔδακον ‘bite’ (< **denḱ-*) offer two more cases of zero-grade **CeNC-* roots, neither producing a full-grade thematic present (†πένθω, †δένκω).[218] A heuristic search might yield some exceptions, but the overall weakness of the apophonic connection between *a* and *en* is noticeable in the Greek verbal system.[219] The lack of the *CeNC-* presents may be considered under the broader phenomenon of the “clunkiness” of ablaut involving (Proto-Greek) *a*,[220] which has been explored in different contexts by Kimball (1988; cf. 2.3 above) in a paper detailing the rise of the secondary ablaut *a* : *ā* in the perfect replacing the inherited *a* : *ō* with roots with an internal **h$_2$* (e.g. ἑστᾱ- <= *ἑστω- < **sestoh$_2$-*). The alternative explanation that early Greek had †χένδω, †λέγχω, and †μένθω but lost them at some point is less convincing, as inherited *CéNC-e/o-* presents such as τέγγω (cf. Lat. *tingō*) and potentially substrate verbs such as πέμπω ‘send’ (EDG *s.v.*) are kept intact.

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[218] A close formal approximation to †πένθω is πενθέω ‘grieve’ (Hom.+), but this is a denominative based on πένθος following an entirely different derivational process from ἔπαθον >> †πένθω.

[219] The observation presented above should not be construed as suggesting that *a* : *en* ablaut is *absent* in Greek, which is clearly not the case. Nominals, for instance, do appear to be more conservative in preserving the pattern, cf. the type βάθος : βένθος ‘depth’. Certain verbs of the shape *Cen-* still show recoverable, if weak, alternation of *a* and *en* in their paradigms, such as κτείνω ‘kill’ (**kt**en**-i̯e/o-* : ἔκτ**α**το), τείνω ‘stretch’ (**t**en**-i̯e/o-* : τέτ**α**κα, ἐτ**ά**θην, etc.). What is proposed concerns the (lack of) productivity of such a pattern regarding roots of the shape *CeNC-*.

[220] For reflex of **o̯* as *a* in Common Greek (the conditioned *o*-reflex in Aeolic and Mycenaean notwithstanding), see recently van Beek 2021: 27-31.

3.3.2.2 Hypothesis II: original THEM. AOR. + NASAL PRES.

We now turn to the alternative hypothesis that the class started with THEM. AOR. + NASAL PRES. despite the fact that many of the λιμπάνω-type presents themselves are demonstrably young. Since the nasal presents (whether with an inherited ancestor or entirely secondary) of the verbs in this group formally all surface as the λιμπάνω-type, an evaluation of this scenario revolves crucially around one's view on the genesis of the Greek double-nasal presents. The various early accounts including that of Thurneysen (1894), Kuiper (1937: 152-7), and Schwyzer (1939: 700f.) all rightly observe that a thematic nasal suffix -(α)νω becomes associated with roots containing an internal nasal (*χενδ-, *λεγχ-) and/or those possessing an inherited nasal-infix present (*λιμπ-, *πυνθ-, etc.), albeit all without exploring the deeper motivation behind that. The fact that the -άνω presents, the vast majority of which are of the double-nasal type, are prominently paired with the thematic aorist is noticed in many modern handbooks but none provides an explanation for the origin of this connection (Schwyzer 1939: 699, Chantraine 1948: 315, Risch 1974: 271, Willi 2018: 211).[221] The latest treatment of the topic by Jasanoff (2022a) advances an ambitious account that aims to unify the double-nasal presents in Graeco-Armenian, Celtic, and Germanic under an innovated stem type that already existed in (late-)PIE. Given the importance of this work and its potential implication for our understanding of the category, an examination of Jasanoff's theory is called for.

The scenario proposed by Jasanoff is centered on the early behavior of the PIE root *g^hed- (cf. fn.215). First, the Lindeman variant of the nasal-infix present *g^h*n̥ned*- (vs. "normal" *g^h-*ne-d*-) was reanalyzed as containing the zero-grade of a neo-root *g^h*end*-, i.e.

[221] Kuiper (1937: 153) suggests "Vielmehr ist die feste Korrelation von Präsentien auf -ανω und Aor. II darauf zurückzuführen, daß beide aus einem athemat. Präsens hervorgegangen sind. Während nun die schwundstufige thematisierte Stammform wie im Aind. (*avidam*, S. 76 sq.) als Aorist Verwendung fand, wurde das neue Präsens durch ein besonderes Formans charakterisiert.". But this particular view of his that both the nasal-infix present and the thematic aorist developed from an original athematic root present seems to be operating on an outdated model of the Indo-European verbal system. As is widely accepted, the thematic aorist, where originating from an athematic formation, is normally based on the root aorist.

g^{h}*n̥-ne-d-*, citing Ved. *bhanákti* 'crush'(< **b*^{h}*n̥-né-g-ti*), which is plausible. He then argues (2022: 95f.) that **CeNC-/CNeC*- roots are inherently clumsy in building/preserving nasal-infix presents because "their weak stem was indistinguishable from the zero grade of the bare root", resulting in the difficulty in deriving the strong stem from, e.g., 3pl. **g*^{h}*n̥dénti* (based on the neo-root **g*^{h}*end-*). This then prompts the "externalization" of the nasal-infix (seen in Ved. *jānắti* 'know' < **ǵn̥h₃-néH-ti* [not from **ǵn̥-né-h₃-ti*]) to provide more cues on the connection between the singular and the plural, whence **g*^{h}*n̥-ne-d-* is substituted by **g*^{h}*n̥d-ne-*(*ti, -si*, etc.) next to the weak stem **g*^{h}*n̥d-* (97). The resulting paradigm later becomes fully thematic with the stem **g*^{h}*n̥d-né/ó-* generalized to the plural, as reflected in Celtic and Germanic (95). However, one may raise criticism on an important point here, namely that the weak stem of a nasal-infix present built to a **CeNC-/CNeC*- root should, purely on a formal level, be **Cn̥-n-C-*, different from the zero-grade of the bare root in **Cn̥C-*.[222] Jasanoff (94 fn.14) brushes the problem away by stating that "I take it for granted that the surface realizations of underlying **g*^{h}*n̥-n-d-* (**b*^{h}*n̥-n-g-*) and **g*^{h}*-n̥-d-*(**b*^{h}*-n̥-g-*) would have been identical", suspecting that "while the two could theoretically have been realized as long [*g^{h}n̥:d-] ([*b^{h}n̥:g-]) and short [*g^{h}n̥d-] ([*b^{h}n̥g-]), respectively, PIE did not otherwise have a length distinction in syllabic liquids or nasals, and there is no reason to believe it would have introduced one here". The assumption, however, that **Cn̥-n-C-* and **Cn̥C-* have the same surface realization and are to be considered interchangeable for the analysis Jasanoff proposes is not a light one and, in fact, may not hold. The precise behavior of **Cn̥nC* in PIE cannot be determined due to the scarcity (non-existence?) of the materials, and past treatments of the problem are inconclusive.[223] In principle, there is no phonological

[222] Normally, according to Schindler's syllabification scheme (1977, also discussed *passim* in Barber 2013, Byrd 2015), **CRRC*- should yield **CRR̥C-*, e.g. **sru̯-tó-* 'flowed' > **srutó-* (Ved. *srutáḥ*, Gk. ῥυτός), **tu̯nk-i̯é/ó-* 'push' > **tu̯n̥k-i̯é/ó-* (Gk. σάπτω, OHG *dūhen*). But Jasanoff (2022a: 94) is correct in pointing out that the nasal-infix often resists becoming syllabic even in an environment where syllabification is the norm, e.g. 3pl. **i̯u-n-g-énti* (Ved. *yuñjánti* 'yoke') instead of the phonologically expected **i̯u̯-n̥-genti*.

[223] Fortson (2009: 51) suggests the possibility of a long **n̄C* from **n̥nC* but also admits such a sequence is likely subject to early degemination, to be held against McCone 1998 (468),

difficulty in assuming that *$C\underset{\circ}{n}nC$- is realized just as the phonological notation indicates, and at least in Greek and Indo-Iranian the outcome of *$C\underset{\circ}{n}nC$- ([CanC-], perhaps < *[Cə̃nC-]< *[CəənnC-]) would regularly be different from that of *$C\underset{\circ}{n}C$- ([CaC-] < *[Cə̃C] < *[CəənC-]).[224] It is worth noting that the assumption that PIE even possesses Sievers- and Lindeman-variants to begin with suggests that it does treat -$\underset{\circ}{R}R$- as phonetically distinguishable from -*R*- and -$\underset{\circ}{R}$- and is sensitive to the change of syllable weight such variants cause.[225] Considering that *$C\underset{\circ}{n}nC$- and *$C\underset{\circ}{n}C$- *are* different in syllable weight, there is very little ground to presume that, e.g., 3pl. *[$g^h\underset{\circ}{n}$n.dén.ti] and *[$g^h\underset{\circ}{n}$.dén.ti] would be acoustically identical to the speaker, although that remains a possibility. The corollary thereof is that the motivation behind Jasanoff's proposed suffixation of the nasal infix in the singular *$g^h\underset{\circ}{n}$-*ne-d-* => *$g^h\underset{\circ}{n}d$-*ne-* is unfounded. This is a severe problem for his analysis because the subsequent argument relies on the coordination of 3sg. *$g^h\underset{\circ}{n}dnéti$: 3pl. *$g^h\underset{\circ}{n}dénti$ (where the sg. forms have undergone suffixation, and the pl. still retains the shape of the infixed stem) as the analogical model whereby, Jasanoff argues (100), other verbs with an "internal nasal" would backfit a suffix *-*né/ó*- to the singular, which then spread back to the plural, resulting in the synchronic double-nasal pattern. This allegedly affects both root presents of *$CeNC$- roots such as OIr. *teinnid* 'cut' (via 3sg. *$tént(s)ti$: 3pl. *$t\underset{\circ}{n}dénti$ => 3sg. *$t\underset{\circ}{n}dnéti$: 3pl. *$t\underset{\circ}{n}dénti$) and original nasal-infix presents such as MW *gwnn* 'know' (via 3sg. *$\underset{\sim}{u}inét(s)ti$: 3pl. $\underset{\sim}{u}indénti$ => 3sg. $\underset{\sim}{u}indnéti$: 3pl. *$\underset{\sim}{u}indénti$). The number of analogical steps involved in the entire process is substantial, and some of the postulated mechanisms are remarkably speculative, especially (1) the back and forth analogical remodeling between the singular and the plural forms of the original nasal-infix present and (2) the wide spread of a pattern based on single, aberrant paradigm in 3sg. *$g^h\underset{\circ}{n}dnéti$: 3pl. *$g^h\underset{\circ}{n}dénti$ with the effect of altering stem

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who operates with a regular *$g^h\underset{\circ}{n}$-*n-d*- > PCelt. *g^h*annd*- and Gk. χανδ- (so also GEW *s.v.* χανδάνω).

[224] On the process of PIE *$\underset{\circ}{n}$ > Gk.-IIr. *-*a*-, cf. Meier-Brügger and Fritz 2021: 109f.

[225] Evidence for the application of Lindeman's Law on nasals (or liquids) – as mostly provided by Edgerton (1943) – is dubious and less conclusive than that for the glides, cf. Sihler 1971, Barber 2013: 43-6,

formation in verbs of other established categories.

The situation in Greek is more uncertain. The predicted *g^{h}n̥dné/ó- would have yielded †χαδνω and, next to ἔχαδον, fallen together with the type ἔταμον : τάμνω, ἔδακον : δάκνω, etc. For this Jasanoff (103) resorts to an analogy ἔλιπον : λιμπάνω :: ἔχαδον : X where X is resolved as χανδάνω, but λιμπάνω itself is supposed to have been created following the analogy of 3pl. *g^{h}n̥dénti : 3sg. *g^{h}n̥dnéti :: 3pl. *$link^{w}$énti : X, where X is resolved as *$link^{w}$néti. Note that this analogy works less effectively in Greek than in Celtic, because in the latter zero-grade *g^{h}n̥d- becomes PCelt. *gand- (also *tn̥d- > *tand- etc.) and converges at some point in stem shape with the inherited weak stems of the nasal-infix present such as *u̯ind- (cf. above) as *CVNC-, providing some support to the hypothesized transfer of the model 3pl. *g^{h}n̥dénti : 3sg. *g^{h}n̥dnéti to nasal presents. In Greek, this process relies solely on the fact that *g^{h}n̥d- and *li-n-k^{w}-, *b^{h}u-n-d^{h}-, etc. are all athematic stems with an internal nasal, which is hardly enough to justify a remodeling of this scale. The bigger claim of Jasanoff's (100) that "*all* active athematic presents with a 3pl. in *-NT-enti (*T* = any stop, not just *$d^{(h)}$) were liable to be fitted out with a backformed analogical 3sg. in *-NT-neti, which eventually gave rise to a full thematic present in *-NT-ne/o-." finds little support in the Greek data. As acknowledged in the paper (100 fn.31), τένδω (according to Jasanoff [97f.] an athematic root present in origin) was not remade into †τανδάνω, to which one may add two additional counterexamples: σπένδω 'pour libation' (no †σπανδάνω; root pres. per LIV^{2} *s.v.* 1.*spend-), ἄγχω 'squeez' (no †ἀγχάνω, cf. LIV^{2} *s.v.* *h_{2}em$\acute{g}^{h}$-).[226] The treatment of the inherited *i̯ungénti (cf. Ved. *yunákti*, etc.) as ζέυγνυμι 'yoke' rather than †ζυνγάνω is particularly damaging as the *CeI̯C- root shape is exactly paralleled by λιμπάνω, πυνθάνομαι, φυγγάνω, etc.

[226] Fortson (2025: 5) mentions λάμπω 'shine' and σφίγγω 'tie tightly' in his criticism of Jasanoff on the same point, although it is unclear whether there were originally root presents in these two cases.

3.3.3 From **CeRH-* roots to **CeRT-* roots

Jasanoff's starting point that the double-nasal presents in Greek have their origins in the inherited nasal-infix present is no doubt correct, but his subsequent reliance on **$g^he(n)d$-* and 3pl. *$g^hn̥dénti$* (and more broadly, presents in **-NT-énti*) as the pivot in my view is misguided and not effective in explaining some of the structural particularities of the Greek data. Returning to the facts themselves in Greek, the most important features of the λιμπάνω-type present may be summarized as follows:

(1) The correspondence with the thematic aorist is *exceptionless*. All double-nasal presents (both Homeric and post Homeric cases) have a corresponding zero-grade thematic aorist in the shape of (-)*CVC*. Where relatively chronology can be established, the present always postdates the aorist (e.g. ἔλαβον [Hom.+] vs. λαμβάνω [Arch.+], θίγον [Arch.+] vs. θιγγάνω 'touch' [A.+]).

(2) An inherited nasal-infix present can be postulated for many members of the type. To the core set already discussed above incl. ἀνδάνω, λιμπάνω, πυνθάνομαι, φυγγάνω, and perhaps κυνθάνω, the post-Homeric θιγγάνω may be added in view of Arm. *dizanem* 'heap up', Lat. *fingō*, OIr. *dingid* 'press' (< *d^hi-n-$ǵ^h$- based on *$d^heiǵ^h$-, cf. Jasanoff 2022a: 101; on *τιχ- => θιγ-, see DELG *s.v.*).[227]

(3) Within the observable history of Greek, the double-nasal presents are a young formation that is still marginal and rare in Homer but sees a moderate degree of

[227] LIV^2 (*s.v.* *$d^heiǵ^h$- fn.7) does not include Gk. θιγγάνω, θίγον due to the phonological difficulty of reconciling the final *-g-* in Greek with **-g^h-*, cf also EDG *s.v*. On the problem of explaining the post-nasal deaspiration in θιγγ (< *d^hing^h-) with "Miller's Law", see recently Nikolaev 2023, who suggests an alternative PIE root *$d^heiǵ/ǵ$- for θιγγάνω, MHG *tîchen* 'mange, handle', and tentatively OIr. *dingid*.

productivity in the later stage of the language.

A simpler version of the prehistory of this category that accounts for these features could be found Greek-internally. The core idea, which is already present in the earlier treatments by Kuiper and Schwyzer (see above), is that the -άνω suffix is ultimately to be connected with an independent nasal suffix *-*ne/o*- (e.g. κάμνω 'toil', βάλλω 'throw', etc.), originally thematized from the weak stem of the nasal-infix presents of *seṭ* roots, i.e. *-*n-H-é/ó*- (Klingenschmitt 1982: 159-61, Rix 1992: 211, Sihler 1995: 518, Willi 2018: 18). The standalone *-*ne/o*- suffix, Fortson recently comments (2025: 4), spread to some non-*seṭ* roots and had "gotten associated with thematic aorists (or their forebears)". That νω-presents in Greek (in general, not limited to the double-nasal type) prominently pairs with the thematic aorist is already observed by Risch (1974: 270), who states "Die Präsentien auf -νω stehen in der Regel neben thematischen Aoristen". Some well-established members of this type include ἔταμον : τάμνω 'cut' < *$temh_1$-,[228] ἔκαμον : κάμνω 'toil' < *$kemh_2$-, ἔβαλον : βάλλω 'throw' < *g^welh_1-, ἔδακον : δάκνω 'bite' < *$denḱ$-, and (less securely) ἐπλόμην : τέλλομαι 'fulfill, go about' < *k^welh_1-,[229] many of which have an underlying root shape *$CeRH$-.

[228] For τάμνω, the standard reconstruction with a root-final *h_1 is adopted (EDG, DELG, GEW all *s.v.*, LIV^2 *s.v.* *$temh_1$-). Cardona (1960: 77f.) operates with *$temh_2$- based on τέμαχος 'slice of fish (or meat)' (in Attic comedy and prose), but in a detailed treatment Forssman (1966) has convincingly shown that the root is *h_1-final on the strength of Pindaric τέτμηνται (*I.* 6.22) and Myc. *te-me-no* (τέμενος 'plot of land'). See fn.43 in Chapter 2 for more references.

[229] LIV^2 (*s.v.* *k^welh_1-) is followed here in connecting (Aeol.) πέλομαι/ἐπλόμην 'become' (occasionally active in Homer) with (περι-)τέλλομαι 'go about' (Hom.+; cf. Cret. τέλλω 'accomplish') under *k^welh_1- along with Ved. *cárati* 'run, move', Lat. *colō* 'care, cultivate'. The πλ- outcome of *$k^wl̥h_1$-*e/o*- (instead of the expected *παλ-) can either be due to an early loss of the laryngeal (so Klingenschmitt 1982: 280f.; via the νεογνός rule if augmented forms were treated like reduplicated stems, cf. Balles 2012) or a secondary substitution under the influence of πέλομαι (after ἐπτόμην : πέτομαι etc.), which is an inherited thematic present, see above. Against a unified root, EDG (*s.vv.*) reconstructs *k^wel- for τέλλομαι and *k^welh_1- for πέλομαι; DELG does not mention the connection between the two verbs.

3.3.3.1 **CeRh$_{1/3}$-* roots as an original source of thematization

This broad structural affinity between the thematic aorist and the thematic nasal present in -νω (and later [-N-T]-ανω) in Greek is significant and likely results from a similar prehistoric path of thematization unique to the **CeRH-* roots, in particular **CeRh$_1$-* ones.

On the one hand, the paradigm of the original athematic nasal-infix present offers – by virtue of laryngeal vocalization (**CR̥-n-h̥$_1$-C- > *CR̥ne-C-* in the weak stem) – multiple "thematic looking" slots in Greek including act. 2pl. **CR̥ne-te*, mid.3sg. **CR̥ne-toi*, 2pl. **CR̥ne-st^{h}e*, 3pl. **CR̥ne-ntoi*, which could have served as the pivotal points leading to the full thematic paradigm.[230] The thematization of the "βάλλω-type" present is treated in full detail in the recent work by Sturm (2021: 26-8).[231] It is worth noting that the three verbs her analysis concludes as "secure instances" (2021: 45) of thematized nasal presents to **h$_1$-*final roots, namely βάλλω, τάμνω, and τέλλω, all produce a thematic aorist in Greek.

On the other hand, the important role played by **CeRH-* roots in the early history of the Greek thematic aorist has been discussed at length by Harđarson (1993: 155-70; cf. 1.2.3). According to Harđarson, the 2sg. and 3sg. of the strong stem of the original amphikinetic root aorist of **CeRh$_1$-* roots, i.e. **CeRe-s* (< **CéRh̥$_1$-s*) and **CeRe-t* (< **CéRh̥$_1$-t*), directly underlie the synchronic *e*-grade thematic aorists such as Arc. ἔζελε (< **[h$_1$e-]-g^{w}elh$_1$-t*) 'throw'.[232] The basis of the synchronic zero-grade thematic aorist of the type ἔταμον, ἔβαλον, etc. comes from two sources. First, the 3pl. **CR̥h$_1$-ént* regularly becomes **CaR-ent* (on **CR̥h$_1$V- > CaRV-*, cf. 2.5), which then underwent the replacement of the inherited **-ent* with **-ont*, whence **CaR-ont*. It is unclear if Harđarson's parenthetical "…(mit **-ont* für uridg. **-ent*)" (160, cf. 155) envisions a replacement of the Cardonian type based on the participle

[230] Theoretically nasal presents to **h$_3$-*final roots should be prone to thematization as well (e.g. act.1sg. **CR̥no-mes* < * *CR̥-n-h̥$_3$-mes*), but this category merged into νῡ/νυ-presents due to the effect of Cowgill's law, see Sihler 1995: 526-7, Sturm 2021: 90-2. On the scope of Cowgill's Law, see Vine 1999.

[231] Less appealing is Harđarson's suggestion (1993: 161 fn.69) that the 3pl. (e.g. **g^{w}l̥nh$_1$-énti*) is the pivot form with a seemingly unmotivated replacement of **-enti* with **-onti*.

[232] The general idea already in Wackernagel 1916: 14.

(e.g. φέροντι : ἔφερον[τ] :: βαλόντι : X, where X is resolved as ἔβαλον[τ]). Such spontaneous replacement of the athematic *-*ent* with thematic *-*ont* is not free of problems, although it remains a possibility (cf. above 1.2.2). What is not explicitly emphasized by Harđarson is that the change *-*ent* => *-*ont* is more plausible in the root aorist paradigm of **CeRH*- roots (than elsewhere, e.g. for **lik*^{w}*ént*) because the 3pl. of **CeRh*$_3$- roots, i.e. **C*$\underset{\circ}{R}$*h*$_3$*-ént*, produces a thematic looking **CoRont* regularly (cf. the "*o*-grade" thematic aorists 3pl. ἔμολον, ἔπορον, etc.) and thereby provides significant support for a 3pl. *-*ont*.[233] It is also worth noting that the specialness of **CeRh*$_1$- and **CeRh*$_3$- roots in the establishment of the thematic paradigm can be indirectly inferred from the fact that while most **CeRh*$_{1/3}$- roots (*R = r, l, m, n*) possess a thematic aorist, a large number (or majority) of the **CeRh*$_2$- roots do not, including (all Hom.+):

- **demh*$_2$- 'tame, subdue': *s*-aor. ἐδάμασα; pres. δάμνημι
- **ḱerh*$_2$- 'mix': *s*-aor. ἐκέρασα; pres. κίρνημι
- **merh*$_2$- 'grab': pres. μάρναμαι
- **pelh*$_2$-: 'come near': *s*-aor. ἐπέλασα, root aor. πλόμην; pres. πίλνημι
- **perh*$_2$- 'sell': *s*-aor. ἐπέρασα; pres. πέρνημι
- **telh*$_2$- 'lift, support': *s*-aor. ἐτάλασ(σ)α, root aor. τλήν

While the system of Harđarson is not perfect, the combined paradigmatic peculiarities of the original root aorist paradigm of **CeRh*$_{1/3}$- roots, offering several potential paths towards a thematic paradigm, suggest they are likely among the earliest roots to generate thematic aorists in Greek.[234] Thus, the close relationship between the early νω-presents and the thematic aorist finds a historical explanation in their shared origins in the **CeRH*- roots.

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[233] That **C*$\underset{\circ}{R}$*h*$_3$*V* > **CoRV*- is phonologically regular under labial environments is discussed in detail in Chapter 2, cf. 2.4, 2.5. For cross-morphemic laryngeal coloring in 3pl. *-*h*$_3$*-ent*, cf. above 2.4.2.1.

[234] One of the insufficiencies of Harđarson's scheme, for instance, is that it does not explain why the zero-grade root becomes the norm for the thematic aorist (both of **CeRH*- roots and elsewhere). In the original athematic root aorist, the full-grade base in the 3sg. (**CeRe-t* < **CeR*$\underset{\circ}{h}$$_1$*-t*) is arguably a stronger, more paradigmatically salient candidate than the 3pl. (**CaR-ent*) to serve as the basis for a new paradigm. For a critical evaluation of Harđarson, see Willi (2018: 333-42), cf. 1.2.3, below fn.242.

3.3.3.2 The spread of thematic aorist to **CeRT-* roots

The coordination of – perhaps originally sporadic – thematic forms in the root aorist next to the nasal present among **CeRh₁-* roots may have inspired and been a catalyst to the creation of the thematic aorist themselves to roots of other shapes (including but perhaps not limited to **CeRT-*) with an accompanying nasal-infix present. Forms such as (**témh̥₁-t* >) **teme-t* and (**tm̥h₁-ént* =>) **tamo-nt* standing next to the still athematic nasal-infix present **tam-nē-ti*/*tam-ne-nti* served as a model to root aorists such as **leiḱ*^{w}*-t*/*lik*^{w}*-ent* (see Step 4 below).[235] The analogically created thematic forms, probably first to **CeIT-* roots (**(e)-lik*^{w}*et*, *(e)-lik*^{w}*ont*, etc.) chose the zero-grade of the original root aorist, initially due to the generalization of a base *CVC-* from their models in **CeRH-* roots (**tem-*, **tam-*, **por-*, etc.), which favors **CIC-* over **CeIC-* for structural conformity.

Under this scenario, the use of the zero-grade root of in ἔλιπον, ἔφυγον, and so on is not paradigmatically determined (i.e. via inherited ablaut), but rather a result of the early spread of **CVC-é/ó-* as the default stem shape for the category. Two observations may speak in favor of this. First, **CeRH-* roots, being (among) the earliest root types to produce the Greek thematic aorist, show the greatest variety of root vocalism including the *e*-grade (Arc. ἔζελε [i.e. *ἔδελε], Att. ἔτεμον, ἐγενόμην), *o*-grade (ἔπορον, ἔμολον, etc.), and zero-grade with synchronic *-a-* (ἔταμον, ἔκαμον, ἔβαλον), suggesting that originally the root gradation is not as clearly a defining feature for the speaker generating a thematic form of the aorist as the phonetic structure of **CVC-*. Second, outside of the **CeRH-* roots, where the root ablaut of

[235] It is assumed that the athematic stem **-nē/ne-* is still morphologically close enough to the infixed **-ne-T-*/*-n-T-* for both to be registered as "ablauting nasal present" to early Greek speakers. There is reason to believe that the thematization of the βάλλω-type present happens fairly late, in view of the athematicity of nasal presents of **h₂-* and **h₃*-final roots even in historic Greek. It is unclear when Greek lost the laryngeals in **VHT* environments, but it is unlikely that stems such as **tm̥-né-h₁-mi, -si, -ti,* etc. would have remained intact next to **temet*, **tamont*, etc., although it is argued in Chapter 2 (see 2.5) that **CR̥HV-* > **CəRV-* possibly already took place in late-PIE.

a Greek thematic aorist deviates from the zero-grade norm, the only synchronic stem shapes found are (*V*)*CeC*- (ὤφελον ‘owe’, ἠγερόμην ‘gather’, ἔτεκον ‘beget’, ἔπεσον ‘fall’, εἷλον ‘take’) and maybe once *CoC*- in γοόν ‘bewail’ (*hapax* Z 500; interpreted as imperfect by Nussbaum 2025). Strictly impermissible are bases of the shape **CeiC*-, **CeuC*-, **CeNC*-, and **CeRC*-.[236]

The advantage of the process proposed above is that it postulates a structural reason for the early spread of the thematic aorist, namely the inherited stem pairing pattern of ROOT AOR. + NASAL PRES. shared between certain roots of the shape **CeRh*$_{1(/3)}$- and **CeRT*-. The former underwent thematization spontaneously due to the phonology of its root-/stem-shape in Greek before exporting the paradigm THEM. AOR. + NASAL PRES. to the latter. In other words, the change **leu̯k*^{w}*-t* => **lik*^{w}*e-t* is structurally supported.

3.3.3.3 The emergence of λείπω etc.

One may posit an intermediate stage where the ἔλιπον-λιμπάνω-λείπω type had acquired a newly created thematic aorist **lik*^{w}*e*- next to the still athematic inherited nasal-infix present **li-ne-k*^{w}*-/li-n-k*^{w}- (and possibly lingering remnants of the old athematic root aorist **leu̯k*^{w}*-/lik*^{w}-).[237] At this point, the induction of the full-grade thematic present (**léu̯k*^{w}*e/o*-, etc.) into

[236] The underlying structure of εὗρον ‘find’ is not entirely clear but generally thought to reflect a modified reduplicated aorist **u̯é-u̯r-e/o*- (LIV^{2} *s.v.* **u̯reh*$_{1}$-) with the loss of the laryngeal after reduplication and a secondary diphthong *-eu*-, see Beckwith 1994, McDonald and Zair 2012: 35 (on possible connection with Osc. ϝουρουστ ‘s/he will have found’), recently in great detail Kostopoulos 2014-5, and in passing Vine 2024: 358, cf. Willi 2018: 77f. for reconstruction of a root **u̯erh*$_{1}$- . The comparison with OIr. preterite 3sg. *-fúair* ‘found’ < **u̯ou̯r*- (Beckwith 1994: 26f.), included in LIV^{2}, is objected to by Schumacher (2004: 681). The potential metacritique on the observed absence of thematic aorists of the shape **CeRT*- that such forms will simply be recognized as an imperfect by modern lexicographers in my view is not valid – a hypothetical †ἔπενθον next to πάσχω would have been as reasonably categorizable as an aorist as, e.g., τέκον next to τίκτω, based on aspectual value reflected in the readings, stem collocation, etc.

[237] Nikolaev (p.c.) brought to my attention the Hesychian gloss λ 1125 λίσσωμεν· ἐάσωμεν

the verbal paradigm can be thought of as primarily driven by the overall tendency in Greek to phase out the nasal-infix present to stop-final roots (i.e. *-né-T-*/*-n-T-*́), which towards the end of its existence is the only infixed stem in Greek after -νᾱ-/-να- (< **-ne-h₂-*/*-n-h₂-*), -νῡ-/-νυ- (<= **-ne-u-*/*-n-u-* and **-ne-h₃-*/*-n-h₃-*), and (the pre-thematized) **-nē-*/*-ne-* (< **-ne-h₁-*/*-n-h₁-*) became fully suffixalized (Sturm 2021:121). In other words, presents of the type **linepti*/**limpenti* – or perhaps with a generalized weak stem **limpti*/**limpenti*[238] – were facing increasing systemic pressure. Such pressure is, of course, not limited to the ancestors of the λιμπάνω-type, but also applied to other inherited **-n(e)-T-* presents. Secure examples include **i̯u-n(e)-g-* 'yoke' (Ved. *yunákti*, Lat. *iungō*, Lith. *jùngiu*) > PGk. **dzun(e)g-* => ζεύγνυμι, **h₂mr̥-n(e)-ǵ-* 'wipe' (Ved. *mr̥ñjata* [RV 9.24.1 = 9.65.26]; cf. LIV² *s.v.* **h₂merǵ-*) > PGk. **amarn(e)g-* => ὀμόργνυμι, **h₃r̥-n(e)-ǵ-* 'stretch' (Ved. *r̥ñjate*, Lith. *rę́žiu*) > PGk. **orag-*/*orneg-*(?) => ὀρέγνυμι, and **ph₂-n(e)-ǵ-* 'fasten' (Lat. *pangō*, Arm. *s-p'acanim* 'gird (oneself)'; on Ved. *pā́paje*, see LIV² *s.v.* **peh₂ǵ-* fn.4) > PGk. **pan(e)ǵ-* => πήγνυμι. This type of remodeling of inherited nasal-infix presents to stop-final roots as Greek nasal present in -νυμι with full-graded root is well documented (cf. Schwyzer 1939: 697, Clackson 1994: 84, Kocharov 2019: 86, Sturm 2021: 120f.). All of the examples cited above are accompanied with an *s*-aorist (ἔζευξα, ὤμορξα, ὦρεξα, ἔπηξα, all Hom.+), which no doubt forms the basis of the new -νυμι presents (Kocharov *ibid.*, Rix 1992: 210, van de Laar 2000: 350).[239] Thus, comparing the ζεύγνυμι-type with the λιμπάνω-type, the fate of the inherited nasal-infix

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'let us allow', which, if to be connected with λείπω (and the root **leı̯kʷ-*), could reflect a **-i̯e/o-* present built directly to the root (aorist stem) *λικ-. The form is mentioned in GEW *s.v.* λείπω and Schwyzer 1939: 692.

[238] Cf. McCone's (1991: 53) remark on the "post-PIE tendency to base new non-ablauting paradigms upon the weak stem", cf. Lat. *linquit*, *iungit*, etc.

[239] There is a general affinity between νυ-presents and the *s*-aorist, cf. the full list of verbs with such a pairing in van de Laar 2000: 348-50. This may simply be due to the overall productivity of the νυ-presents. As Rix (1992: 210) notes, presents in -νυμι are also found as a secondary formation next to inherited present presents, e.g. μείγνυμι 'mix' next to Hom. μίσγω (< **mig-ské/ó-*, cf. Lat. *misceō* 'id.'), ἕννυμι 'clothe' next to Hom. ἕστο (< **u̯es-*, cf. Ved. *váste* 'be clothed', Hitt. *wēšta* 'id.').

presents to stop-final roots seem broadly to be separate along the lines of the aorist available to them – between *s*-aorist >> νυμι-present and thematic aorist >> *-N-T-*άνω present.

After the *λιμπμι-type presents became paradigmatically disfavored (as a synchronically "irregular" conjugation), speakers seek to furnish an alternative formation based on the most salient stem available, namely the aorist, at the time already thematized as λιπέ/ό-. There could be two possible contributors that lead to the introduction of the thematic present into the paradigm for that purpose. First, as mentioned above, another major stem pairing pattern that involves the thematic aorist is THEM. AOR. + THEM. PRES. of the type ἔσχον (< *$sǵ^h$-*é/ó-*) : ἔχω (< *$ségʰ$-*e/o-*), accounting synchronically for over 20 verbs (not counting λείπω, etc.).[240] The group consists prominently of roots of the shape **CeLT-*/*CLeT-* (e.g. ἔτραφον : τρέφω, ἔτραπον : τρέπω, ἔπραθον : πέρθω, ἔδρακον : δέρκομαι) and **CeC(H)-* (e.g. ἐσπόμην : ἕπομαι, ἔσχον : ἔχω, ἐπτόμην : πέτομαι) but also a few items in *(*V*)*CeI̯T-* such as ἤριπον : ἐρείπω 'ruin, tear down'. It shows a distinct stem collocation from the type discussed so far that often features a concurrent *s*-aorist (ἔθρεψα, ἔτρεψα, ἔπερσα, etc.), a category which almost never appears alongside the pattern THEM. AOR. + NASAL PRES. Further investigation on the provenance of the thematic aorist in this group is needed (cf. 4.3), but some have the profile of being old formations. For instance, for (ἔνι)σπον/(ἐνν)έπω 'say', the root thematic present has an exact correspondence between imp. ἔννεπε and Lat. imp. *inseque*/*insece* 'Say! Tell!' (Enn. *Ann*. 10.322, Andr. 1.1+), while the thematic aorist shows up in both -σπον and Lat. *inquit* (< *$en\text{-}sk^w\text{-}e/o$-; cf. LIV^2 *s.v.* 2.*$sek^w$-, EDLIL *s.v. īnsece*/*inquam*). On the same item, recently Rothstein-Dowden (2023: 58 fn.96) raised the possibility that (some of) the thematic aorists may have arisen from a phonological constraint against light monosyllables in verbal forms such as *h_2e-conjugation 3sg. *sk^w-*e*, which is then remodeled as *sk^w-*e-t* after the *mi*-conjugation (for similar ideas, cf. Jasanoff 2017; but). It is at any rate conceivable that some pairs in this group are old and already exist to serve as

[240] N.B. the list of "Thematic root present : non-sigmatic aorist pairs in Greek" given recently by Willi (2018: 433f.) is incomplete.

the model when λιπέ/ό- is used as the basis for the derivation of a new present in light of the decline of *λιμπτι.

The second source for the full-grade thematic stem (λείπε/ο-) is the subjunctive of the original root aorist (*λειπ-ε/ο-). The subjunctive origin for some of the Greek thematic presents is already suggested by Kuiper (1937: 156) and similar explanations have been used to explain certain thematic verbs in Germanic (see above 3.2.4).[241] Whether the root aorist subjunctive played a role in the formation of some of the (secondary) thematic presents in individual branches and how (or if) such a mechanism is to be connected with the hypothesized, much older nominal origin behind the PIE thematic present (Watkins 1969: 106) is beyond the scope of the current study.[242] For the case at hand, however, since it is probable that the thematic aorist and the root aorist coexisted for a period of time, the subjunctive forms – still retrievable in the grammar (e.g. **leiˀkʷ-e-ti*.) – may have provided additional support to the analogical process stated above.

3.3.3.4 Remodeling of *λίμπμι => λιμπάνω etc.

Once the thematic presents of the type λείπω, φεύγω, πεύθομαι, etc. are established, they quickly became the dominant type given its productivity and prevalence in the Greek verbal system while the old athematic nasal-infix presents (*λίμπμι, *φυγγμι, *πυνθμι, etc.) survive as relic by-forms. The final step in the scenario proposed here is not fundamentally different from the early accounts of Schwyzer *et al.* (cf. above) and the recent brief scheme offered by Fortson (2025: 4), namely the attachment of the newly created thematic νω-suffix – initially having emerged in *$CeRh_1$- roots (cf. above) – to the now derelict athematic nasal stems, resulting in the attested double-nasal present via a Sievers-Edgerton type of process as also

[241] See Vine (1998: 76 fn.184 with ref.) for more examples of thematic presents that have been explained as root aorist subjunctive in origin.
[242] The reader is referred to Willi (2018: 184-96) for a detailed evaluation of Watkins' theory and the recent discussion by Villanueva Svensson (2021). Cf. above 1.2 and fn.13.

adopted in Jasanoff's proposal (2022a: 102), i.e. *λίμπνω > λιμπάνω.[243] The spread of the νω-suffix from *$CeRh_1$- roots including τάμνω and βάλλω to *λιμπμι etc. is well motivated given that now the model of THEM. AOR. + νω-PRES. provided by ἔταμον/τάμνω, ἔβαλον/βάλλω, etc. lends itself well to being the basis for ἔλιπον to "modernize" the outdated *λιμπμι, which may explain why *λίμπμι, *φύγγμι, etc. did not end up as †λείπνυμι, †φεύγνυμι, etc. following the pattern seen in ζεύγνυμι (cf. fn.245).

3.3.4 Conspectus

The process proposed above may be summarized as follows:

Step 1:

PIE possesses certain *$CeRH$- and *$CeRT$- roots that form ROOT AOR. + NASAL PRES.[244]

*$CeRH$- ROOTS, e.g.:

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[243] Crucially, the spread of -νω and the subsequent epenthesis resulting in -ανω after heavy syllables proposed here happens at a much later stage than Jasanoff's *-*NT-ne/o*- > *-*NT-n̥ne/o*- which is construed as a pre-Graeco-Armenian development.

[244] Our starting point is not a coincidence given that (1) both *$CeRH$- and *$CeRC$- are highly frequent root shapes in PIE that build a nasal-infix present (accounting for roughly half of all such roots, based on the appendices in LIV^2 712f.) and (2) the pairing of a root aorist and a nasal present, in general, can be considered typical in PIE (Strunk 1967). The expected contrast in the markedness (or a lack thereof) of the stem between the present and the aorist is a core idea of the "Hoffmann-Strunk" model on the genesis of tense and aspect (Hoffmann 1970, Strunk 1994, summarized in Willi 2018: 28-34), whereby (in a simplified version) telic roots are expected to yield root aorists (because of the inherent affinity of telicity with perfectivity) and characterized presents, and the opposite (i.e. root presents next to characterized aorists) is expected for atelic roots. To what extent such a system is valid in PIE and preserved in the daughter languages cannot be evaluated here. A majority of the roots (of the shape *$CeRC$- and *$CeRH$-) relevant for the current discussion do exhibit telicity in their typical usage in Greek (cf. 3.2-8 on the event type of individual items) incl. ἐπυθόμην 'learn, ascertain', ἔφυγον 'escape', ἔλιπον 'leave', ἔλαχον 'obtain by lot, receive', ἔταμον 'cut (off), fell', and ἔβαλον 'throw'.

témh$_1$-t, *t̥mh$_1$-ént* : *t̥m-né-h$_1$-ti*, *t̥m-n-h$_1$-énti* (*telh$_1$- ‘cut’)
g^{w}élh$_1$-t, *g^{w}l̥h$_1$-ént* : *g^{w}l̥-né-h$_1$-ti*, *g^{w}l̥-n-h$_1$-énti* (*g^{w}elh$_1$- ‘throw’)
d^{h}érh$_3$-t, *d^{h}r̥h$_3$-ént* : *d^{h}r̥-né-h$_3$-ti, *d^{h}r̥-n-h$_3$-énti (*d^{h}erh$_3$- ‘leap’)
pérh$_3$-t, *pr̥h$_3$-ént* : *pr̥-né-h$_3$-ti, *pr̥-n-h$_3$-énti (*perh$_3$- ‘provide’)
(…)

*CeRT- ROOTS, e.g.:
léi̯k^w-t, *lik^w-ént : *li-né-k^w-ti*, *li-n-k^w-énti (*lei̯k^w- ‘leave’)
b^{h}eu̯g-t, *b^{h}ug-ént : *b^{h}u-né-g-ti*, *b^{h}u-n-g-énti (*b^{h}eu̯g- ‘free [oneself]’)
b^{h}eu̯d^h-t, *b^{h}ud^h-ént : *b^{h}u-né-d^h-ti*, *b^{h}u-n-d^h-énti (*b^{h}eu̯d^h- ‘wake up’)
(…)

Step 2:

Pre-Greek vocalizes *C$\underset{\circ}{H}_i$C > *CE$_i$C and resolves the sequence *C$\underset{\circ}{R}$HV- into *CaRV/CoRV-.

Loss of laryngeal in *VHC environment (> *V̄C).

*CeRH- ROOTS, e.g.:
téme-t, *tam-ént : *t̥m-né́-ti*, *t̥m-n-énti (*telh$_1$- ‘cut’)
g^{w}éle-t, *g^{w}al-ént : *g^{w}l̥-né́-ti*, *g^{w}l̥-n-énti (*g^{w}elh$_1$- ‘throw’)
d^{h}éro-t, *d^{h}ar-ónt : *d^{h}r̥-nó́-ti, *d^{h}r̥-n-ónti (*d^{h}erh$_3$- ‘leap’)
péro-t, *por-ónt : *pr̥-nó́-ti*, *pr̥-n-ónti (*perh$_3$- ‘provide’)
(…)

*CeRT- ROOTS, e.g.:
léi̯k^w-t, *lik^w-ént : *li-né-k^w-ti*, *li-n-k^w-énti (*leik^w- ‘leave’)
b^{h}eu̯g-t, *b^{h}ug-ént : *b^{h}u-né-g-ti*, *b^{h}u-n-g-énti (*b^{h}eu̯g- ‘free [oneself]’)
b^{h}eu̯d^h-t, *b^{h}ud^h-ént : *b^{h}u-né-d^h-ti*, *b^{h}u-n-d^h-énti (*b^{h}eu̯d^h- ‘wake up’)
(…)

Step 3:

(1) The reanalysis of **CéRe-t* as a thematic form and (2) the spread of 3pl. *-ont* in **CR̥h₃-ónt* lead to the emergence of a thematic conjugation (with undetermined root vocalism) in **CeRh₁/₃-* roots.

**CeRh₁/₃-* ROOTS, e.g.:

>> ****tem-é/ó-, * tam-é/ó-*** : **tm̥-né-ti*, **tm̥-n-énti*
>> ****gʷel-é/ó-, *gʷal-é/ó-*** : **gʷl̥-né-ti*, **gʷl̥-n-énti*
>> ****dʰer-é/ó-, *dʰar-é/ó-*** : **dʰr̥-né-ti, *dʰr̥-n-énti*
>> ****per-é/ó-, *por-é/ó-*** : **pr̥-nó-ti*, **pr̥-n-ónti*

**CeRT-* ROOTS, e.g.:

**léi̯kʷ-t*, **likʷ-ént* : **li-né-kʷ-ti*, **li-n-kʷ-énti*
**bʰeu̯g-t*, **bʰug-ént* : **bʰu-né-g-ti*, **bʰu-n-g-énti*
**bʰeu̯dʰ-t*, **bʰudʰ-ént* : **bʰu-né-dʰ-ti*, **bʰu-n-dʰ-énti*

Step 4:

**CeRT-* roots start forming thematic forms in the aorist based on analogy with the model *CVC-é/ó-* aorist + ablauting athematic nasal present set by the **CeRh₁/₃-* roots[245]:

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[245] Among other instances of (potentially) inherited NASAL PRES. + ROOT AOR. that did *not* result in the thematization of the aorist following the model of **CeRh₁/₃-* roots, two important groups need to be addressed. The first group is **CeRh₂-* roots, whose aorist mostly becomes sigmatized, e.g. ἐδάμασα 'tame' (< **demh₂-*), ἐκέρασα 'mix' (< **ḱerh₂-*), ἐπέρασα 'sell' (< **perh₂-*). In theory, they formally are the closest to the **CeRh₁/₃-* models, appearing at one point with, e.g. for **perh₂-* 'sell' (πέρνημι), aor. 3sg. **pera-t* (< **pérr̥₂-t*), 3pl. **para-nt* (< **pr̥h₂-ént*), pres. 3sg. **pr̥-nā-ti* (< **pr̥-né-h₂-ti*), 3pl. **pr̥-n-anti* (< **pr̥-n-h₂-énti*). Willi (2018: 339), in particular, called out this discrepancy (although he considers both **CeRh₂-* and **CeRh₃-* to be the outliers.):

> Leaving aside the forms with **-h₁-,* why were those with **-h₂/₃-* sigmatised at all, if the strategy envisaged by Cardona and Harđarson for the creation of thematic aorists was available? Why was the sigmatic ancestor of, say, ἐκέρασ(σ)ε and ἐστόρεσε built if the synchronically awkward **(e-)kera-t* and **(e-)stero-t* could have been discontinued by modeling a paradigm around the 3pl. forms [**(e-)kar-ant* =>] **(e-)kar-ont* and **(e-)star-ont* (yielding historical 3sg. †ἔκαρε and †ἔσταρε)?

*$CeRh_{1/3}$- ROOTS, e.g.:

*tem-é/ó-, * tam-é/ó-	:	*tm̥-né-ti, *tm̥-n-énti
*g^{w}el-é/ó-, *g^{w}al-é/ó-	:	*g^{w}l̥-né-ti, *g^{w}l̥-n-énti
*d^{h}er-é/ó-, *d^{h}ar-é/ó-	:	*d^{h}r̥-né-ti, *d^{h}r̥-n-énti

---

As has been shown in Chapter 2 (2.7), with the exception of ἐστόρεσα all *CoRe-s*- aorists in Greek have secondary origins and do not continue *$CeRh_3$-*s*- with the alleged metathesis of *-*eRo*- > -*oRe*-. This leaves us with *$CeRh_2$- roots. The reason for their escaping thematization can be attributed to the fact that while roots with *h_1- or *h_3- have *some* elements in their paradigm interpretable as being thematic – for *$CeRh_1$- roots, 2/3sg.*CeRe-s*, -*t*; for *$CeRh_3$- roots, 3pl. **Ca/oR-ont* – the aorist paradigm of *$CeRh_2$- has consistent predesinential *a*-vocalism, making thematization less likely. Willi's point about the strategy involving 3pl. *-*ent* => -*ont* being also available to *h_2-final roots misses the fact that at one point, e.g., **temet* (< *$tém h_1$-t*) and **tament* (< *$tm̥h_1$-ént*) are in the same paradigm, regardless how the root vocalism levelled out in favor of *tam*- in the end. Thus, the substitution of *-*ent* for -*ont* (whether through the Cardonian path or under the influence of *$CR̥h_3$-ónt*) is facilitated by the cues provided by the thematic-looking 2/3sg. forms, a factor absent in, e.g. **pera-t* (< *$pérh_2$-t*), 3pl. **para-nt* (< *$pr̥h_2$-ént*). A corollary is that sigmatisation of *$CeRh_2$- roots postdates the thematization of *$CeRh_{1/3}$- roots, at a time when the old, "odd looking" forms such as **teme-m*, **teme-nt*, *d^{h}aro-s*, **poro-t*, etc. no longer existed in the fully thematized paradigms of **tem-e/o*-, **tam-e/o*-, **por-e/o*-, etc. whereas **perat*, *perant*, etc. still were liable to be reformed. The second group of inherited nasal-infix presents potentially standing next to a root present that does not produce a thematic aorist following the scheme proposed here for *$leik^{w}$- etc. is the few νυμι-presents built to **(H)CeRT*- roots including ζεύγνυμι and ὀμόργνυμι (both with comparative support, cf. above). One expects †ἔζυγον and †ὤμοργον (or †ὤμαργον, cf. fn.20) if the same ROOT AOR. + NASAL PRES. model had been available for them. The preempting factor for these items is likely the early sigmatization (in Greek or late-PIE). In other words, at the time when Step 4 takes place, ζεύγνυμι etc. had stem pairing pattern of *$dzeu̯k$-*s*- and *dzu-*n(e)*-*g*- (if the nasal-infix present has not already been remodeled after the aorist into *$dzeu̯g$-*nū*/*nu*-). As shown earlier, a great majority of νυμι-presents, whether inherited or secondary, pair with an *s*-aorist. The origin of the affinity between the two stem formations is unclear. For ζεύγνυμι and ὀμόργνυμι, ἔζευξα is distantly paralleled by Lat. *iūnxī* (<= *$iūxī$ < *$i̯eu̯g$-*s*-) and ὤμορξα by Ved. 2sg. *amārkṣīṣ* 'wipe' (on the form see LIV^2 *s.v.* *$h_2merǵ$- fn.4). We are reminded of the fact that the ἔλιπον-λιμπάνω type (and their analogical ancestors ἔταμον/τάμνω, ἔβαλον/βάλλω, etc.) never pairs with an *s*-aorist. The patterns seen in ζεύγνυμι : ἔζευξα and λιμπάνω : ἔλιπον mark two distinct paths of stem interaction between the inherited nasal present and the accompanying aorist.

**per-é/ó-, *por-é/ó-* : **pr̥-nṓ-ti, *pr̥-n-ónti*

⇩

**CeRT-* ROOTS, e.g.:

leu̯ik^{w}-/lik^{w}-* >> ****lik^{w}-é/ó-* : **li-né-k^{w}-ti, *li-n-k^{w}-énti*

b^{h}eu̯g-/b^{h}ug-* >> ****b^{h}ug-é/ó-* : **b^{h}u-né-g-ti, *b^{h}u-n-g-énti*

b^{h}eu̯d^{h}-/b^{h}ud^{h}-* >> ****b^{h}ud^{h}-é/ó-* : **b^{h}u-né-d^{h}-ti, *b^{h}u-n-d^{h}-énti*

Step 5:

The athematic nasal-infix presents to **CeRT-* roots as the only infixed stem become obsolete, prompting the creation of a concurrent root thematic present, facilitated by the influence of the abundant roots showing aor. *R(Ø)-é/ó-* and pres. *R(é)-e/o-*. Before this stage, **CN̥C-* had already realized as *CaC-*.

**Ce(R)T-/C(R)eT-* ROOTS, e.g.:

tr̥p-é/ó-* : **trép-e/o-* (τρέπω) (trep-* 'turn')

sk^{w}-é/ó-* : **sék^{w}-e/o-* ([ἐν]έπω) (sek^{w}-* 'say')

(…)

⇩

**CeRT-* ROOTS, e.g.:

lik^{w}-é/ó-* : >> ****léi̯k^{w}-e/o-* **(λείπω)**

b^{h}ug-é/ó-* : >> ****b^{h}éu̯g-e/o-* **(φεύγω)**

b^{h}ud^{h}-é/ó-* : >> ****b^{h}éu̯d^{h}-e/o-* **(πεύθομαι)**

The expected full-grade thematic present of the two instances of **CeNT-* roots had not gained a strong enough foothold in Greek to survive in the extant materials due to the inactiveness of *a/en* ablaut:

g^{h}ad-é/ó-* : >> ?? (†χένδω) (g^{h}end-* 'get')

lag^{h}-é/ó-* : >> ?? (†λέγχω) (leng^{h}-* 'obtain by lot')

Step 6:

The nasal present to **CeRh₁*- roots is thematized (via 3sg. **CaR-n-e-toi*, etc.; see 3.3.3.1), offering the model THEM. AOR. + νω-PRES. whereby the suffix -νω spread to ἔλιπον/*λίμπμι etc.

ἔταμνον	:	*τάμνημι	=>	τάμ-νω		
ἔβαλον	:	*βάλλημι	=>	βάλ-λω		
⇩						
ἔλιπον	:	*λιμπμι	=>	*λίμπ-νω	>	λιμπάνω
ἐπυθόμην	:	*πύνθμαι	=>	*πύνθ-νομαι	>	πυνθάνομαι
ἔχαδον	:	*χάνδμι	=>	*χάνδ-νω	>	χανδάνω

Table 3.4 The development of the ἔλιπον-λιμπάνω-λείπω type (Step 1-6)

That *initially* the nasal present forms the basis of this process, i.e. **-NT-mi* => **-NT-n̥ne/o-*, rather than directly from the aorist via a "double-marking" rule, i.e. *CVT-é/ó-* => *CV-**N**-T-**n̥n**e/o-*, can be seen from the case of ἔδακον (Hom.+) : δάκνω (Thgn.+). The phonetically unproblematic [(†)]δαγκάνω (not attested until grammarians from the 1st cent. CE) is absent in the early materials probably due to the fact that the root **denḱ-* (LIV[2] *s.v.*) lacks an inherited nasal-infix present, and thus the newly generalized -νω suffix is attached directly to the aorist stem based on the pattern ἔταμον : τάμνω.[246] After the pattern ἔλιπον : λιμπάνω is established, a semi-productive synchronic derivational rule: aor. *CVT-e/o-* >> pres. *CV-N-T-ane/o-*, can then be extended to roots (or stems) that historically do not have a nasal present such as

[246] LIV[2] (*s.v.*) adds the possibility of an inherited nasal-*suffix* present 3pl. (**dn̥ḱ-n̥u̯-énti*) => **dn̥ḱ-n̥u̯-onti* => *δακάν(Ϝ)ω, but the subsequent replacement of *δακάνω with δάκνω seems unmotivated to me. Rix (1992: 210f.) cites 3pl. τίνουσι (< **kʷi-nu̯-onti* <= **kʷi-nu̯-enti*; cf. Ion. τῐ́νουσι) 'pay a price' as an example of a type of thematization underlying the νω-present. This type of process involving the loss of **u̯* must be considered relatively late. The more common outcome of thematization of original (or secondary) *-nū/nu-* presents in Greek is the present in -νυω, ὄρνυμι => ὄρνυω 'rise', τάνυμι => τάνυω 'stretch', cf. Schwyzer 1939: 698, Chantraine 1948: 473, Tucker 1990: 395, Sturm 2021: 99-103.

ἔλαθον >> λα-**ν**-θ-**άνω**. The double-nasal presents are created late enough that only a few of them were adopted by the bards and entered the epic diction (and those that did were still marginal there). The scenario proposed above is necessarily speculative. But it offers the advantages of providing a historical explanation to (1) the affinity between the double-nasal present and the thematic aorist (both having been analogically influenced by **CeRh*$_{1/3}$- roots where thematization of both stem formations originated) and (2) the paradoxical chronology between the synchronically young but well-supported on the basis of comparative evidence nasal present and the thematic present, which lacks external parallels but is the old stem variant within Greek.

Chapter 4
Concluding Remarks

In the preceding chapters I have attempted to clarify two problems in the history of the Greek thematic aorist: the origin of the vocalism in the so-called "*o*-grade" formations, and the mechanisms underlying the coordination patterns that connect the zero-grade thematic aorist with nasal and full-grade thematic presents. The main findings and arguments are summarized below.

4.1 Conclusion I: *$C_{([+labial])}\underset{\circ}{R}h_3V$- > *$C_{([+labial])}oRV$-

The analyses in Chapter 2 demonstrated that the aberrant *o*-grade thematic aorists are best understood not as products of isolated metatheses of *-*eRo*- > -*oRe*-. It was argued that such a process, whether starting from the future (e.g. *θερό-(σ)ομαι > θορέ-(σ)ομαι, as per Ruipérez 1950) or the 2sg. and 3sg of the root aorist (e.g. *ἔ-θερο-τ > ἔ-θορε-τ, Harðarson 1993: 169f.), finds little independent support in Greek and would have to operate in a sporadic, lexically restricted manner. Likewise, the analogical accounts by Kuryłowicz (1956: 207f.) based on the ablaut model *CRā : CaR* set by *$CeRh_2$- roots depend on paradigmatic relations that are weakly attested or structurally implausible. By contrast, the analysis developed here proposed that the *o*-vocalism is the regular Greek outcome of *$C\underset{\circ}{R}h_3V$- sequences, once the phonology of *h_3 in this environment is properly understood. In a detailed examination of the two alleged counterexamples of *$C\underset{\circ}{R}h_3V$- > *$CoRV$-, I demonstrated that (1) Pi. πεπαρεῖν based on its literary context is to be kept separate from ἔπορον and the root *$perh_3$-, even if a satisfying etymology is still lacking; and (2) ἐάλων is best interpreted as a root aorist *$\underset{\circ}{u}leh_3$-, as the existence of the state II full-grade and an intransitive, "stative"-like semantics ('die', 'fall') intrinsic to the root (i.e. not by the addition

of *-*eh*$_1$-) are fully supported by the Anatolian evidence. In a tangential but necessary digression on laryngeal coloring across morpheme boundaries, I looked at the tendencies of its operation (or the lack thereof) and illustrated the idea a stative *$\underset{\circ}{\hat{u}}$*lh*$_3$-*oh*$_1$- is not likely to survive PIE. In a more tentative attempt to expand the explanatory scope of our *$C\underset{\circ}{R}h_3V$ > *CoR-, it was suggested that two *o*-grade presents βόλομαι and ὄρομαι may find alternative interpretation as zero-grade *tudáti* presents under the proposed development. Finally, observing the data we came to the observation that the preceding [+labial] might be a conditioning factor of the *o*-vocalism outcomes and that such a conditioned reflex of *$\underset{\circ}{R}$ or *$\underset{\circ}{R}H$ is paralleled both Greek-internally and in Indo-Iranian.

4.2 Conclusion II: the history and origin of ἔλιπον-λιμπάνω-λείπω etc.

In Chapter 3 we examined how the coordination between the thematic aorist, the nasal present, and the full-grade thematic present arose in Greek. The philological survey revealed that (1) the thematic aorist often is the best represented stem and occupies a central position in these paradigms, (2) the double-nasal presents have relatively weak attestational records in the early materials, but many including λιμπάνω, φυγγάνω, πυνθάνομαι, and possibly ἀνδάνω and κυνθάνω are supported with comparative evidence of an inherited nasal-infix present, and (3) with the exception of ἀνδάνω the thematic presents in this group, though often isolated in Indo-European, are the standard, dominant present type in Homer, only later superseded by the double-nasal present in some cases (πεύθομαι, λήθω). I then presented a theory on the genesis of the stem distribution pattern of ἔλιπον-λιμπάνω-λείπω based on the analogical spread of the thematized forms from *$CeRh_{1/3}$- roots. There thematization happened early in the aorist due to the behaviors of the laryngeals in certain forms in the root aorist paradigm. Sharing the original stem pairing of ROOT AOR. + NASAL PRES., *$CeRT$- roots then analogically remodeled their root aorist into a thematic one (e.g. *lik^we-*t* : *li-ne-k^w-*ti*) following the model of *$CeRh_{1/3}$- roots (*tame-t* : *tam-nē-ti*). The thematic present is

proposed as a secondary, albeit archaic, innovation in Greek driven by the obsolescence of the nasal-infix presents to stop-final roots (such as *λίμπμι). I suggested that the model for ἔλιπον >> λείπω is the ample stock of verbs with the pairing of THEM. AOR. + ROOT THEM. PRES. of the type ἔσχον : ἔχω. The emergence of the double-nasal formation is construed as a relatively late, final step after the thematization of the nasal-infix present to *h_1-final roots was completed and the pattern of THEM. AOR. + νω-present (e.g. ἔταμον : τάμνω) established.

4.3 Final thoughts

The broader theoretical goal of the investigation of the "ἔλιπον-λιμπάνω-λείπω" triads is to situate the quest for the origins and spread of the Greek thematic aorist (and potentially other verbal stems) not just in the local peculiarities of the individual paradigms – whether they are the analogical 3pl. *-*ent* => *-*ont* envisioned by Cardona or the chameleonic nature of the **C(e)RH*- root structure in Harđarson's system – but also in the stem pairing patterns as reflected in the historical materials. In a crude comparison, as the famous maxim in distributional semantics goes "you shall know a word by the company it keeps" (Firth 1957: 11), one may say that much can be known or inferred about the prehistory of a verbal stem by the close examination of what stem formations it coexists with. Although differing from this study in the interpretation of the Greek data, Rau's recent treatment (2024) on the close relationship between the thematic aorist and the nasal present across Greek, Slavic, and Indo-Iranian marks additional progress in that direction. What remains to be investigated next is the other major stem pairing pattern in Greek involving the thematic aorist, namely THEM. AOR. + ROOT THEM. PRES (e.g. ἔσχον : ἔχω). As briefly mentioned earlier (cf. 3.3.2.1, 3.3.3.3), the type is highly prolific since Homer and consists of some of the canonical cases of the intransitive use of the active thematic aorist (in otherwise transitive verbs) such as ἔτραπον : τρέπομαι 'turn (intr.)'. Many verbs in this group have a polymorphic (sometimes

contrastive in transitivity) *s*-aorist, e.g. ἔτραπον : ἔτρεψα, ἔπραθον : ἔπερσα ‘ruin’, etc., which is entirely absent among verbs possessing the thematic aorist and the νω-present. Atelicity is significantly more prominently represented here than the ἔλιπον-λιμπάνω-λείπω type as well (cf. fn.244), e.g. ἔτραφον ‘nurture’, ἔτραπον ‘turn’, ἔδρακον ‘look’, ἐσπόμην ‘follow’, ἔσχον ‘hold’, ἤρυγον ‘belch’, etc. Certain root shapes such as **CeT(H)*- and **CeRT*- are common in this category but are not found in the group examined in the present study. These peculiarities combined raise the possibility that a separate “ἔτραπον-τρέπω-ἔτρεψα” pattern might exist as a distinct group with a different prehistory from the ἔλιπον-λιμπάνω-λείπω type. Further research is required to understand how such a pattern arose and the mechanism behind the origin of the polymorphic thematic and sigmatic aorist in these verbs.

* * *

Abbreviations:

DELG = Chantraine, Pierre. 1980.
DMic = Aura Jorro, Francisco. 1985.
EDG = Beekes, Robert S. P. 2010.
EDHIL = Kloekhorst, Alwin. 2008.
eDiAna = Digital Philological-Etymological Dictionary of the Minor Ancient Anatolian Corpus Languages. https://www.ediana.gwi.uni-muenchen.de/.
EDLIL = de Vaan, Michiel. 2008
EWAia = Mayrhofer, Manfred. 1986-2001.
GEW = Frisk, Hjalmar. 1960-72.
IEW = Pokorny, Julius. 1959.
IG = *Inscriptiones Graecae*. 1877-. Berlin.
LEW = Walde, Alois, and J. B. Hofmann. 1938-1954.
LfgrE = *Lexikon des frühgriechischen Epos*. 1944-2010. Göttingen: Vandenhoeck & Ruprecht.
LIV^2 = Rix, Helmut. 2001.
LSJ = Liddell, Henry G., Robert Scott, Henry S. Jones et al. 1996.
NIL = Wodtko, Dagmar S., Britta Sofie Irslinger, and Carolin Schneider. 2008.

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